POST-PUBLICATION NOTE

Profits, Scale Economies, and the Gains from Trade and Industrial Policy

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Introduction

In Lashkaripour and Lugovskyy (2023), we characterize unilaterally optimal trade and industrial policies in a multi-country, multi-industry general equilibrium model with scale economies, profits, and input–output linkages. We also quantify the gains from these policies for 43 economies.

This note updates the paper in four respects. First, we summarize the existence and uniqueness result of Lashkaripour (2026) for the free-entry version of the model, together with the nested algorithm that its proof delivers (Section 1). Second, we update the second-best policy formulas of Theorem 2 and the third-best tariff formula of Theorem 3 by removing the approximations used in their proofs under restricted entry and deriving their counterparts under free entry (Section 2, with proofs in Appendix A). Section 2 also reports an updated version of Table 4, in which the new formulas replace the numerical optimization behind the free-entry columns. Third, we restore a term omitted from the conditional inverse export supply elasticity $\omega_{ji,k}$, which enters the optimal import tax of Theorem 1; with the term restored, we update the first-order approximation of $\omega_{ji,k}$ in equation (10) (Section 3). The restored term leaves every quantitative result and conclusion of the paper unchanged (Table 2 and Figure 4). Fourth, we update the first-best policy formulas of Theorem 4, which allow for input–output linkages, by restoring the omitted term in $\omega_{ji,k}$, allowing foreign wages to adjust, and removing the fixed-wage pass-through approximation in the export-subsidy wedge. The architecture of the theorem and its domestic subsidy are unchanged, the published formulas deliver over 99.9 percent of the optimal welfare gains in simulated economies, and we amend footnote 40 accordingly (Section 4, with the proof in Appendix A).

1 Existence and uniqueness of equilibrium

Lashkaripour (2026) establishes the existence and global uniqueness of equilibrium in the free-entry, Cobb–Douglas version of the model of Section I of the paper, without taxes and without input–output linkages. The result holds for any number of countries, for bilateral trade costs that need not be symmetric, and for trade and scale elasticities that differ across industries, provided that the

product of the trade and scale elasticities is below one in every industry, $(\sigma_k - 1)\mu_k < 1$. Under this restriction, Kucheryavyy, Lyn, and Rodríguez-Clare (2023) show that this class of models is well-behaved: given wages, the labor allocation is unique, and under frictionless trade the equilibrium is unique. The new result covers the general case with trade costs. This section states the equilibrium system in shares, the result, the proof in Steps 1–7, and the nested algorithm in exact hat algebra that the proof delivers.

The equilibrium system in shares. Country i has L_i workers, wage w_i, income $Y_i = w_i L_i$, and Cobb–Douglas expenditure shares $e_{i,k}$ with $\sum_k e_{i,k} = 1$. Let $\rho_{i,k} \equiv L_{i,k}/L_i$ denote the share of country i's labor employed in industry k, and let

$$\alpha_k \equiv (\sigma_k - 1)\mu_k \tag{1}$$

denote the product of the trade elasticity and the scale elasticity. Under free entry the number of firms is proportional to employment, so the price index of industry k goods from i in market n is proportional to $\tau_{in,k}\, w_i\, L_{i,k}^{-\mu_k}$, and the expenditure share is

$$\lambda_{in,k} = \frac{\Xi_{in,k}\, L_{i,k}^{\alpha_k}\, w_i^{1-\sigma_k}}{\sum_j \Xi_{jn,k}\, L_{j,k}^{\alpha_k}\, w_j^{1-\sigma_k}}, \tag{2}$$

where $\Xi_{in,k} > 0$ collects $\tau_{in,k}^{1-\sigma_k}$ and the constant entry and unit labor cost terms, and depends neither on wages nor on employment. Labor market clearing in industry k of country i requires that the wage bill equal sales,

$$w_i L_{i,k} = \sum_n \lambda_{in,k}\, e_{n,k}\, Y_n \quad \text{whenever } L_{i,k} > 0, \qquad \sum_k L_{i,k} = L_i. \tag{3}$$

Normalize world income to one, $\sum_i Y_i = 1$. Substituting $L_{i,k} = L_i \rho_{i,k}$ and $w_i = Y_i/L_i$ into (2) expresses the trade share in terms of incomes and employment shares,

$$\lambda_{in,k}(Y,\rho) = \frac{\Xi_{in,k}\, L_i^{\alpha_k+\sigma_k-1}\, \rho_{i,k}^{\alpha_k}\, Y_i^{1-\sigma_k}}{\sum_j \Xi_{jn,k}\, L_j^{\alpha_k+\sigma_k-1}\, \rho_{j,k}^{\alpha_k}\, Y_j^{1-\sigma_k}}. \tag{4}$$

Define the market clearing residual of (3) and the residual per unit of employment share,

$$H_{i,k}(Y,\rho) \equiv Y_i \rho_{i,k} - \sum_n \lambda_{in,k}(Y,\rho)\, e_{n,k}\, Y_n, \tag{5}$$

$$G_{i,k}(Y,\rho) \equiv \frac{H_{i,k}(Y,\rho)}{\rho_{i,k}} = Y_i - \frac{1}{\rho_{i,k}} \sum_n \lambda_{in,k}(Y,\rho)\, e_{n,k}\, Y_n, \tag{6}$$

so that $G_{i,k}$ equals L_i times the wage net of sales per worker in industry k. We define $G_{i,k}$ at $\rho_{i,k} = 0$ as its limit as $\rho_{i,k} \downarrow 0$, holding Y and the other shares fixed. An equilibrium is a pair (Y,ρ) with

$\sum_i Y_i = 1$, $\sum_k \rho_{i,k} = 1$ for every i, and, for every i and k,

$$G_{i,k}(Y,\rho) \geq 0, \qquad \rho_{i,k} \geq 0, \qquad \rho_{i,k}\, G_{i,k}(Y,\rho) = 0. \tag{7}$$

Condition (7) is complementary slackness: an industry either employs workers, with sales per worker equal to the wage, or is shut down, with sales per worker below the wage. Wages and employment follow from $w_i = Y_i / L_i$ and $L_{i,k} = L_i \rho_{i,k}$.

THEOREM (LASHKARIPOUR 2026)

Let $L_i > 0$, $e_{n,k} > 0$, $\Xi_{in,k} > 0$, $\sigma_k > 1$, and

$$0 \leq \alpha_k < 1 \qquad \text{for every industry } k. \tag{8}$$

Then an equilibrium exists; every equilibrium has $L_{i,k} > 0$ for every country i and industry k; and any two equilibria have the same employment $L_{i,k}$, the same trade shares $\lambda_{in,k}$, and wages that differ by a common factor. Equivalently, (7) has exactly one solution (Y,ρ). The bound in (8) is sharp: with $\alpha_k = 1$ in some industry, the set of equilibria can be a continuum, or every equilibrium can have $L_{i,k} = 0$ for some i and k.

The proof in Steps 1–7. Each step is a lemma of Lashkaripour (2026). The condition $\alpha_k < 1$ is used in Steps 1, 3, and 6.

Step 1 (no equilibrium at a corner). Fix $Y > 0$ and the employment shares of the other industries. By (4),

$$\frac{\lambda_{in,k}(Y,\rho)}{\rho_{i,k}} = \frac{\Xi_{in,k}\, L_i^{\alpha_k+\sigma_k-1}\, Y_i^{1-\sigma_k}\, \rho_{i,k}^{\alpha_k-1}}{\sum_j \Xi_{jn,k}\, L_j^{\alpha_k+\sigma_k-1}\, \rho_{j,k}^{\alpha_k}\, Y_j^{1-\sigma_k}}. \tag{9}$$

As $\rho_{i,k} \downarrow 0$ the numerator goes to infinity because $\alpha_k < 1$, while the denominator is bounded above; the ratio therefore goes to infinity (it equals $\rho_{i,k}^{-1}$ when no other country employs labor in industry k). Hence $G_{i,k}(Y,\rho) = -\infty$ whenever $\rho_{i,k} = 0$: in an industry with no employment, sales per worker exceed the wage by any margin, so (7) fails. Every equilibrium has $\rho_{i,k} > 0$ for all i and k.

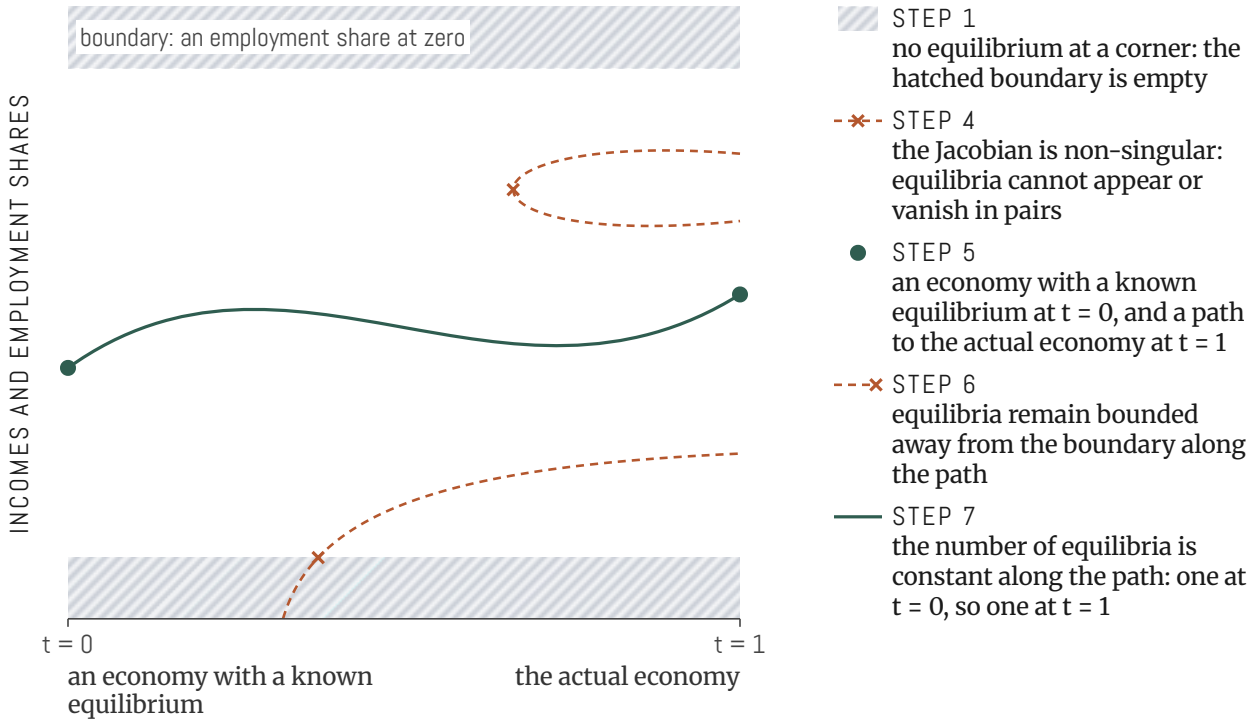
Step 2 (interior equivalence and Walras's law). By Step 1, (Y,ρ) solves (7) if and only if $\rho > 0$ and $H_{i,k}(Y,\rho) = 0$ for all i and k. These are NK equations in the $NK - 1$ variables left free by $\sum_i Y_i = 1$ and $\sum_k \rho_{i,k} = 1$, but one of them is redundant, because $\sum_i \lambda_{in,k} = 1$ and $\sum_k \rho_{i,k} = \sum_k e_{i,k} = 1$ give, at every interior point,

$$\sum_i \sum_k H_{i,k}(Y,\rho) = \sum_i Y_i - \sum_k \sum_n e_{n,k} Y_n = 0. \tag{10}$$

Step 3 (two matrices per industry). At any interior point let $X_{in,k} \equiv \lambda_{in,k}\, e_{n,k}\, Y_n > 0$ denote bilateral sales and $X_{i,k} \equiv \sum_n X_{in,k}$ industry sales, and define the $N \times N$ matrices

$$[A_k]_{in} \equiv \frac{X_{in,k}}{X_{i,k}}, \qquad [\Lambda_k]_{in} \equiv \lambda_{in,k}, \qquad P_k \equiv A_k \Lambda_k^{\top}, \qquad B_k \equiv (1-\alpha_k) I + \alpha_k P_k. \tag{11}$$

Figure 1: *The proof of existence and uniqueness: a path from a known equilibrium to the actual economy*



Notes: The horizontal axis is the path parameter t of Step 5, from an economy with a known equilibrium ($t = 0$) to the actual economy ($t = 1$); the vertical axis stands for incomes and employment shares. The hatched bands are the boundary, where an employment share is zero; Step 1 rules out equilibria there. The solid line is the equilibrium followed along the path. The dashed lines are the two events that Steps 4 and 6 exclude: a pair of equilibria appearing or vanishing, and an equilibrium reaching the boundary. Step 7 then counts the equilibria along the path: one at $t = 0$, so one at $t = 1$. The steps are those of the proof below.

A_k averages a destination vector over markets with exporter i's sales weights; $\Lambda_k^\top$ averages a source vector over exporters with market n's expenditure shares. P_k is positive and its rows sum to one. With $M_k \equiv \operatorname{diag}(X_{i,k})$ and $E_k \equiv \operatorname{diag}(e_{n,k}Y_n)$,

$$M_k P_k = \Lambda_k E_k \Lambda_k^\top, \tag{12}$$

so P_k is symmetric in the inner product $\langle f, h\rangle_k \equiv f^\top M_k h = \sum_i X_{i,k} f_i h_i$, its eigenvalues lie in $[0, 1]$, and $P_k f = f$ only if f is constant across countries. The eigenvalues of B_k therefore lie in $[1 - \alpha_k, 1]$: B_k is invertible because $\alpha_k < 1$. Two inequalities follow, for any source vector f and destination vector g:

$$\left\langle f, B_k^{-1}(f - A_k g)\right\rangle_k \geq \tfrac{1}{2}\left(f^\top M_k f - g^\top E_k g\right), \tag{13}$$

$$\left\langle f, B_k^{-1}(I - P_k) f\right\rangle_k \geq 0, \quad \text{with equality if and only if } P_k f = f. \tag{14}$$

Step 4 (the Jacobian is non-singular at every equilibrium). Consider log changes $u_i \equiv d \ln Y_i$ and

$v_{i,k} \equiv d \ln \rho_{i,k}$ that remain within the constraints,

$$\sum_i Y_i u_i = 0, \qquad \sum_k \rho_{i,k} v_{i,k} = 0 \quad \text{for every } i. \tag{15}$$

Differentiating (4),

$$d \ln \lambda_{in,k} = \alpha_k v_{i,k} - (\sigma_k - 1) u_i - \sum_j \lambda_{jn,k} \big(\alpha_k v_{j,k} - (\sigma_k - 1) u_j \big). \tag{16}$$

The log change of the left side of (3) is $u_i + v_{i,k}$, and that of the right side is the sales-weighted average of $d \ln \lambda_{in,k} + u_n$ over markets. A change (u, v) at which the derivative of (3) is zero sets the two equal; in matrix form, with $v_k \equiv (v_{i,k})_i$ and $P_k = A_k \Lambda_k^\top$,

$$B_k v_k + (I - A_k) u + (\sigma_k - 1)(I - P_k) u = 0, \tag{17}$$

so $v_k = -B_k^{-1}[(I - A_k)u + (\sigma_k - 1)(I - P_k)u]$. Because $M_k = \operatorname{diag}(Y_i \rho_{i,k})$ at an equilibrium, the second condition in (15), weighted by $Y_i u_i$ and added over i, gives $\sum_k \langle u, v_k \rangle_k = 0$. Substituting v_k and applying (13) with $f = g = u$ and (14),

$$\begin{aligned} 0 &= \sum_k \Big\langle u, B_k^{-1} \big[(I - A_k) u + (\sigma_k - 1)(I - P_k) u \big] \Big\rangle_k \\ &\geq \tfrac{1}{2} \sum_k \big(u^\top M_k u - u^\top E_k u \big) + \sum_k (\sigma_k - 1) \big\langle u, B_k^{-1} (I - P_k) u \big\rangle_k . \end{aligned} \tag{18}$$

The first sum is zero because $\sum_k Y_i \rho_{i,k} = Y_i = \sum_k e_{i,k} Y_i$, and every term of the second sum is non-negative since $\sigma_k - 1 > 0$. Hence $P_k u = u$ for every k, so u is constant across countries, and the first condition in (15) makes it zero; then (17) gives $v_k = 0$. One of the NK equations is redundant by (10) and can be omitted; the Jacobian of the remaining $NK - 1$ equations with respect to the $NK - 1$ free variables is then non-singular at every equilibrium.

Step 5 (an economy with a known equilibrium, and a path to the actual one). The economy with the same L_i but $\Xi_{in,k} = 1$, $e_{n,k} = 1/K$, $\sigma_k = 2$, and $\alpha_k = 0$ has the unique equilibrium

$$Y_i = \frac{\sqrt{L_i}}{\sum_j \sqrt{L_j}}, \qquad \rho_{i,k} = \frac{1}{K}. \tag{19}$$

With these parameters (4) becomes $\lambda_{in,k} = L_i Y_i^{-1} / \sum_j L_j Y_j^{-1}$ for every n and k, so market clearing gives $Y_i \rho_{i,k} = \frac{1}{K} L_i Y_i^{-1} / \sum_j L_j Y_j^{-1}$ for every k; hence $\rho_{i,k} = 1/K$ and, adding over k, $Y_i^2 \propto L_i$. A path from this economy to the actual one is, for $t \in [0, 1]$,

$$\begin{aligned} \Xi_{in,k}(t) &= \Xi_{in,k}^t, \qquad & e_{n,k}(t) &= \frac{1 - t}{K} + t\, e_{n,k}, \\ \sigma_k(t) &= 2(1 - t) + t\, \sigma_k, \qquad & \alpha_k(t) &= t\, \alpha_k, \end{aligned} \tag{20}$$

along which the parameters satisfy the conditions of the Theorem at every t and $\max_{t,k} \alpha_k(t) = \max_k \alpha_k < 1$.

Step 6 (equilibria remain bounded away from the boundary along the path). There are constants $\underline{Y}, \underline{\rho} > 0$, which depend only on the path, such that every equilibrium at every t satisfies $Y_i \geq \underline{Y}$ and $\rho_{i,k} \geq \underline{\rho}$. The bound on incomes is proved first. Given that bound, and because $\Xi_{in,k}(t)$, L_i, and $Y_i \leq 1$ are bounded, (4) gives $\lambda_{in,k} \geq c\, \rho_{i,k}^{\alpha_k}$ for a constant $c > 0$; with $\underline{e} \equiv \min_{t,n,k} e_{n,k}(t) > 0$, market clearing then gives

$$Y_i \rho_{i,k} = \sum_n \lambda_{in,k}\, e_{n,k}\, Y_n \geq c\, \underline{e}\, \rho_{i,k}^{\alpha_k} \sum_n Y_n = c\, \underline{e}\, \rho_{i,k}^{\alpha_k}, \tag{21}$$

and $Y_i \leq 1$ yields $\rho_{i,k}^{1-\alpha_k} \geq c\, \underline{e}$, that is, $\rho_{i,k} \geq (c\, \underline{e})^{1/(1-\alpha_k)}$. The bound keeps $\rho_{i,k}$ away from zero only because $1 - \alpha_k > 0$.

Step 7 (the number of equilibria is constant along the path). Along (20) the equilibria lie in a compact subset of the interior (Step 6), the Jacobian is non-singular at each of them (Step 4), and the parameters have a continuous derivative in t. Hence the number of equilibria is finite and does not change with t. It is one at $t = 0$ by (19), so it is one at $t = 1$. Step 2 turns this interior solution into an equilibrium (7), and Step 1 rules out any other. Finally, if all wages are scaled by $s > 0$, every term $\Xi_{in,k} L_{i,k}^{\alpha_k} w_i^{1-\sigma_k}$ in (2) is scaled by $s^{1-\sigma_k}$ and (Y, ρ) is unchanged; conversely, equal (Y, ρ) means equal $L_{i,k}$, equal $\lambda_{in,k}$, and proportional wages.

At $\alpha_k = 1$ the three uses of the condition fail at once: the ratio in (9) remains finite when another country employs labor in industry k, so corner equilibria become possible; $B_k = P_k$ can be singular; and (21) no longer keeps $\rho_{i,k}$ away from zero. Lashkaripour (2026) gives examples with a continuum of equilibria and with only corner equilibria.

A nested algorithm in exact hat algebra. The proof delivers a procedure for computing counterfactual equilibria. Let $\hat{z} = z^*/z$ denote the change in a variable between the observed equilibrium and the counterfactual one. The data are the observed trade shares $\lambda_{in,k} > 0$, incomes Y_i normalized so that $\sum_i Y_i = 1$, and expenditure shares $e_{n,k}$. The observed employment shares follow from the data and market clearing (3),

$$\rho_{i,k} = \frac{1}{Y_i} \sum_n \lambda_{in,k}\, e_{n,k}\, Y_n. \tag{22}$$

The shock is a change in trade costs, $\hat{\tau}_{in,k}$, which affects (2) through $\hat{\Xi}_{in,k} = \hat{\tau}_{in,k}^{1-\sigma_k}$; the same applies to any other shock to $\Xi_{in,k}$. Labor endowments are fixed, so $Y_i^* = \hat{w}_i Y_i$, and the unknowns are $\hat{w}_i$ and $\rho_{i,k}^*$, with $\hat{\rho}_{i,k} = \rho_{i,k}^*/\rho_{i,k}$. Expressing (2) in changes,

$$\lambda_{in,k}^* = \frac{\lambda_{in,k}\, \hat{\Xi}_{in,k}\, \hat{\rho}_{i,k}^{\alpha_k}\, \hat{w}_i^{1-\sigma_k}}{\sum_j \lambda_{jn,k}\, \hat{\Xi}_{jn,k}\, \hat{\rho}_{j,k}^{\alpha_k}\, \hat{w}_j^{1-\sigma_k}}, \tag{23}$$

and the counterfactual equilibrium solves

$$\hat{w}_i Y_i\, \rho_{i,k}^* = \sum_n \lambda_{in,k}^*\, e_{n,k}\, \hat{w}_n Y_n, \qquad \sum_k \rho_{i,k}^* = 1, \qquad \sum_i \hat{w}_i Y_i = 1. \tag{24}$$

System (23)–(24) is the equilibrium system (4)–(7) of an economy with unit labor endowments and pricing terms $\lambda_{in,k}\hat{\Xi}_{in,k}\rho_{i,k}^{-\alpha_k}Y_i^{\sigma_k-1} > 0$. By the Theorem, its solution exists, is unique, and has $\rho_{i,k}^* > 0$ for all i and k, so the complementary slackness conditions (7) need not be imposed.

For values $(\hat{w}, \rho^*)$ that need not solve (24), define the counterfactual sales

$$X_{in,k}^* \equiv \lambda_{in,k}^* e_{n,k}\hat{w}_n Y_n, \qquad X_{i,k}^* \equiv \sum_n X_{in,k}^*, \qquad X_i^* \equiv \sum_k X_{i,k}^*. \tag{25}$$

The identity $\sum_i X_{in,k}^* = e_{n,k}\hat{w}_n Y_n$ holds at every such point; $X_{i,k}^* = \hat{w}_i Y_i \rho_{i,k}^*$ holds only at the solution. The matrices of (11) are computed from $X_{in,k}^*$ and $\lambda_{in,k}^*$ at the given point. The algorithm has an inner and an outer loop.

Inner loop (employment shares given wages). Given $\hat{w}$, update

$$\rho_{i,k}^* \longleftarrow \frac{X_{i,k}^*(\hat{w}, \rho^*)}{X_i^*(\hat{w}, \rho^*)}. \tag{26}$$

At a solution of (26), employment shares equal sales shares, $X_{i,k}^* = \rho_{i,k}^* X_i^*$. Differentiating (23) and (25) as in (16), with $u \equiv d\ln\hat{w}$ and $v_k \equiv (d\ln\rho_{i,k}^*)_i$,

$$d\ln X_{\cdot,k}^* = \alpha_k(I - P_k)\, v_k + D_k\, u, \qquad D_k \equiv A_k - (\sigma_k - 1)(I - P_k). \tag{27}$$

The derivative of the update (26) with respect to v_k is $\alpha_k(I - P_k)$ followed by the normalization that imposes $\sum_k \rho_{i,k}^* = 1$; the eigenvalues of $\alpha_k(I - P_k)$ lie in $[0, \alpha_k]$ and the normalization does not increase distances. Hence the inner loop is a contraction near its solution with rate $\max_k \alpha_k$, and it is exact after one update when every $\alpha_k = 0$.

Outer loop (wages). Let $\rho^*(\hat{w})$ denote the solution of the inner loop and define the country-level residual

$$G_i(\hat{w}) \equiv \ln X_i^*\big(\hat{w}, \rho^*(\hat{w})\big) - \ln\big(\hat{w}_i Y_i\big). \tag{28}$$

$G(\hat{w}) = 0$ if and only if $(\hat{w}, \rho^*(\hat{w}))$ solves (24). The numeraire in (24) is needed for two reasons: G does not change when $\hat{w}$ is scaled by a constant, and, by Walras's law, $\sum_i X_i^* = \sum_i \hat{w}_i Y_i$, so one component of G is redundant. Differentiating (26) at its solution gives $v_k + d\ln X^* = d\ln X_{\cdot,k}^*$; substituting (27) gives $B_k v_k = D_k u - d\ln X^*$, and imposing $\sum_k \rho_{i,k}^* v_{i,k} = 0$ yields the Jacobian of G in closed form,

$$dG = \big(\Phi^{-1}\Psi - I\big)\, du, \qquad \Phi \equiv \sum_k \mathrm{diag}(\rho_{\cdot,k}^*)\, B_k^{-1}, \qquad \Psi \equiv \sum_k \mathrm{diag}(\rho_{\cdot,k}^*)\, B_k^{-1} D_k. \tag{29}$$

Φ is invertible at every interior point; $(\Psi - \Phi)\mathbf{1} = 0$ because $D_k\mathbf{1} = \mathbf{1} = B_k\mathbf{1}$; and at the solution $(\Psi - \Phi)x = 0$ only if x is proportional to $\mathbf{1}$, by Step 4. The Newton step therefore solves

$$(\Psi - \Phi)\,\Delta u = -\Phi\, G(\hat{w}), \qquad \text{with one row replaced by } \sum_i \hat{w}_i Y_i\, \Delta u_i = 0, \tag{30}$$

and applies the update $\hat{w}_i \longleftarrow \hat{w}_i \exp(\theta\,\Delta u_i)$ with a step size $\theta \in (0,1]$ that reduces $\max_i |G_i|$, followed by the normalization $\sum_i \hat{w}_i Y_i = 1$. Each step costs K linear solves of size N for the B_k^{-1} and one for (30); no numerical derivatives are used, and the Newton iteration converges rapidly near the solution.

Converging globally. The algorithm first applies Newton's method from $\hat{w} = \mathbf{1}$ at the full shock. If it fails, the algorithm follows the path

$$\hat{\Xi}_{in,k}(t) = \hat{\Xi}^t_{in,k}, \qquad t \in [0,1], \tag{31}$$

which starts at the observed equilibrium: $\hat{w} = \mathbf{1}$ and $\rho^* = \rho$ at $t = 0$. For every t the pricing terms $\lambda_{in,k}\hat{\Xi}^t_{in,k}$ are positive and $\alpha_k < 1$, so Steps 4, 6, and 7 apply: the solution $(\hat{w}(t), \rho^*(t))$ is unique, interior, continuous in t, and has a non-singular Jacobian. The algorithm increases t in steps, solving the nested problem at each t from the solution at the previous one, decreasing the step when Newton fails and increasing it otherwise. Because the solution path cannot leave the interior and the Jacobian cannot become singular, the procedure is certain to reach $t = 1$.

The log form and the complementary slackness test. At any point, the per-unit slack of (6) in hat notation and the log residual of (26) are related by

$$G_{i,k} \equiv \hat{w}_i Y_i - \frac{X^*_{i,k}}{\rho^*_{i,k}}, \qquad \ln \frac{X^*_{i,k}}{\hat{w}_i Y_i\, \rho^*_{i,k}} = \ln\left(1 - \frac{G_{i,k}}{\hat{w}_i Y_i}\right), \tag{32}$$

which is zero if and only if $G_{i,k} = 0$, positive if and only if sales per worker exceed the wage ($G_{i,k} < 0$), and goes to $+\infty$ as $\rho^*_{i,k} \downarrow 0$, because $X^*_{i,k} \propto (\rho^*_{i,k})^{\alpha_k}$ and $\alpha_k < 1$, as in Step 1. The level residual $\hat{w}_i Y_i \rho^*_{i,k} - X^*_{i,k}$, in contrast, vanishes as $\rho^*_{i,k} \downarrow 0$, so for a solver in levels $\rho^*_{i,k} = 0$ solves the equation $0 = 0$ without being an equilibrium. The log form (26)–(28) rules this out. Because the solution of (24) is unique, the point the algorithm delivers is the counterfactual equilibrium, and no search over starting points is needed. Each part of the algorithm depends on $1 - \alpha_k$: the inner rate is $\max_k \alpha_k$, the condition number of B_k is at most $1/(1-\alpha_k)$, and the bound of Step 6 is $\rho_{i,k} \geq (c\,\underline{e})^{1/(1-\alpha_k)}$. The calculations of Section 2 use this nested structure, with the inner loop in employment shares and the outer Newton step in wages, in the full quantitative model with taxes.

2 Updated second- and third-best policy formulas

2.1 Second-best optimal tariffs

Theorem 2 of the paper (p. 2778) characterizes the second-best policy when domestic industrial subsidies are unavailable. The policy consists of import tariffs and export subsidies. The proof of Theorem 2 treats entry as restricted and uses a first-order approximation around $\mu_k \approx \bar{\mu}_i$ (online Appendix I, footnote 29). Under this approximation, the average markup that enters the formulas is the sales-weighted average $\bar{\mu}_i$ of equation (5), and the inverse export supply elasticities are evaluated with equation (10), the first-order approximation of $\omega_{ji,k}$ updated in Section 3. The

free-entry second-best results of the paper (Table 4) were computed by numerical optimization, because no formula was available. Below we state Theorem 2 under restricted entry without the approximation and its free-entry counterpart, Theorem 2′. Both theorems take foreign governments as passive (no foreign taxes) and characterize regular interior optima up to the uniform tax shifter of Lerner symmetry. They do not claim that the optimum is unique. Each theorem is followed by the Cobb–Douglas case used in the quantitative analysis and by the closed form with two countries. Section 2.2 gives the corresponding update of Theorem 3, the third best with import tariffs only, under restricted and under free entry.

THEOREM 2 (RESTRICTED ENTRY, WITHOUT APPROXIMATIONS)

Suppose entry is restricted, the government is unable (or unwilling) to apply domestic industrial subsidies, foreign governments impose no taxes, and relative foreign effective wages are held constant, as stipulated in the paper's derivation. At any regular interior optimum, the second-best import tariffs and export subsidies are, up to a uniform tax shifter $\bar{t}_i \in \mathbb{R}_+$, implicitly given by

$$\begin{aligned} 1 + t_{ji}^{**} &= (1+\bar{t}_i)(\mathbf{1}+\Omega_{ji}) \oslash \left[\mathbf{1} + E_{-ii}^{-1} E_{-ii}^{(ii)} \left(1 - \tfrac{1+\tilde{\mu}_i}{1+\mu_k}\right)_k\right], \\ 1 + x_{ij}^{**} &= -(1+\bar{t}_i)\left[E_{ij}^{-1} E_{ij}^{(-ij)} (\mathbf{1}+\Omega_{-ii})\right] \odot \left(\tfrac{1+\mu_k}{1+\tilde{\mu}_i}\right)_k, \end{aligned} \tag{33}$$

where $\tilde{\mu}_i$ is the demand-weighted average markup of economy i, which replaces the sales-weighted average markup $\bar{\mu}_i$ of equation (5),

$$\tilde{\mu}_i = \frac{\sum_k v_{i,k}\, \mu_k/(1+\mu_k)}{\sum_k v_{i,k}/(1+\mu_k)}, \qquad v_i = e_{ii} - \left(E_{-ii}^{-1} E_{-ii}^{(ii)}\right)^\top e_{-ii},$$

with $e_{ii} = (e_{i,k}\lambda_{ii,k})_k$ and $e_{-ii} = (e_{i,k}\lambda_{ni,k})_{n\neq i,k}$ the expenditure shares on domestic and imported goods. The vector $\Omega_{ji} = [\omega_{ji,k}]_k$ collects the inverse export supply elasticities. They solve the import-valuation and labor-market clearing conditions of the exporting economies at constant relative foreign wages, a linear system in foreign sales shares, markups, and demand elasticities that is given in Appendix A and has the closed form below with two countries. The matrices E_{-ii}, $E_{-ii}^{(ii)}$, E_{ij}, and $E_{ij}^{(-ij)}$ are the Marshallian demand elasticities defined in Theorem 2.[1]

Replacing the average markup $\tilde{\mu}_i$ with $\bar{\mu}_i$, and $\omega_{ji,k}$ with its equation (10) value, recovers Theorem 2 as published. Within each exporting partner, the inverse export supply elasticity is affine in $1/(1+\mu_k)$ across industries, so it is the same in every industry when markups are uniform.

[1] The general display assumes homothetic preferences across industries, under which the expenditure-weighted elasticity blocks of the derivation coincide with the E matrices of the paper's Definition (D1). With nonhomothetic preferences, the blocks are replaced by their expenditure-weighted transposes, and the shares in v_i are scaled by income elasticities (Appendix A). The Cobb–Douglas case below is the one used in the quantitative analysis.

Cobb–Douglas preferences. With Cobb–Douglas preferences, the formulas of Theorem 2 become

$$1+t^{**}_{ji,k} = \frac{1+(\sigma_k-1)\lambda_{ii,k}}{1+\frac{1+\tilde{\mu}_i}{1+\mu_k}(\sigma_k-1)\lambda_{ii,k}}\,(1+t^{*}_{ji,k}), \tag{34}$$
$$1+x^{**}_{ij,k} = \frac{1+\mu_k}{1+\tilde{\mu}_i}\,(1+x^{*}_{ij,k}),$$

where the first-best components are

$$1+t^{*}_{ji,k} = (1+\omega_{ji,k})(1+\bar{t}_i), \qquad 1+x^{*}_{ij,k} = \frac{(\sigma_k-1)\sum_{n\neq i}[(1+\omega_{ni,k})\lambda_{nj,k}]}{1+(\sigma_k-1)(1-\lambda_{ij,k})}(1+\bar{t}_i),$$

and

$$\tilde{\mu}_i \equiv \frac{\sum_k \vartheta_{i,k}\mu_k}{\sum_k \vartheta_{i,k}}, \qquad \vartheta_{i,k} \equiv \frac{e_{i,k}\lambda_{ii,k}\sigma_k}{[1+(\sigma_k-1)\lambda_{ii,k}](1+\mu_k)}.$$

The weights $\vartheta_{i,k}$ tilt the sales weights of $\bar{\mu}_i$ by the factor $\sigma_k/[1+(\sigma_k-1)\lambda_{ii,k}]$. Replacing $\tilde{\mu}_i$ with $\bar{\mu}_i$ reproduces the Cobb–Douglas display of the paper on p. 2779.

Two countries. With two countries (i and j), the inverse export supply elasticity has the closed form

$$\omega_{ji,k} = -\Big(1-\frac{1+\bar{\mu}_j}{1+\mu_k}\Big)\frac{1-\sum_g \dfrac{e_{j,g}\sigma_g\lambda_{jj,g}}{1+(\sigma_g-1)\lambda_{jj,g}}}{1-\sum_g \dfrac{e_{j,g}\sigma_g\lambda_{jj,g}}{1+(\sigma_g-1)\lambda_{jj,g}}\Big(1-\dfrac{1+\bar{\mu}_j}{1+\mu_g}\Big)},$$

and the second-best policy is

$$1+t^{**}_{ji,k} = (1+\bar{t}_i)(1+\omega_{ji,k})\,\frac{1+(\sigma_k-1)\lambda_{ii,k}}{1+\frac{1+\tilde{\mu}_i}{1+\mu_k}(\sigma_k-1)\lambda_{ii,k}}, \tag{35}$$
$$1+x^{**}_{ij,k} = (1+\bar{t}_i)(1+\omega_{ji,k})\,\frac{1+\mu_k}{1+\tilde{\mu}_i}\,\frac{(\sigma_k-1)\lambda_{jj,k}}{1+(\sigma_k-1)\lambda_{jj,k}}.$$

In two- and three-country simulations with Cobb–Douglas, CES, and Stone–Geary preferences, the restricted-entry formulas above reproduce the numerically optimized second-best policy to machine precision.

THEOREM 2′ (FREE ENTRY)

Suppose entry is free, the government is unable (or unwilling) to apply domestic industrial subsidies, and foreign governments impose no taxes; all wages adjust. At any regular interior optimum, the second-best import tariffs and export subsidies are, up to a uniform tax shifter $\bar{t}_i \in \mathbb{R}_+$, implicitly given by

$$\begin{aligned} 1 + t_{ji}^{**} &= (1+\bar{t}_i)(\mathbf{1}+\Omega_{ji}) \oslash \left[\mathbf{1} + E_{-ii}^{-1} E_{-ii}^{(ii)} \left(1 - u_{i,k}\right)_k\right], \\ 1 + x_{ij}^{**} &= -(1+\bar{t}_i)\left[E_{ij}^{-1} E_{ij}^{(-ij)} (\mathbf{1}+\Omega_{-ii})\right] \oslash \left(u_{i,k}\right)_k, \end{aligned} \tag{36}$$

where

$$u_i = \mathbf{1} - H_i^{-1}(\mu - \tilde{\mu}_i \mathbf{1}), \qquad \tilde{\mu}_i = \frac{v_i^\top H_i^{-1} \mu}{v_i^\top H_i^{-1} \mathbf{1}},$$

$$H_i = \operatorname{diag}(\mathbf{1}+\mu) + \operatorname{diag}(\mu \odot r_{ii})\left[E_{ii} - E_{ii}^{(-ii)} E_{-ii}^{-1} E_{-ii}^{(ii)}\right], \qquad v_i = e_{ii} - \left(E_{-ii}^{-1} E_{-ii}^{(ii)}\right)^\top e_{-ii},$$

$\mu = (\mu_k)_k$ is the vector of scale elasticities, $r_{ii} = (r_{ii,k})_k$ the vector of domestic revenue shares, and E_{ii} and $E_{ii}^{(-ii)}$ the elasticities of domestic demand for domestic goods with respect to domestic and import prices. The vector $\Omega_{ji} = [\omega_{ji,k}]_k$ collects the inverse export supply elasticities. They solve the foreign free-entry and labor-market clearing conditions, a linear system in foreign scale elasticities, revenue shares, and demand elasticities that is given in Appendix A and has the closed form below with two countries.[2]

In these formulas, $\omega_{ji,k}$ is the inverse export supply elasticity under free entry, and $u_{i,k}$ is the domestic scale wedge, which accounts for the effect of trade-policy-induced changes in domestic production scale on domestic consumer prices. The average scale elasticity $\tilde{\mu}_i$ replaces $\bar{\mu}_i$, which is zero under free entry because profits are zero. Replacing $u_{i,k}$ with the restricted-entry markup ratio $(1+\bar{\mu}_i)/(1+\mu_k)$ and evaluating $\omega_{ji,k}$ with equation (10) recovers Theorem 2 as published.

Cobb–Douglas preferences. With Cobb–Douglas preferences, the formulas of Theorem 2′ become

$$\begin{aligned} 1 + t_{ji,k}^{**} &= \frac{1 + (\sigma_k - 1)\lambda_{ii,k}}{1 + u_{i,k}(\sigma_k - 1)\lambda_{ii,k}} \left(1 + t_{ji,k}^{*}\right), \\ 1 + x_{ij,k}^{**} &= \frac{1}{u_{i,k}} \left(1 + x_{ij,k}^{*}\right), \end{aligned} \tag{37}$$

[2] As in Theorem 2, the general display assumes homothetic preferences across industries; Appendix A gives the nonhomothetic case.

with the same first-best components as under restricted entry (with the free-entry $\omega_{ji,k}$), and

$$u_{i,k} = \frac{1+\tilde{\mu}_i-\kappa_{i,k}}{1+\mu_k-\kappa_{i,k}}, \qquad \kappa_{i,k} \equiv \frac{\mu_k\sigma_k r_{ii,k}}{1+(\sigma_k-1)\lambda_{ii,k}},$$
$$\tilde{\mu}_i \equiv \frac{\sum_k \vartheta_{i,k}\mu_k}{\sum_k \vartheta_{i,k}}, \qquad \vartheta_{i,k} \equiv \frac{e_{i,k}\lambda_{ii,k}\sigma_k}{[1+(\sigma_k-1)\lambda_{ii,k}](1+\mu_k-\kappa_{i,k})}.$$

Here $\kappa_{i,k}$ is the elasticity of industry k's domestic consumer price index with respect to trade-policy-induced firm entry. With $\kappa_{i,k}=0$ and $\bar{\mu}_i$ in place of $\tilde{\mu}_i$, the wedge returns to the corresponding term of the display on p. 2779.

Two countries. With two countries, the inverse export supply elasticity under free entry has the closed form

$$1+\omega_{ji,k} = \frac{1/(1+\mu_k-\kappa_{j,k})}{\sum_g \rho_{j,g}/(1+\mu_g-\kappa_{j,g})}, \qquad \kappa_{j,k} = \frac{\mu_k\sigma_k r_{jj,k}}{1+(\sigma_k-1)\lambda_{jj,k}},$$

where $\rho_{j,g}$ is industry g's share of total revenue in country j, and the second-best policy is

$$1+t^{**}_{ji,k} = \frac{1+\bar{t}_i}{1+\mu_k-\kappa_{j,k}}\,\frac{1+(\sigma_k-1)\lambda_{ii,k}}{1+u_{i,k}(\sigma_k-1)\lambda_{ii,k}}, \tag{38}$$
$$1+x^{**}_{ij,k} = \frac{1+\bar{t}_i}{1+\mu_k-\kappa_{j,k}}\,\frac{(\sigma_k-1)\lambda_{jj,k}}{u_{i,k}\,[1+(\sigma_k-1)\lambda_{jj,k}]},$$

so that

$$\frac{1+t^{**}_{ji,k}}{1+x^{**}_{ij,k}} = \frac{u_{i,k}[1+(\sigma_k-1)\lambda_{ii,k}]}{1+u_{i,k}(\sigma_k-1)\lambda_{ii,k}}\Big(1+\frac{1}{(\sigma_k-1)\lambda_{jj,k}}\Big),$$

which reduces to the Gros (1987) formula with one industry ($u_i=1$), with uniform scale elasticities ($\tilde{\mu}_i=\mu$), and under constant returns ($\mu\to 0$). In all two- and three-country free-entry environments we have simulated (Cobb–Douglas, CES, Stone–Geary), Theorem 2′ reproduces the numerically optimized second-best policy and welfare gains to machine precision.

2.2 Third-best optimal tariffs

Theorem 3 of the paper (p. 2780) characterizes the third best, in which import tariffs are the only instrument. The third-best tariff is the import tariff of Theorem 2 scaled by a uniform shifter $1+\bar{\tau}^*_i$, where $\bar{\tau}^*_i$, the inverse of the elasticity of international demand for economy i's labor, stands in for the unavailable export taxes. The proof of Theorem 3 (online Appendix J) uses the approximations of Theorem 2 and two more: the response of foreign demand to the home wage is evaluated at fixed foreign wages and incomes, and home income is treated as proportional to the home wage. Because export taxes are unavailable, however, there is no uniform tax shifter to normalize: the level of the tariff vector is determined by the income and labor-market clearing conditions of the

foreign economies, in which every foreign wage and income adjusts. In addition, the published matrix displays of Theorems 2 (p. 2778) and 3 (p. 2780) print the domestic markup ratio inverted relative to the first-order conditions—whereas the Cobb–Douglas display on p. 2779 has the right orientation—and the display of Theorem 3 sums the unit term of the shifter over two industry indices rather than one (Remarks of Sections A.2 and A.4). The statements below remove all of these approximations and correct the displays, first under restricted entry and then, in Theorem 3′, under free entry.

THEOREM 3 (RESTRICTED ENTRY, WITHOUT APPROXIMATIONS)

Suppose entry is restricted, the government is unable (or unwilling) to apply domestic industrial subsidies or export taxes, and foreign governments impose no taxes. At any regular interior optimum, the third-best import tariffs are given by

$$1 + t_{ji}^{***} = (1 + \bar{\tau}_i^*)(\mathbf{1} + \Omega_{ji}) \oslash \left[\mathbf{1} + E_{-ii}^{-1} E_{-ii}^{(ii)} \left(1 - \tfrac{1+\tilde{\mu}_i}{1+\mu_k}\right)_k\right], \tag{39}$$

where $\tilde{\mu}_i$ is the demand-weighted average markup of Theorem 2 and E_{-ii} and $E_{-ii}^{(ii)}$ are its Marshallian demand elasticity matrices. The uniform shifter $1 + \bar{\tau}_i^*$ and the inverse export supply elasticities $\Omega_{ji} = [\omega_{ji,k}]_k$ enter only through their product, which is affine in $1/(1 + \mu_k)$ within each exporting economy. Its coefficients are the multipliers of the labor-market clearing conditions and income identities of the foreign economies, in which every foreign wage and income adjusts; they solve a linear system in expenditure shares, markups, and demand elasticities that is given in Appendix A and has the closed form below with two countries. Because export taxes are unavailable, the level of the tariff vector is determined.

Replacing $\tilde{\mu}_i$ with $\bar{\mu}_i$, $\omega_{ji,k}$ with its equation (10) value, and $\bar{\tau}_i^*$ with its published value recovers Theorem 3, with the corrected relative markup ratio. In the published statement the shifter is computed at fixed foreign wages and incomes and multiplies the $\omega_{ji,k}$ of Theorem 2; here the two are determined jointly.

Cobb–Douglas preferences. With Cobb–Douglas preferences, the formula of Theorem 3 becomes

$$1 + t_{ji,k}^{***} = (1 + \bar{\tau}_i^*)(1 + \omega_{ji,k}) \, \frac{1 + (\sigma_k - 1)\lambda_{ii,k}}{1 + \frac{1+\tilde{\mu}_i}{1+\mu_k}(\sigma_k - 1)\lambda_{ii,k}}, \tag{40}$$

with $\tilde{\mu}_i$ as in Theorem 2. This is the second-best tariff of equation (34) with $1 + \bar{\tau}_i^*$ in place of $1 + \bar{t}_i$ and with the $\omega_{ji,k}$ of this theorem.

Two countries. With two countries (i and j), the product of the shifter and the inverse export supply elasticity has the closed form

$$(1+\bar{\tau}_i^*)(1+\omega_{ji,k}) = (1+\tilde{\mu}_i)\left[\sum_g \frac{\chi_{ij,g}}{1+\mu_g} + \Lambda_{ij}\left(\frac{1}{1+\mu_k} + \frac{\sum_g \frac{e_{j,g}\lambda_{jj,g}}{1+\mu_g}}{\sum_g e_{j,g}\lambda_{ij,g}}\right)\right], \tag{41}$$

where $\chi_{ij,g} \equiv e_{j,g}\lambda_{ij,g} / \sum_s e_{j,s}\lambda_{ij,s}$ is the share of industry g in economy i's exports, $\bar{e}_{ij}$ is the elasticity of international demand for economy i's labor, the object that defines the published shifter, namely $1+\bar{\tau}_i^* = 1 + 1/\bar{e}_{ij}$, and

$$\bar{e}_{ij} \equiv \sum_g \chi_{ij,g}(\sigma_g - 1)\lambda_{jj,g}, \qquad \Lambda_{ij} \equiv \frac{(1-\bar{e}_{ij})\sum_g \frac{\chi_{ij,g}}{1+\mu_g} + \sum_g \frac{\chi_{ij,g}(\sigma_g-1)\lambda_{jj,g}}{1+\mu_g}}{\sum_g \frac{\chi_{ij,g}(\sigma_g-1)\lambda_{jj,g}}{1+\mu_g} + \bar{e}_{ij}\frac{\sum_g e_{j,g}\lambda_{jj,g}/(1+\mu_g)}{\sum_g e_{j,g}\lambda_{ij,g}}}.$$

Normalizing the revenue-weighted average of $\omega_{ji,k}$ across industries to zero as in Theorem 2, $\sum_k \rho_{j,k}\omega_{ji,k} = 0$, splits the product into $1+\bar{\tau}_i^*$, which equals the right-hand side of (41) with $1/(1+\tilde{\mu}_j)$ in place of $1/(1+\mu_k)$, and

$$\omega_{ji,k} = \frac{(1+\tilde{\mu}_i)\Lambda_{ij}}{1+\bar{\tau}_i^*}\left(\frac{1}{1+\mu_k} - \frac{1}{1+\tilde{\mu}_j}\right),$$

proportional to $1/(1+\mu_k) - 1/(1+\tilde{\mu}_j)$ as in Theorem 2. The tariff is then given by equation (40). With uniform markups, Λ_{ij} equals the export share $\sum_g e_{j,g}\lambda_{ij,g}$ divided by $\bar{e}_{ij}$, $\omega_{ji,k} = 0$, and the tariff is $1 + t_{ji,k}^{***} = 1 + 1/\bar{e}_{ij}$ in every industry: the published formula, with the shifter written as in equation (J.5) of the online appendix. Appendix A shows that this holds for any homothetic preferences, with $\bar{e}_{ij} = -\sum_g \chi_{ij,g}\left(1 + \sum_s \varepsilon_{ij,g}^{(ij,s)}\right)$. In two- and three-country simulations with Cobb–Douglas, CES, and Stone–Geary preferences, the formula of Theorem 3 reproduces the numerically optimized third-best tariffs to machine precision.

Under free entry the third-best tariff keeps the form of Theorem 2′, with $1+\bar{\tau}_i^*$ in place of $1+\bar{t}_i$. The difference is in the domestic scale wedge u_i: with export taxes available, the first-order conditions for export prices absorb the effect of domestic industry scale on export prices; without them, that effect enters the home free-entry conditions.

THEOREM 3′ (FREE ENTRY)

Suppose entry is free, the government is unable (or unwilling) to apply domestic industrial subsidies or export taxes, and foreign governments impose no taxes; all wages adjust. At any regular interior optimum, the third-best import tariffs are given by

$$1 + t_{ji}^{***} = (1 + \bar{\tau}_i^*)(\mathbf{1} + \Omega_{ji}) \oslash \left[\mathbf{1} + E_{-ii}^{-1} E_{-ii}^{(ii)} (1 - u_{i,k})_k\right], \tag{42}$$

where $u_i = (u_{i,k})_k$ is the domestic scale wedge and E_{-ii} and $E_{-ii}^{(ii)}$ are the Marshallian demand elasticity matrices of Theorem 2. The uniform shifter $1 + \bar{\tau}_i^*$ and the inverse export supply elasticities $\Omega_{ji} = [\omega_{ji,k}]_k$ enter only through their product, the vector of multipliers of economy j's free-entry conditions. Without export taxes, the home free-entry conditions carry export-market terms, and u_i is determined jointly with these products rather than by home-market elasticities and domestic revenue shares alone, as in Theorem 2′. Together they solve the free-entry and labor-market clearing conditions of every economy and the home income identity, a linear system in scale elasticities, revenue and expenditure shares, and demand elasticities that is given in Appendix A and has $2K + 2$ equations with two countries. Because export taxes are unavailable, the level of the tariff vector is determined.

Replacing $u_{i,k}$ with $(1 + \tilde{\mu}_i)/(1 + \mu_k)$ returns the display of Theorem 3. When export taxes are available, their first-order conditions set the export-market terms to zero, and the system separates into the domestic block of Theorem 2′, which determines u_i, and a foreign block that determines the products only up to the uniform shifter $1 + \bar{t}_i$.

Cobb–Douglas preferences. With Cobb–Douglas preferences, the formula of Theorem 3′ becomes

$$1 + t_{ji,k}^{***} = (1 + \bar{\tau}_i^*)(1 + \omega_{ji,k}) \frac{1 + (\sigma_k - 1)\lambda_{ii,k}}{1 + u_{i,k}(\sigma_k - 1)\lambda_{ii,k}}, \tag{43}$$

the second-best tariff of equation (37) with $(1 + \bar{\tau}_i^*)(1 + \omega_{ji,k})$ in place of its first-best component and with the $u_{i,k}$ of this theorem.

Two countries. With two countries (i and j), let $u_{j,k} \equiv (1 + \bar{\tau}_i^*)(1 + \omega_{ji,k})$ be the multiplier of economy j's free-entry condition in industry k; the tariff is equation (43). The wedges $u_{i,k}$ and $u_{j,k}$ and the multipliers h_i and h_j of the two labor-market clearing conditions, with $h_i = 1 + \tilde{\mu}_i$, solve the $2K + 2$ linear equations

$$\begin{aligned} h_i &= (1 + \mu_k - \kappa_{i,k})u_{i,k} + \kappa_{i,k} - \mu_k r_{ij,k}\left[u_{i,k} + (\sigma_k - 1)\lambda_{jj,k}(u_{i,k} - u_{j,k})\right], \\ h_j &= (1 + \mu_k r_{ji,k})u_{j,k} + \mu_k r_{jj,k}(\sigma_k - 1)\lambda_{ij,k}(u_{i,k} - u_{j,k}), \end{aligned} \tag{44}$$

for every industry k, and

$$\sum_k \frac{e_{i,k}\lambda_{ii,k}\sigma_k}{1+(\sigma_k-1)\lambda_{ii,k}}(u_{i,k}-1)=0, \qquad \sum_k e_{j,k}\lambda_{ij,k}\left[u_{i,k}+(\sigma_k-1)\lambda_{jj,k}(u_{i,k}-u_{j,k})\right]=0,$$

with $\kappa_{i,k}$ as in Theorem 2′ and $r_{ij,k}$ the share of market j in economy i's revenue in industry k. The two equations of (44) are the free-entry conditions of economies i and j; the last two are economy i's income identity and economy j's labor-market clearing condition. The terms in brackets are the export-market terms. With export taxes available they are zero, and the home equations give the scale wedge $u_{i,k}$ of Theorem 2′. With one industry, the income identity gives $u_i = 1$, labor-market clearing gives $u_j = 1 + 1/[(\sigma-1)\lambda_{jj}]$, and the tariff is $1 + t^{***}_{ji} = 1 + 1/[(\sigma-1)\lambda_{jj}]$, its value under restricted entry: scale economies leave the one-industry tariff unchanged. As $\mu_k \to 0$ in every industry, $u_{i,k} \to 1$, $u_{j,k} \to 1 + 1/\bar{e}_{ij}$, and the tariff is uniform at $1 + 1/\bar{e}_{ij}$, the published formula. In all two- and three-country free-entry environments we have simulated (Cobb–Douglas, CES, Stone–Geary), Theorem 3′ reproduces the numerically optimized third-best tariffs to machine precision.

2.3 Table 4 with the updated formulas

Table 1 recomputes Table 4 of the paper using a replication package built on the updated formulas, keeping the paper's model, data, and calibration. The restricted-entry first- and second-best columns are computed as in the paper, with equation (10) as published. Four columns are updated: the restricted-entry third-best column uses the updated Theorem 3 of Section 2.2; the free-entry first-best column solves Theorem 1 as a fixed point using $\omega_{ji,k}$ from the updated equation (10) of Section 3; and the free-entry second- and third-best columns solve Theorems 2′ and 3′ as fixed points in the 44-country model, replacing the numerical optimization behind the published columns.

Third-best policy outcomes under restricted entry. In the paper, the restricted-entry third-best column of Table 4 was computed from the published formula of Theorem 3. Because the updated Theorem 3 removes the approximations of that formula and corrects the inverted markup ratio, it delivers the exact third-best optimum and therefore yields higher welfare gains than those reported in the paper. The increase is small: the mean welfare gain across the 43 economies rises from 0.47 to 0.50 percent, reinforcing the paper's conclusion that third-best import tariffs are a poor substitute for corrective subsidies.

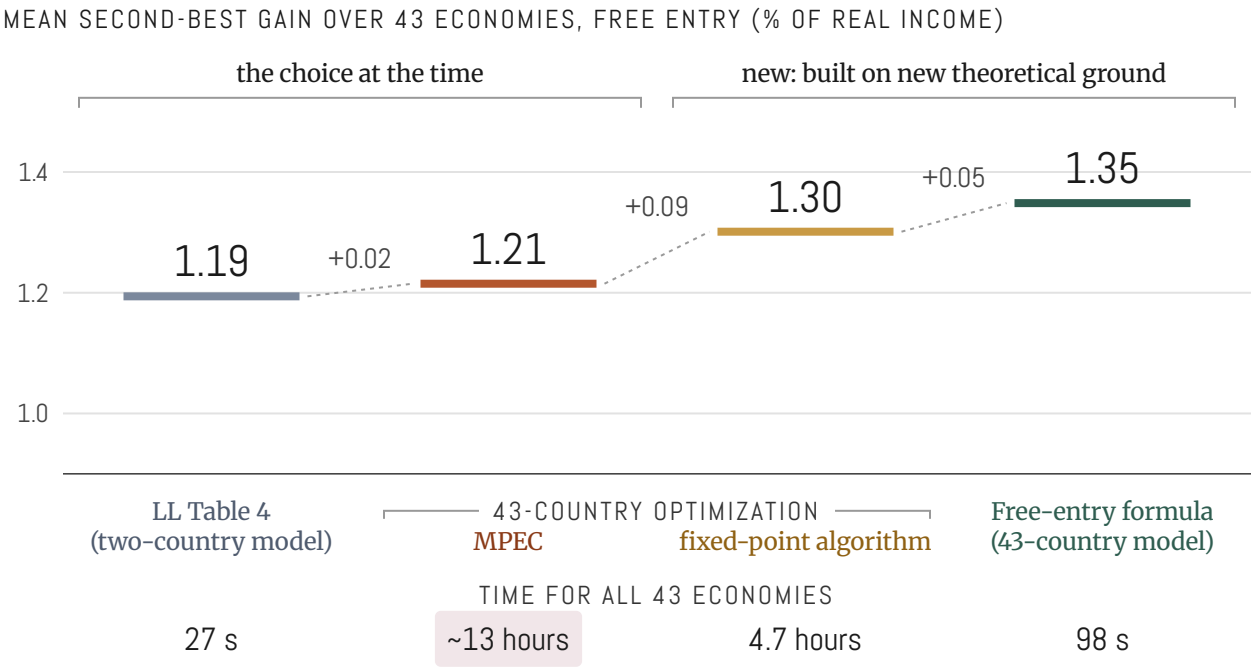
Second- and third-best policy outcomes under free entry. In the paper, the free-entry second- and third-best columns of Table 4 were computed by numerical optimization in a two-country version of the model, each economy against the rest of the world, because no formulas were available. Solving Theorems 2′ and 3′ as fixed points in the 44-country model raises the mean welfare gain from 1.19 to 1.35 percent for the second-best policy and from 0.63 to 0.69 percent for the third-best policy; the published free-entry gains are therefore conservative. For the second-best policy, fixed-point iteration of Theorem 2′ from the small-open-economy start (zero import tariffs and an export tax

Table 1: *The Gains from Non-Cooperative Policies and the Consequences of Retaliation*

Country	Restricted Entry				Free Entry			
	1st-best	2nd-best trade tax	3rd-best import tax	post retaliation	1st-best	2nd-best trade tax	3rd-best import tax	post retaliation
AUS	0.90%	0.21%	0.18% (0.14)	−0.83%	2.30% (2.30)	0.89% (0.58)	0.34% (0.35)	2.85%
AUT	1.31%	0.66%	0.54% (0.45)	−1.67%	2.13% (2.12)	1.14% (1.11)	0.65% (0.58)	−2.34%
BEL	1.31%	0.69%	0.55% (0.50)	−3.07%	1.76% (1.66)	1.10% (0.90)	0.74% (0.55)	−4.13%
BGR	2.01%	0.63%	0.52% (0.52)	0.04%	6.51% (5.48)	1.82% (1.88)	0.78% (0.83)	0.23%
BRA	1.87%	0.23%	0.20% (0.17)	0.93%	4.31% (4.31)	1.21% (0.64)	0.32% (0.36)	3.21%
CAN	1.72%	0.57%	0.49% (0.45)	−1.10%	3.42% (3.42)	1.59% (1.19)	0.71% (0.46)	0.34%
CHE	1.04%	0.56%	0.48% (0.46)	−0.92%	1.28% (1.28)	0.77% (0.76)	0.57% (0.53)	−1.26%
CHN	1.76%	0.27%	0.24% (0.24)	1.43%	3.69% (3.70)	0.81% (0.39)	0.28% (0.28)	2.90%
CYP	1.75%	0.68%	0.64% (0.61)	−4.20%	5.82% (5.83)	2.89% (1.56)	1.44% (1.43)	−10.88%
CZE	1.68%	0.88%	0.65% (0.54)	−1.70%	2.87% (2.86)	1.72% (1.60)	0.83% (0.73)	−2.13%
DEU	1.71%	0.78%	0.62% (0.49)	−0.37%	2.81% (2.60)	1.34% (1.34)	0.71% (0.65)	−0.10%
DNK	1.22%	0.58%	0.50% (0.48)	−2.41%	1.99% (1.94)	0.88% (0.91)	0.57% (0.49)	−4.78%
ESP	1.53%	0.52%	0.44% (0.39)	−0.15%	2.52% (2.51)	1.05% (1.03)	0.53% (0.49)	0.29%
EST	1.18%	0.62%	0.48% (0.45)	−4.86%	2.70% (2.69)	1.45% (1.42)	0.64% (0.56)	−7.90%
FIN	1.40%	0.54%	0.45% (0.25)	−0.76%	2.03% (2.00)	0.77% (0.83)	0.50% (0.47)	−1.08%
FRA	1.20%	0.45%	0.37% (0.34)	−0.78%	2.51% (2.52)	1.20% (1.10)	0.55% (0.50)	0.01%
GBR	1.09%	0.48%	0.42% (0.42)	−0.65%	2.12% (2.10)	1.03% (1.03)	0.56% (0.58)	0.17%
GRC	1.82%	0.59%	0.53% (0.53)	−0.20%	2.76% (2.75)	1.03% (1.08)	0.69% (0.68)	−0.04%
HRV	1.03%	0.55%	0.47% (0.45)	−2.27%	1.80% (1.79)	0.76% (0.73)	0.55% (0.52)	−2.99%
HUN	2.25%	1.06%	0.76% (0.73)	−2.85%	4.18% (4.08)	2.35% (2.28)	1.10% (0.96)	−3.33%
IDN	2.00%	0.36%	0.32% (0.26)	−0.37%	4.82% (4.81)	1.60% (1.45)	0.52% (0.50)	2.42%
IND	1.81%	0.35%	0.31% (0.31)	1.60%	4.20% (4.20)	1.05% (1.10)	0.35% (0.37)	2.82%
IRL	0.86%	0.68%	0.55% (0.53)	−2.07%	1.48% (1.49)	0.93% (0.89)	0.47% (0.40)	−3.32%
ITA	1.50%	0.46%	0.40% (0.26)	0.36%	2.76% (2.75)	1.09% (0.95)	0.50% (0.48)	0.80%
JPN	1.48%	0.32%	0.25% (0.22)	0.31%	2.97% (2.97)	1.04% (0.75)	0.40% (0.42)	1.55%
KOR	2.12%	0.65%	0.51% (0.49)	0.14%	4.43% (4.37)	1.54% (1.62)	0.69% (0.73)	1.25%
LTU	2.50%	0.93%	0.77% (0.76)	−1.38%	3.63% (3.54)	1.41% (1.26)	0.93% (0.85)	−2.47%
LUX	0.93%	0.81%	0.79% (0.78)	−3.59%	1.68% (1.52)	1.29% (1.15)	1.03% (1.01)	−4.90%
LVA	0.91%	0.54%	0.46% (0.44)	−4.26%	1.62% (1.57)	0.92% (0.80)	0.67% (0.46)	−4.85%
MEX	2.24%	0.60%	0.47% (0.44)	−0.92%	4.96% (4.96)	2.05% (1.36)	0.83% (0.76)	1.96%
MLT	1.35%	0.92%	0.85% (0.82)	−3.75%	2.28% (2.08)	1.60% (1.39)	1.16% (1.06)	−4.81%
NLD	1.35%	0.66%	0.55% (0.54)	−3.44%	1.92% (1.95)	1.01% (0.98)	0.73% (0.59)	−3.73%
NOR	1.19%	0.40%	0.35% (0.27)	−0.74%	2.03% (2.02)	0.81% (0.81)	0.45% (0.41)	−0.33%
POL	2.19%	0.76%	0.66% (0.65)	−0.26%	4.03% (4.03)	1.80% (1.62)	0.86% (0.81)	0.80%
PRT	2.04%	0.74%	0.66% (0.65)	−0.36%	3.92% (3.92)	1.88% (1.72)	0.96% (0.80)	0.73%
ROU	2.05%	0.77%	0.68% (0.67)	−0.42%	3.98% (3.92)	1.89% (1.72)	1.01% (0.98)	1.15%
RUS	2.39%	0.32%	0.28% (0.27)	0.84%	5.26% (5.25)	1.50% (1.39)	0.38% (0.37)	2.80%
SVK	2.01%	1.06%	0.81% (0.79)	−2.17%	3.51% (3.50)	2.13% (2.09)	1.18% (1.08)	−2.73%
SVN	1.42%	0.87%	0.72% (0.67)	−3.09%	2.98% (1.35)	1.62% (1.20)	0.97% (0.90)	−4.92%
SWE	1.21%	0.62%	0.52% (0.45)	−0.95%	1.64% (1.54)	0.85% (0.77)	0.64% (0.49)	−1.48%
TUR	1.43%	0.46%	0.39% (0.32)	−1.52%	3.55% (3.52)	1.29% (1.34)	0.56% (0.61)	−6.33%
TWN	2.18%	0.69%	0.56% (0.56)	−0.94%	4.98% (4.92)	1.88% (1.85)	0.85% (0.79)	0.81%
USA	1.53%	0.32%	0.27% (0.27)	0.69%	3.06% (3.04)	1.01% (0.80)	0.33% (0.30)	2.15%
Average	1.59%	0.60%	0.50% (0.47)	−1.23%	3.14% (3.05)	1.35% (1.19)	0.69% (0.63)	−1.20%

Notes: The data source is the 2014 World Input-Output Database (WIOD, Timmer et al. 2015). The *1st-best* policy is characterized by Theorem 1; *2nd-best trade taxes* are characterized by Theorem 2; and *3rd-best import taxes* are characterized by Theorem 3. *Post retaliation* corresponds to a situation where the home country sets its 1st-best policies and the rest of the world retaliates. Four columns are updated. Under restricted entry, 3rd-best import taxes are given by the updated Theorem 3. Under free entry, the 1st-best policy is the fixed point of Theorem 1 with $\omega_{ji,k}$ from the updated equation (10); 2nd-best trade taxes and 3rd-best import taxes are the fixed points of Theorems 2′ and 3′, computed for each economy in the full 44-region model; and each equilibrium is solved by the nested algorithm. The four updated columns are shaded; the small gray figure in parentheses is the value reported in Lashkaripour and Lugovskyy (2023).

Figure 2: *Four ways to compute the free-entry second best of Table 4*

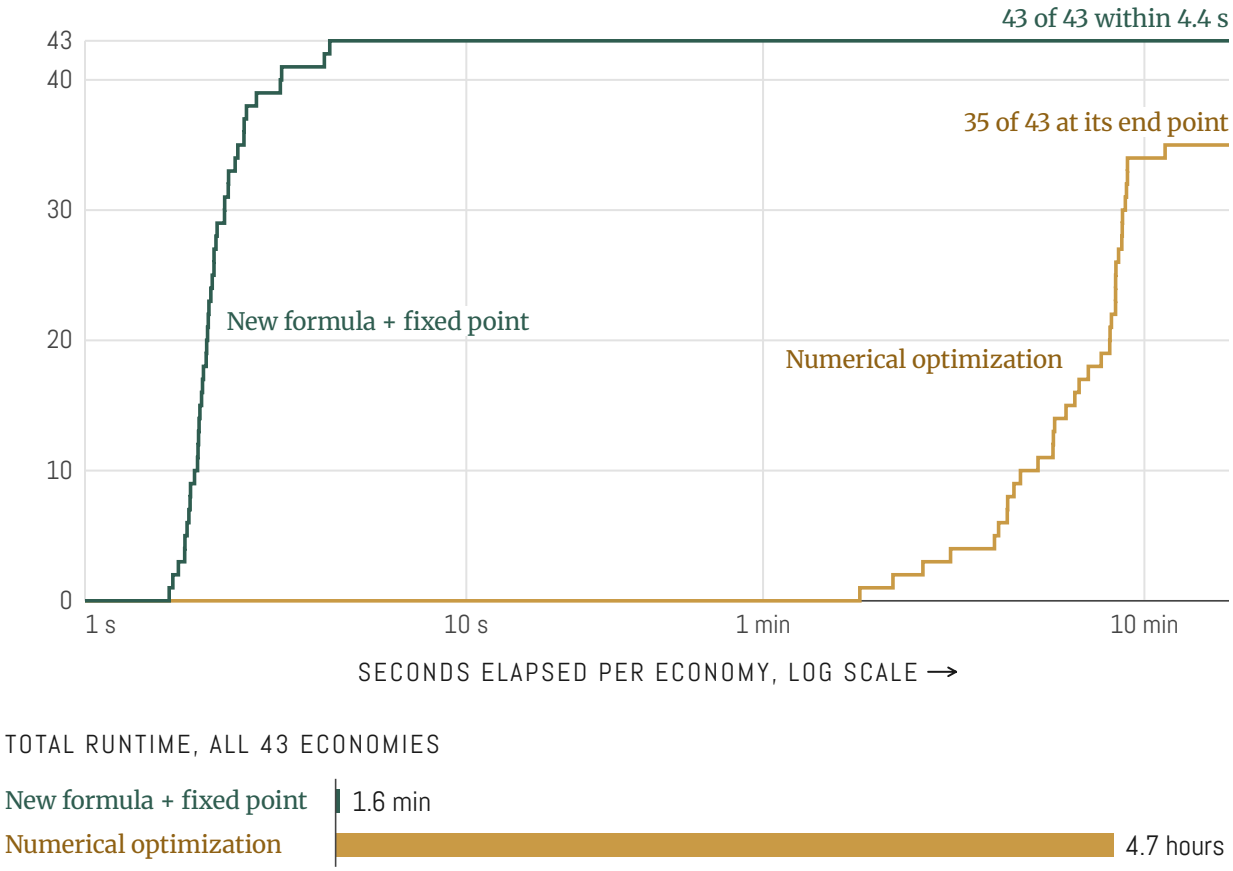


Notes: This figure compares four ways to compute the free-entry second-best gains of Table 4 (import tariffs and export taxes). Each value is the mean gain over the 43 economies, in percent of real income. LL Table 4 is the published column, computed with the replication code, which treats each economy against the rest of the world in a two-country model. The other three use the full model with all 43 economies and the rest of the world. MPEC is the optimization method of the replication code (MATLAB's fmincon), stopped at 150 iterations; it was run for four economies, and the gain and time shown are predicted for all 43 from those runs. The fixed-point algorithm is numerical optimization with the nested equilibrium solver, run to its end point. The free-entry formula is Theorem 2′ solved as a fixed point; it gives the highest gain, 1.35 percent, in 98 seconds. Times are for all 43 economies on one core.

factor of $\sigma_k/(\sigma_k - 1)$ in each industry) converges in all 43 economies, with a median of 20 iterations and a few seconds per economy. By contrast, numerical optimization of the 44-country model with a nested equilibrium solver and its exact gradient, run to its end point, takes 4.7 hours for the 43 economies; the fixed point of the formula agrees with that end point to within 0.00001 percentage points in 33 economies and ends above it in the other 10. Figure 2 compares the four computations of the free-entry second best, and Figure 3 shows, economy by economy, how fast the formula and the optimizer reach the highest gain found.

First-best policy outcomes under free entry. The paper's replication code computes the free-entry first-best column of Table 4 by applying MATLAB's fsolve to the equilibrium conditions in levels at the taxes of equation (12). As Step 1 of Section 1 shows, industry shutdown is never an equilibrium, because sales per worker exceed the wage by any margin as employment approaches zero. In levels, however, zero employment trivially satisfies the industry market-clearing condition, so fsolve can stop at a shutdown point in smaller economies even when the interior equilibrium exists. Our updated procedure avoids this trap by solving each equilibrium with the nested algorithm of

Figure 3: *The new formula against numerical optimization: precision and runtime*



Notes: Free-entry second best (import tariffs and export taxes) of each of the 43 economies of Table 4 against the rest of the world in the 44-country model, computed in two ways: Theorem 2′ iterated as a fixed point ("new formula + fixed point") and numerical optimization with the nested equilibrium solver and its exact gradient, run to its end point. Upper panel: the number of economies whose gain is within 0.01 percentage points of the highest gain found for that economy, against the time elapsed per economy (log scale). The formula reaches that gain in all 43 economies within 4.4 seconds. The optimizer reaches it in 35 economies at its end point, after a median of 6.8 minutes per economy; in the other 8 its end point is 0.04 to 0.89 percentage points below the formula. Lower panel: total runtime over the 43 economies, 1.6 minutes against 4.7 hours. Wall-clock times on one machine.

Section 1: because the algorithm solves the equilibrium conditions in logs, industry shutdown no longer satisfies the system. Using this solver at each iteration, we iterate the Theorem 1 formulas to a fixed point with $\omega_{ji,k}$ from the updated equation (10) of Section 3, starting from the equation (12) taxes.

The results for Bulgaria and Slovenia show why this matters. For Bulgaria, fsolve converges to a point where two industries shut down—electronics and electrical equipment (1.9 percent of observed Bulgarian sales) and furniture and other manufacturing—and yields the published welfare gain of 5.48 percent. At the same tax rates, the nested algorithm finds the true equilibrium,

where every industry is active, electronics and electrical equipment accounts for 8.6 percent of Bulgarian sales, and the equilibrium conditions in levels hold as well. The welfare gain at this equilibrium is 6.40 percent, and rises to 6.51 percent under the updated tax formula. For Slovenia, which is modeled against an aggregated rest of the world, fsolve terminates without solving the system near a point where the coke and refined petroleum industry shuts down, recording a welfare gain of 1.35 percent. At those same taxes, the nested algorithm finds the equilibrium, where the welfare gain is 2.98 percent under both the original and updated policies. Correcting the solutions for these two economies accounts for two-thirds (0.06 percentage points) of the column's 0.09 percentage-point average increase, from 3.05 to 3.14 percent.

3 An omitted term in the conditional inverse export supply elasticity

The elasticity and the omitted term. Theorem 1 of the paper characterizes country i's unilaterally optimal import tax, export subsidy, and domestic subsidy. The import tax on good ji,k (origin j, destination i, industry k) is $1+t^*_{ji,k} = (1+\omega_{ji,k})(1+\bar{t}_i)$. Here $\omega_{ji,k}$ is the conditional inverse export supply elasticity, defined on p. 2773 as

$$\omega_{ji,k} \equiv \frac{1}{r_{ji,k}\,\rho_{j,k}} \sum_{g\in\mathbb{K}} \left[\frac{\dot{w}_i L_i}{\dot{w}_j L_j}\,\rho_{i,g} \left(\frac{\partial \ln P_{ii,g}}{\partial \ln Q_{ji,k}} \right)_{\mathbf{w},\mathbf{Y},\tilde{\mathbb{P}}_i} + \sum_{n\neq i} \frac{\dot{w}_n L_n}{\dot{w}_j L_j}\, r_{ni,g}\,\rho_{n,g} \left(\frac{\partial \ln P_{ni,g}}{\partial \ln Q_{ji,k}} \right)_{\mathbf{w},\mathbf{Y},\tilde{\mathbb{P}}_i} \right]. \tag{45}$$

In this expression, $r_{ni,g}$ and $\rho_{n,g}$ are the good-specific and industry-wide sales shares of origin n. The paper describes $\omega_{ji,k}$ as "how the producer prices linked to economy i respond to a change in $Q_{ji,k}$, holding $\tilde{\mathbb{P}}_i$ and the entire vector of wage and income levels constant" (p. 2773). The bracket collects two responses: the producer prices of country i's own goods (first term) and the producer prices of its imports (second term).

An import tax on good ji,k alters the sales of foreign firms and, in the presence of scale economies, their producer prices. Buyers in each foreign market $n \neq i$ then substitute between those foreign varieties and country i's exports. This substitution alters country i's export quantities to destination n, $Q_{in,g}$, and the associated export tax revenues. We refer to these quantity responses as extraterritorial demand effects and define them as

$$\Phi_{nj,k} \equiv \sum_{\ell\neq i} \frac{\partial \ln Q_{in,g}}{\partial \ln P_{\ell n,k}} \left(\frac{\partial \ln P_{\ell n,k}}{\partial \ln Q_{ji,k}} \right)_{\mathbf{w},\mathbf{Y},\tilde{\mathbb{P}}_i}. \tag{46}$$

Extraterritorial demand effects enter $\omega_{ji,k}$ through three channels. The first two operate through the producer-price derivatives $\partial \ln P_{ii,g}/\partial \ln Q_{ji,k}$ and $\partial \ln P_{ni,g}/\partial \ln Q_{ji,k}$ in the bracket of (45); the paper accounts for both, as spelled out in online Appendix E of the paper. The third is the direct effect of the export-quantity response $\Phi_{nj,k}$ on country i's export-tax revenue, valued at the export-tax wedge $(\tilde{P}_{in,g} - P_{in,g})/P_{in,g}$. The definition in (45) omits this third term. With the term restored, the

complete elasticity is

$$\tilde{\omega}_{ji,k} \equiv \frac{1}{r_{ji,k}\,\rho_{j,k}} \sum_{g \in \mathbb{K}} \left[\frac{\dot{w}_i L_i}{\dot{w}_j L_j}\,\rho_{i,g} \left(\frac{\partial \ln P_{ii,g}}{\partial \ln Q_{ji,k}} \right)_{\mathbf{w},\mathbf{Y},\tilde{\mathbb{P}}_i} + \sum_{n \neq i} \frac{\dot{w}_n L_n}{\dot{w}_j L_j}\, r_{ni,g}\,\rho_{n,g} \left(\frac{\partial \ln P_{ni,g}}{\partial \ln Q_{ji,k}} \right)_{\mathbf{w},\mathbf{Y},\tilde{\mathbb{P}}_i} - \sum_{n \neq i} \frac{\dot{w}_i L_i}{\dot{w}_j L_j}\,\rho_{i,g}\, r_{in,g}\, \frac{\tilde{P}_{in,g} - P_{in,g}}{P_{in,g}}\,\Phi_{nj,k} \right]. \tag{47}$$

The restored term is shown in red. In Theorem 1, $\tilde{\omega}_{ji,k}$ takes the place of $\omega_{ji,k}$. Farrokhi and Lashkaripour (2025, footnote 18 and equation 21) note this channel in a related model. Bai, Lu, and Wang (2025) make an important contribution to the analysis of optimal trade policy in general equilibrium. They derive exact formulas for optimal policy in multi-country models with a wide class of supply-side specifications, including the free-entry version of our model, and these formulas include the extraterritorial demand effects that (47) restores.

The updated equation (10). Equation (10) of the paper (p. 2774) gives $\omega_{ji,k}$ to a first-order approximation, as a function of sales shares, scale elasticities, and Marshallian demand elasticities. The approximation uses the approximate inverse of a diagonally dominant matrix from Wu et al. (2013), as described in footnote 22 of the paper. With the omitted term restored, the approximation becomes, under free entry,

$$\tilde{\omega}_{ji,k} \approx \frac{-\frac{\mu_k}{1+\mu_k} r_{ji,k}}{1 + \frac{\mu_k}{1+\mu_k} \sum_{\iota \neq i} r_{j\iota,k}\varepsilon_{j\iota,k}} \left[1 - \frac{w_i L_i}{w_j L_j} \sum_{n \neq i} \frac{\rho_{i,k} r_{in,k}}{\rho_{j,k} r_{ji,k}} \left(\frac{\mu_k}{1+\mu_k} + \frac{\tilde{P}_{in,k} - P_{in,k}}{P_{in,k}} \right) \varepsilon^{(jn,k)}_{in,k} \right], \tag{48}$$

and, under restricted entry,

$$\begin{aligned} \tilde{\omega}_{ji,k} \approx & - \frac{\theta_{j,k} \sum_g r_{ji,g}\rho_{j,g}}{1 + \sum_g \sum_{\iota \neq i} \theta_{j,g}\, r_{j\iota,g}\, \rho_{j,g}\, \varepsilon_{j\iota,g}} \\ & \times \left[1 - \frac{\dot{w}_i L_i}{\dot{w}_j L_j} \sum_{n \neq i} \sum_g \frac{\rho_{i,g} r_{in,g}}{\sum_s \rho_{j,s} r_{ji,s}} \left(1 - \frac{1+\overline{\mu}_i}{1+\mu_g} + \frac{\tilde{P}_{in,g} - P_{in,g}}{P_{in,g}} \right) \varepsilon^{(jn,g)}_{in,g} \right], \end{aligned} \tag{49}$$

where $\theta_{n,g} \equiv 1 - (1+\overline{\mu}_n)/(1+\mu_g)$ and all other symbols are as defined in the paper. As in (47), red marks the contribution of the restored term. These expressions hold for an arbitrary number of countries and cover both entry regimes.

Effect on the welfare results. The restored term is inconsequential for the paper's welfare results. Table 2 reports the average first-best welfare gains across the 43 economies of Table 4 of the paper under each version of $\omega_{ji,k}$. To isolate the effect of the elasticity, we compute them with the paper's replication code, data, and solver (fsolve), changing only $\omega_{ji,k}$. Under restricted entry, the average gain is 1.593 percent with equation (10) and with its update. Under free entry, the update moves the average from 3.063 to 3.064 percent. Relative to equation (10), the update changes no economy's gain by more than 0.0003 percentage points under restricted entry and 0.012 percentage points

Table 2: *Average first-best welfare gains under each $\omega_{ji,k}$ (percent)*

	MEAN WELFARE GAINS, FIRST BEST	
	Restricted entry	Free entry
$\omega_{ji,k} \approx 0$, equation (12)		3.051%
$\omega_{ji,k}$, equation (10)	1.593%	3.063%
$\tilde{\omega}_{ji,k}$, updated equation (10)	**1.593%**	**3.064%**

The update moves each average by 0.001 percentage points or less

Notes: This table reports the average welfare gains from the first-best policy across the 43 economies of Table 4 of the paper, computed with the paper's replication code and data. The rows differ only in $\omega_{ji,k}$. The first row sets $\omega_{ji,k} \approx 0$, as in equation (12); under free entry, this is the average published in Table 4. The second row uses equation (10); under restricted entry, this is the average published in Table 4. The third row uses the updated equation (10). Relative to equation (10), the update changes no economy's gain by more than 0.0003 percentage points under restricted entry and 0.012 percentage points under free entry.

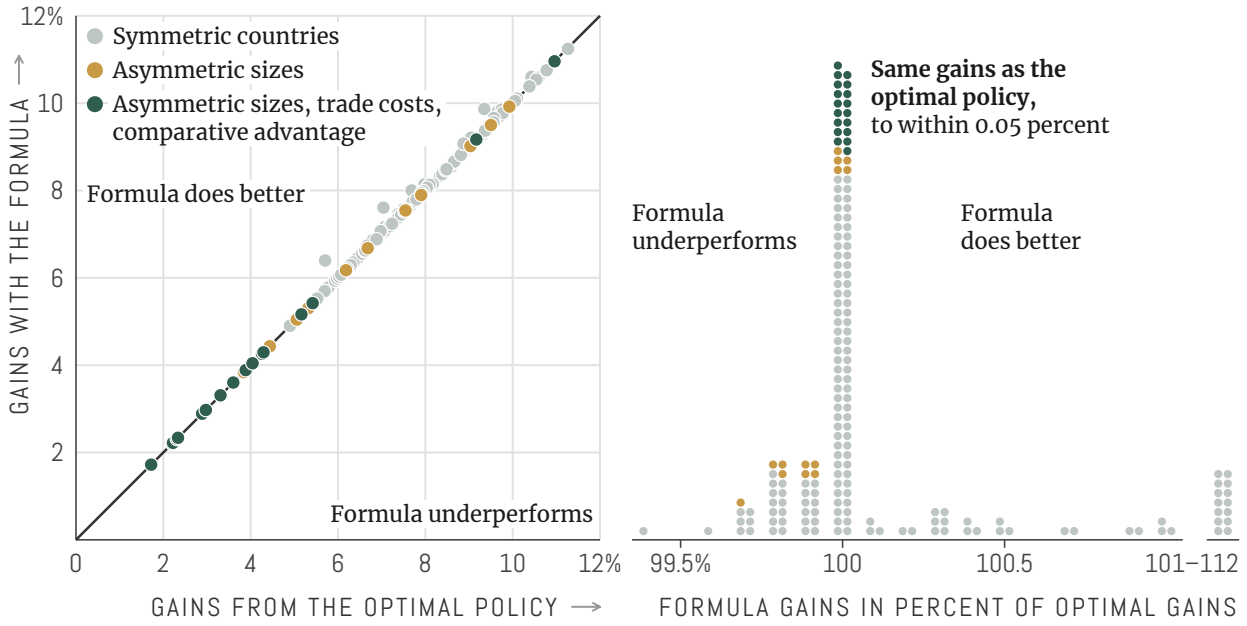
under free entry. Setting $\omega_{ji,k} \approx 0$, as in equation (12) of the paper, gives the free-entry average published in Table 4, 3.051 percent. The replication package of the paper already lets users set $\omega_{ji,k} = 0$ or evaluate it with equation (10), with nearly identical results. Replacing fsolve with the nested algorithm gives the 3.14 percent free-entry average reported in Section 2.3.

Figure 4 compares the policy given by the published formula, Theorem 1 with equation (10), with the optimal policy in 178 simulated economies. The economies follow three designs: the symmetric design of online Appendix H of the paper, with 2, 5, or 20 countries; 5 countries of asymmetric size; and 5 countries of asymmetric size with trade costs and comparative advantage. We leave out 12 draws that did not converge. In the left panel, the horizontal axis shows the welfare gain from the optimal policy found by numerical search (MPEC), and the vertical axis shows the gain with the policy given by the published formula; the right panel plots the ratio of the two. In most economies, the formula delivers the gains of the optimal policy to within 0.05 percent, and it never delivers less than 99.4 percent of them. Where the formula does better, the numerical search stopped at an inferior local solution. The formula is as accurate in the asymmetric designs as in the symmetric one. None of the paper's quantitative results or conclusions change.

4 Theorem 4 without Approximation

Theorem 4 of the paper (p. 2785) extends the first-best policy of Theorem 1 to an environment in which production uses intermediate inputs. As in Theorem 1, the policy consists of a domestic industrial subsidy, an import tariff, and an export subsidy. The proof of Theorem 4 in online Appendix K evaluates the conditional inverse export supply elasticity $\omega_{ji,k}$ at fixed foreign wages, without the export-wedge term restored in Section 3, and approximates the input-linkage adjustment in the export subsidy by the direct fixed-wage pass-through $\Lambda_{ij,k}$, which measures the share of exported inputs that returns to economy i embodied in its imports. Below we state Theorem 4

Figure 4: *Symmetric and asymmetric economies: LL formula vs exact optimum*



Notes: This figure compares the welfare gains from the published formula (Theorem 1 with equation (10)) with the gains from the optimal policy, found by numerical optimization (MPEC), in simulated economies. The economies come from three designs: 2, 5 or 20 symmetric countries, as in online Appendix H of the paper; 5 countries of different sizes; and 5 countries of different sizes with trade costs and comparative advantage. The left panel plots the two gains for each economy. The right panel shows the formula's gains in percent of the optimal gains; each dot is one economy. In every economy, the formula delivers at least 99.4 percent of the optimal gains. Where the formula does better, numerical optimization stopped at a worse local optimum. Not shown are 12 draws in which the solver for the formula's policy did not reach a point with positive entry in every industry.

without these approximations, restoring the export-wedge term in the inverse export supply elasticity and allowing foreign wages and foreign industry scale to adjust in general equilibrium. The update leaves the architecture of Theorem 4 and its domestic subsidy unchanged: the import tax keeps the form of Theorem 1, with the general-equilibrium elasticity $\tilde{\omega}_{ji,k}$ in place of $\omega_{ji,k}$, and the export subsidy keeps its form, with the general-equilibrium input-linkage wedge $\tilde{\Lambda}_{ij,k}$ in place of $\Lambda_{ij,k}$.

THEOREM 4 (INPUT–OUTPUT LINKAGES, FREE ENTRY)

Suppose entry is free, production uses intermediate inputs as in equation (13) of the paper, taxes apply to a good irrespective of its use, and foreign governments impose no taxes; all wages adjust. At any regular interior optimum, the first-best policy is, up to the uniform tax shifters $\bar{t}_i \in \mathbb{R}_+$ and $\bar{s}_i^{\mathcal{C}}$, implicitly given by

$$\begin{aligned} \text{[domestic subsidy]} \qquad & 1 + s_{i,k}^* = (1+\mu_k)\,(1+\bar{s}_i^{\mathcal{C}})^{-1} \\ \text{[import tax]} \qquad & 1 + t_{ji,k}^* = (1+\tilde{\omega}_{ji,k})\,(1+\bar{t}_i)\,(1+\bar{s}_i^{\mathcal{C}}) \\ \text{[export subsidy]} \qquad & 1 + x_{ij}^* = -E_{ij}^{-1}\Big[E_{ij}^{(-ij)}\,(1+t_i^*) + \tilde{\Lambda}_{ij}\,(1+\bar{t}_i)\,(1+\bar{s}_i^{\mathcal{C}})\Big] \end{aligned} \tag{50}$$

where $E_{ij} \equiv E_{ij}^{(ij)}$ and $E_{ij}^{(-ij)} \equiv [E_{ij}^{(nj)}]_{n\neq i}$ are the demand elasticity matrices of Definition 1 of the paper, computed for total demand, final plus intermediate, at fixed consumer income and fixed industry revenues, in the expenditure-weighted form of footnote 26 of the paper:

$$\big[E_{m'n}^{(mn)}\big]_{k,g} \equiv \frac{e_{mn,g}}{e_{m'n,k}}\,\frac{\partial \ln Q_{mn,g}}{\partial \ln \tilde{P}_{m'n,k}}, \qquad e_{mn,k} \equiv \frac{\tilde{P}_{mn,k}Q_{mn,k}}{\mathcal{E}_n},$$

with $\mathcal{E}_n$ gross expenditure in market n: row k refers to the good $m'n,k$ whose price changes, column g to the good mn,g whose quantity responds;[3] the elements of $\tilde{\Lambda}_{ij} \equiv [\tilde{\Lambda}_{ij,k}]_k$ measure the value to country i of the input linkages of good ij,k,

$$\tilde{\Lambda}_{mn,k} \equiv \frac{1}{(1+\bar{t}_i)(1+\bar{s}_i^{\mathcal{C}})}\sum_g (1+\mu_g)\,\beta_{mn,kg}\,\nu_{n,g}, \qquad \beta_{mn,kg} \equiv \frac{\alpha_{n,g}^{m,k}\,\mathcal{Y}_{n,g}}{\tilde{P}_{mn,k}Q_{mn,k}},$$

where $\beta_{mn,kg}$ is the fraction of good mn,k's sales absorbed by industry g in n, $\mathcal{Y}_{n,g}$ is the gross revenue of industry n,g, and $\nu_{n,g}$, the value to country i of one unit of revenue of foreign industry n,g, solves the linear system

$$\begin{aligned} \nu_{n,g} &= (1+t_{ni,g}^*) + \sum_{n'\neq i} r_{nn',g}\,\Gamma_{nn',g}, \\ \Gamma_{mn} &\equiv E_{mn}^{(in)}(1+x_{in}^*) + E_{mn}^{(-in)}(1+t_i^*) + \tilde{\Lambda}_{mn}(1+\bar{t}_i)(1+\bar{s}_i^{\mathcal{C}}) \qquad (m \neq i), \end{aligned}$$

with $r_{nn',g} \equiv P_{nn',g}Q_{nn',g}/\mathcal{Y}_{n,g}$ the revenue shares and $E_{mn}^{(-in)} \equiv [E_{mn}^{(m'n)}]_{m'\neq i}$, so that $\Gamma_{mn,g}$ values at country i's tax factors the demand responses in market n to the price of good mn,g, together with that good's input linkages ($\Gamma_{in} = 0$ is the export line itself); and $\tilde{\omega}_{ji,k}$ is the general-equilibrium inverse export supply elasticity,

$$\tilde{\omega}_{ji,k} \equiv \sum_{n\neq i}\sum_g \frac{P_{ni,g}Q_{ni,g}}{P_{ji,k}Q_{ji,k}}\,\frac{\partial \ln P_{ni,g}}{\partial \ln Q_{ji,k}} \;+\; \sum_{n\neq i}\sum_g \frac{\tilde{P}_{in,g}Q_{in,g}}{P_{ji,k}Q_{ji,k}}\Big(\frac{1+x_{in,g}^*}{1+\bar{t}_i} - 1\Big)\frac{\partial \ln Q_{in,g}}{\partial \ln Q_{ji,k}},$$

whose derivatives run through the foreign equilibrium (free entry, input–output demand, and labor-market clearing) at fixed export prices $\tilde{P}_{in,g}$ and fixed other import quantities. The shifter $\bar{s}_i^{\mathcal{C}}$ is a uniform consumption tax: it assumes a positive value if the taxed item is used only in final consumption and zero otherwise, and the statistics $\tilde{\Lambda}_{ij}$ and $\tilde{\omega}_{ji,k}$ are evaluated at $\bar{s}_i^{\mathcal{C}} = 0$.

The updated formulas differ from the published statement through $\tilde{\omega}_{ji,k}$ and $\tilde{\Lambda}_{ij,k}$. In the import tax, $\tilde{\omega}_{ji,k}$ incorporates the second term above, which is the input–output counterpart of the export-wedge term restored in Section 3: a change in imports shifts foreign wages and scale and, through them, foreign demand for country i's exports, valued at the export wedge $(1 + x^*_{in,g})/(1 + \bar{t}_i) - 1$. Evaluating $\tilde{\omega}_{ji,k}$ at fixed foreign wages and without this second term reduces it to the elasticity $\omega_{ji,k}$ of the paper. In the export subsidy, $\tilde{\Lambda}_{ij,k}$ generalizes the direct pass-through wedge $\Lambda_{ij,k} = \sum_g \beta_{ij,kg} r_{ji,g}$ of the paper. When foreign wages and scale adjust under free entry, an input-cost change alters the price of the using industry j, g by the scale factor $1 + \mu_g$, and a unit of that industry's revenue has general-equilibrium value $\nu_{j,g}$ to country i, which adds to the tariff valuation of the reimported share $r_{ji,g}(1 + t^*_{ji,g})$ the indirect demand and input-linkage effects $\sum_{n' \neq i} r_{jn',g}\left[(1 + t^*_{ji,g}) + \Gamma_{jn',g}\right]$ in other markets. Dropping the scale factor $1 + \mu_g$ and replacing $\nu_{j,g}$ with the direct reimported valuation $(1 + \bar{t}_i)(1 + \bar{s}_i^{\mathcal{C}})\, r_{ji,g}$ reduces $\tilde{\Lambda}_{ij,k}$ to $\Lambda_{ij,k}$ and recovers Theorem 4 as published.

Cobb–Douglas preferences. With Cobb–Douglas preferences across industries, CES demand across origins within an industry (elasticity σ_k), and the same CES aggregator for final and intermediate use, the formulas of Theorem 4 become

$$
\begin{aligned}
1 + t^*_{ji,k} &= (1 + \tilde{\omega}_{ji,k})\,(1 + \bar{t}_i), \\
1 + x^*_{ij,k} &= (1 + \bar{t}_i)\,\frac{(\sigma_k - 1)\sum_{n \neq i}(1 + \tilde{\omega}_{ni,k})\,\lambda_{nj,k} + \tilde{\Lambda}_{j,k}}{1 + (\sigma_k - 1)(1 - \lambda_{ij,k})},
\end{aligned}
\tag{51}
$$

with

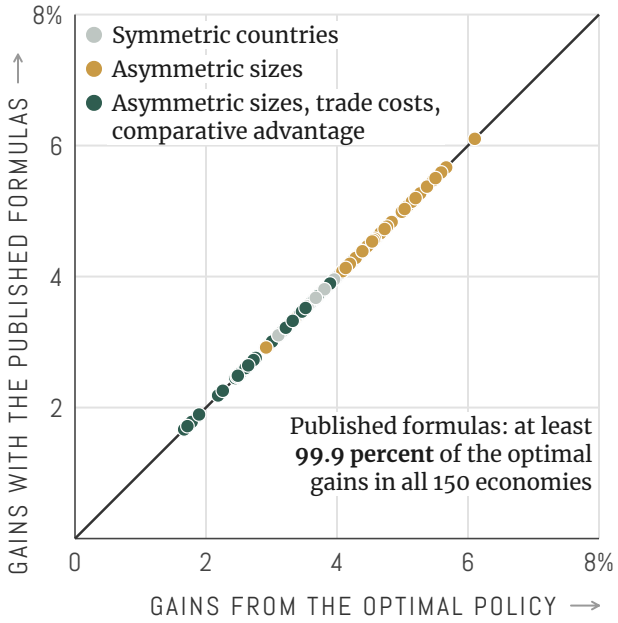
$$
\tilde{\Lambda}_{j,k} = \frac{1}{1 + \bar{t}_i}\sum_g (1 + \mu_g)\,\alpha^k_{j,g}\,\frac{\mathcal{Y}_{j,g}}{\mathcal{E}_{j,k}}\,\nu_{j,g}, \qquad \nu_{j,g} = 1 + t^*_{ji,g} + \sigma_g \sum_{n \neq i} r_{jn,g}\,(x^*_{in,g} - t^*_{ji,g}),
$$

where $\alpha^k_{j,g}$ is industry j, g's spending on industry-k inputs per unit of revenue, $\mathcal{E}_{j,k}$ is gross spending on industry-k goods in market j, and $\lambda_{nj,k}$ are the origin shares. The input-linkage value no longer depends on the origin of the input, and $\nu_{j,g}$ is explicit. Relative to the Cobb–Douglas export formula in equation (11) of the paper, the numerator gains the term $\tilde{\Lambda}_{j,k}$.

[3] With symmetric income elasticities, $e_{mn,g}\,\partial \ln Q_{mn,g}/\partial \ln \tilde{P}_{m'n,k} = e_{m'n,k}\,\partial \ln Q_{m'n,k}/\partial \ln \tilde{P}_{mn,g}$ and the blocks coincide with Definition 1, $[E^{(mn)}_{m'n}]_{k,g} = \partial \ln Q_{m'n,k}/\partial \ln \tilde{P}_{mn,g}$. The first-order conditions deliver the expenditure-weighted form, for the own block E_{ij} (as footnote 26 states) and for the cross blocks $E^{(nj)}_{ij}$ alike.

Precision of Approximated Published Formula. Figure 5 shows that the fixed-wage and direct pass-through approximations in the published Theorem 4 have a negligible effect on welfare. It compares the gains of country i from the optimal policy, which Theorem 4 above characterizes, with its gains from the policy that satisfies the published formulas at its own equilibrium, in 150 simulated two-country economies with ten industries, free entry, and input–output linkages.[4] The economies follow the three designs of Figure 4, with two countries in each: symmetric countries, as in online Appendix H of the paper; countries of different sizes; and countries of different sizes with trade costs and comparative advantage. In every economy, the published formulas deliver at least 99.9 percent of the optimal gains (99.98 percent at the median), and the largest shortfall is 0.003 percentage points.

Figure 5: *Optimal policy under input–output linkages: LL formula vs exact optimum*



Notes: This figure compares the welfare gains of country i from the policy that satisfies the published formulas of Theorem 4 at its own equilibrium (vertical axis) with the gains from the optimal policy, found by numerical optimization (horizontal axis). Gains are in percent relative to the equilibrium without taxes or subsidies; each dot is one economy. The 150 economies have two countries and ten industries, with free entry and input–output linkages; final demand and the input bundle are Cobb–Douglas across industries, with equal weights, and CES across origins within an industry. Fifty economies come from each of three designs: symmetric countries, as in online Appendix H of the paper; countries of different sizes; and countries of different sizes with trade costs and comparative advantage. In each industry, σ_k is drawn from $[2.5, 6]$ and the share of revenue spent on inputs from $[0.2, 0.6]$; the scale elasticity μ_k is drawn below a bound that keeps the free-entry equilibrium interior and unique (median 0.09). The policy that satisfies the published formulas is found by iterating between the formulas and the equilibrium until the policy changes by less than 10^{-12}. In every economy, the published formulas deliver at least 99.9 percent of the optimal gains.

[4]The simulation is `rebuttal_evidence/theorem-4-correction/bform_check/scatter/simulate.py`, with output `results.txt` in the same folder; it uses the model and optimizer of the verification script.

Two countries, one industry. A closed-form special case with two countries and one industry illustrates why the published pass-through approximation is so close to the general-equilibrium wedge. Let $\bar{\alpha}_j$ be the foreign input share per unit of revenue, λ_{ij} the share of country i's good in foreign spending, and $\lambda_{jj} = 1 - \lambda_{ij}$. The ratio of the two trade taxes, which is independent of the Lerner shifter, is

$$\frac{1+t^*}{1+x^*} - 1 = \frac{1-(1+\mu)\bar{\alpha}_j}{(1+\mu)\bar{\alpha}_j\lambda_{ij} + \left[1-(1+\mu)\bar{\alpha}_j\right](\sigma-1)\lambda_{jj}}, \tag{52}$$

whereas the direct pass-through approximation $\Lambda_{ij} = \bar{\alpha}_j\lambda_{ij}$ of the published formula gives $(1 - \bar{\alpha}_j\lambda_{ij})/[(\sigma-1)\lambda_{jj} + \bar{\alpha}_j\lambda_{ij}]$. Without scale economies ($\mu = 0$), the general-equilibrium ratio simplifies to $1/[(\sigma-1)\lambda_{jj} + \bar{\alpha}_j\lambda_{ij}/(1-\bar{\alpha}_j)]$, which replaces the first-round reimport share $\bar{\alpha}_j\lambda_{ij}$ with the cumulative round-trip share $\bar{\alpha}_j\lambda_{ij}/(1-\bar{\alpha}_j)$, and the net wedge vanishes when $(1+\mu)\bar{\alpha}_j = 1$. Normalizing the import tax to zero, the optimal export tax is $R/(1+R)$, with R the ratio in (52). With $\sigma = 4$ and λ_{jj} calibrated to the median country in the 2014 WIOD sample, the net wedge in percent is:

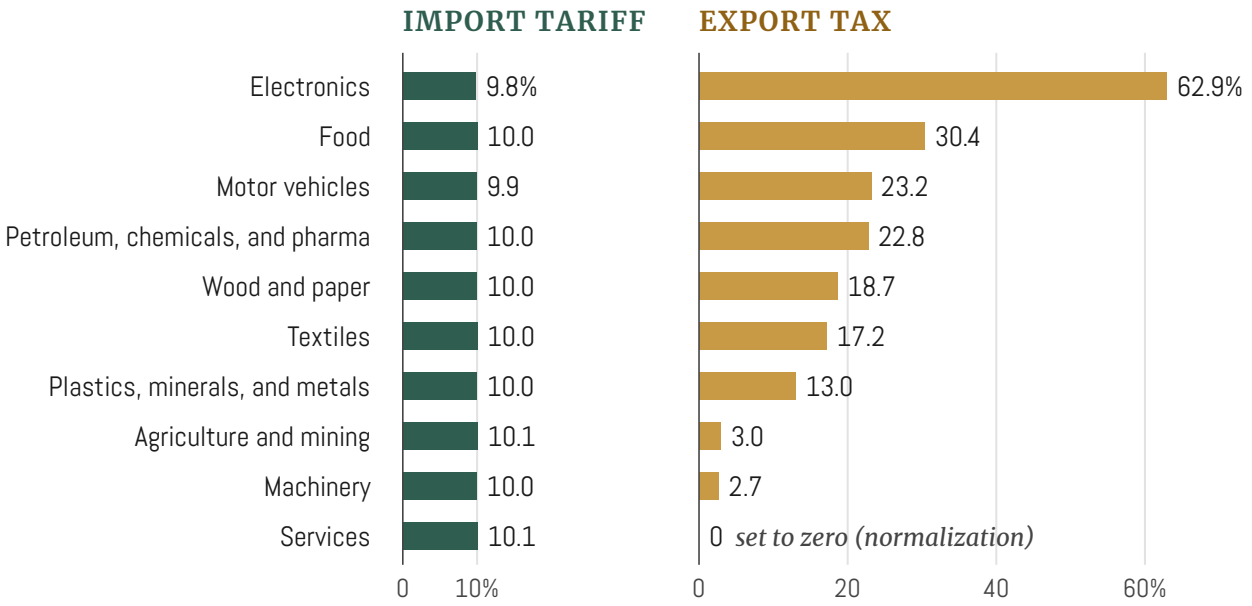
$\bar{\alpha}_j$	0	0.2	0.4
published	33.383	33.370	33.357
corrected, $\mu = 0$	33.383	33.379	33.372
corrected, $\mu = 0.25$	33.383	33.378	33.367
corrected, $\mu = 0.5$	33.383	33.376	33.358

At $\bar{\alpha}_j = 0$, without intermediate inputs, both expressions reduce to the formula of Gros (1987). Across the parameter values in the table, the approximated and exact wedges differ by at most 0.016 percentage points; the gap widens only as $(1+\mu)\bar{\alpha}_j$ approaches one.

Uniformity of import taxes (footnote 40). Footnote 40 of the paper (p. 2786) reads: "In the absence of scale economies or profits, $\omega_{ji,k} = 0$. So, Theorem 1 posits that optimal import tariffs are uniform by choice of $\bar{s}_i^{\mathcal{C}} = 0$ in the multi-industry Armington and Eaton-Kortum models (see online Appendix M)." The heading on p. 2776 reads "Optimal Tariffs Are Uniform, Absent Scale Economies or Profits." When $\omega_{ji,k}$ is evaluated at fixed foreign wages and without the export-wedge term restored in Section 3, producer prices do not respond to import quantities under constant returns ($\mu_k = 0$), so $\omega_{ji,k} = 0$ and optimal import tariffs are strictly uniform even with input–output linkages. Once the export-wedge term is included in $\tilde{\omega}_{ji,k}$ and foreign wages and input demands adjust, an import tariff on good ji,k alters foreign input demand and wages, which shifts foreign demand for country i's taxed exports through the second term of $\tilde{\omega}_{ji,k}$. Because this term does not vanish in general when intermediate inputs are traded, optimal import tariffs under input–output linkages and constant returns are approximately rather than strictly uniform across sectors. In footnote 40, the heading on p. 2776, footnote 3, and the sentence preceding footnote 29 on p. 2777, "uniform" should therefore read "approximately uniform" when production uses intermediate inputs.

The departure from uniformity is negligible in practice. We illustrate this in a two-region model

Figure 6: *Optimal US import tariffs and export taxes by sector under constant returns to scale*



Notes: This figure shows the optimal US import tariff (left) and export tax (right) by sector, found by numerical optimization. The model has two regions, the United States and the rest of the world, and ten sectors with input-output linkages and constant returns to scale. It is calibrated to the 2020 OECD ICIO tables, with sector-specific trade elasticities. The rest of the world sets no policy. Only relative tax rates are determined (Lerner symmetry), so the export tax on Services is set to zero. The import tariff lies between 9.8 and 10.1 percent in every sector, while the export tax ranges from 0 to 63 percent.

of the United States and the rest of the world, with ten sectors, input–output linkages, and constant returns to scale.[5] Figure 6 shows the optimal US policy by sector: the import tariff varies between 9.8 and 10.1 percent across the ten sectors, while the export taxes range from 0 to 63 percent. As Theorem 4 of the paper characterizes, the cross-sectoral variation in optimal trade policy under input–output linkages is carried by the export instruments. Table 3 compares welfare at this policy with welfare at the optimal uniform import tariff, which we obtain by re-solving the problem with one import tariff rate for all sectors and the export taxes still chosen optimally by sector. US welfare rises by 0.45577 percent under the unconstrained optimal tariff schedule and by 0.45576 percent under the uniform import tariff. The difference is 0.000012 percentage points, so a uniform import tariff captures 99.997 percent of the gains. The differences for the rest of the world and for the world as a whole are even smaller.

Remarks. (a) *Balance form of the import line.* Multiplying the import line by $P_{ji,k}Q_{ji,k}$ and using the trade-balance identity gives the equivalent condition

$$(1+t^{*}_{ji,k})\,P_{ji,k}Q_{ji,k} = \sum_{n\neq i}\sum_{g}(1+x^{*}_{in,g})\,\tilde{P}_{in,g}Q_{in,g}\,\frac{\partial \ln Q_{in,g}}{\partial \ln Q_{ji,k}},$$

[5]The model is calibrated to the 2020 OECD Inter-Country Input-Output (ICIO) tables (2023 edition), with sector-specific trade elasticities. The rest of the world sets no policy, and US tax revenue is rebated lump sum.

Table 3: *Welfare changes at the optimal US policy against the rest of the world under constant returns to scale*

	UNCONSTRAINED-OPTIMAL TARIFF	UNIFORM TARIFF	DIFFERENCE (POINTS)
United States	+0.456%	+0.456%	−0.000012
Rest of the world	−0.207%	−0.207%	+0.000005
World	−0.078%	−0.078%	+0.000002

Difference: uniform tariff minus unconstrained–optimal tariff, in percentage points.

Notes: This table reports welfare changes at the optimal US policy in the model of Figure 6, in percent relative to the calibrated equilibrium. The unconstrained-optimal tariff sets a separate import tariff for each sector, as in Figure 6. The uniform tariff sets one import tariff for all sectors (10.04 percent at the optimum). In both cases, export taxes are set optimally by sector. The uniform tariff lowers the US gain by 0.000012 percentage points. World welfare is income-weighted.

with the same derivatives as in $\tilde{\omega}_{ji,k}$: the tax-inclusive cost of the marginal import equals the tax-inclusive value of the exports it generates. It contains neither $\bar{t}_i$ nor $\tilde{\omega}_{ji,k}$ and is the form the proof establishes first; the import line of Theorem 4 is its rearrangement.

(b) *Import line without general-equilibrium derivatives.* The multipliers of the foreign equilibrium give a second equivalent form,

$$1 + t^*_{ni,k} = \sum_{m,g} \alpha^{m,g}_{n,k}\, v_{mn,g} + \beta_{n,k}\, \bar{v}_n - \mu_k\, \nu_{n,k} + \beta_{n,k} \sum_s \ell_{n,s}\, (1 + \mu_s)\, \nu_{n,s},$$

where $v_{mn,g}$ equals $1 + x^*_{in,g}$ for $m = i$ and $1 + t^*_{mi,g}$ otherwise, $\bar{v}_n \equiv \sum_{m,g} v_{mn,g}\, \tilde{P}_{mn,g}\, \partial Q_{mn,g}/\partial Y_n$ values market n's marginal spending, $\beta_{n,k}$ is the labor share of industry n,k, and $\ell_{n,k} \equiv \beta_{n,k}\mathcal{Y}_{n,k}/Y_n$ its employment share. The tariff on good ni,k equals country i's valuation of the inputs and labor income behind a unit of that industry's revenue, net of the scale effect $\mu_k \nu_{n,k}$. With two countries, one industry and no inputs it gives $(1 + t^*)/(1 + x^*) = 1 + 1/[(\sigma - 1)\lambda_{jj}]$ for every μ. This form departs from the architecture of the paper's theorems; we use it as a check on the import line.

(c) *Small open economy.* Footnote 39 of the paper survives. When country i's shares in foreign markets vanish, its import taxes become uniform ($\tilde{\omega}_{ji,k} \to 0$), and Marshallian adding-up gives

$$\sum_{m,s} \frac{e_{mn,s}}{e_{m'n,k}} \frac{\partial \ln Q_{mn,s}}{\partial \ln \tilde{P}_{m'n,k}} = -1, \qquad \text{hence} \qquad \Gamma_{nn',k} \to -(1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i) + \sum_g (1 + \mu_g)\, \beta_{nn',kg}\, \nu_{n',g},$$

so $\nu = (1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i)\, r_{\cdot i} + M\nu$, where $r_{\cdot i} \equiv [r_{ni,k}]$ collects the shares of foreign industries' revenue earned in country i, which vanish, and M is the matrix of $(1 + \mu_g)$-weighted shares of industry n,k's revenue earned from input sales to industry n',g. When $I - M$ is invertible, $\nu = (I - M)^{-1}(1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i)\, r_{\cdot i} \to 0$, hence $\tilde{\Lambda}_{ij} \to 0$, and the export line reduces to the paper's Theorem 1 for a small open economy. In the two-country case $M \to (1 + \mu)\bar{\alpha}_j$, so $I - M$ is invertible in the limit exactly

when $(1+\mu)\bar{\alpha}_j \neq 1$, the point at which the net wedge in (52) vanishes.

A Proofs

A.1. Notation and demand identities

Throughout, i is the policy-setting economy, $n, m, j, \ell \in \mathbb{C}$ index countries, and $k, g \in \mathbb{K}$ index industries. Consumer prices in market j are $\tilde{P}_{nj,k}$; producer prices are $P_{nj,k}$. Marshallian demand in market j is $Q_{nj,k} = \mathcal{D}_{nj,k}(Y_j, \tilde{P}_j)$, where Y_j is total expenditure and $\tilde{P}_j$ collects all consumer prices in j. We use the expenditure shares, price elasticities, and income elasticities

$$e_{nj,k} \equiv \frac{\tilde{P}_{nj,k} Q_{nj,k}}{Y_j}, \qquad \varepsilon_{nj,k}^{(mj,g)} \equiv \frac{\partial \ln Q_{nj,k}}{\partial \ln \tilde{P}_{mj,g}}, \qquad \eta_{nj,k} \equiv \frac{\partial \ln Q_{nj,k}}{\partial \ln Y_j}. \tag{A.1}$$

Under Assumption A1 (nested CES within industries), $e_{nj,k} = e_{j,k}\lambda_{nj,k}$, where $e_{j,k}$ is the expenditure share of industry k in market j and $\lambda_{nj,k}$ the within-industry share of origin n. For the exporting side we use the revenue shares $r_{nj,k} \equiv P_{nj,k}Q_{nj,k} / \sum_\ell P_{n\ell,k}Q_{n\ell,k}$ and $\rho_{n,k} \equiv \sum_\ell P_{n\ell,k}Q_{n\ell,k} / \sum_{\ell,g} P_{n\ell,g}Q_{n\ell,g}$ defined on p. 2773. Any demand system that is derived from utility maximization satisfies, in every market j, the Engel aggregation, Cournot aggregation, homogeneity, and Slutsky symmetry identities

$$\sum_{n,k} e_{nj,k}\eta_{nj,k} = 1, \qquad e_{mj,g} + \sum_{n,k} e_{nj,k}\varepsilon_{nj,k}^{(mj,g)} = 0, \qquad \eta_{nj,k} + \sum_{m,g} \varepsilon_{nj,k}^{(mj,g)} = 0, \tag{A.2}$$

$$e_{nj,k}\big(\varepsilon_{nj,k}^{(mj,g)} + \eta_{nj,k}e_{mj,g}\big) = e_{mj,g}\big(\varepsilon_{mj,g}^{(nj,k)} + \eta_{mj,g}e_{nj,k}\big). \tag{A.3}$$

The compensated (Hicksian) elasticities are $\varepsilon_{nj,k}^{H,(mj,g)} \equiv \varepsilon_{nj,k}^{(mj,g)} + \eta_{nj,k}e_{mj,g}$; by (A.2) they satisfy $\sum_{m,g} \varepsilon_{nj,k}^{H,(mj,g)} = 0$, and by (A.3) the matrix $[e_{nj,k}\varepsilon_{nj,k}^{H,(mj,g)}]$ is symmetric.

Elasticity matrices. Following Definition (D1) of the paper, $E_{nj}^{(mj)} \equiv [\varepsilon_{nj,k}^{(mj,g)}]_{k,g}$ is the $K \times K$ matrix of elasticities of demand for origin-n goods with respect to origin-m prices in market j; $E_{nj} \equiv E_{nj}^{(nj)}$; $E_{nj}^{(-ij)} \equiv [E_{nj}^{(mj)}]_{m \neq i}$ is $K \times (N-1)K$; $E_{-ij} \equiv [E_{nj}^{(mj)}]_{n \neq i, m \neq i}$ is the square $(N-1)K$ matrix of own and cross elasticities among the non-i origins in market j; and $E_{-ij}^{(ij)} \equiv [E_{nj}^{(ij)}]_{n \neq i}$ is $(N-1)K \times K$. The proof works with the *expenditure-weighted transposes*

$$\big(\tilde{E}_{nj}^{(mj)}\big)_{kg} \equiv \frac{e_{mj,g}}{e_{nj,k}}\,\varepsilon_{mj,g}^{(nj,k)} = \varepsilon_{nj,k}^{(mj,g)} + e_{mj,g}\big(\eta_{nj,k} - \eta_{mj,g}\big), \tag{A.4}$$

where the second equality is (A.3). Hence $\tilde{E}_{nj}^{(mj)} = E_{nj}^{(mj)}$ whenever all income elasticities in market j are equal, in particular under homothetic preferences, where $\eta \equiv 1$. The blocks $\tilde{E}_{-ij}$, $\tilde{E}_{-ij}^{(ij)}$, $\tilde{E}_{ij}$, $\tilde{E}_{ij}^{(-ij)}$ are assembled from (A.4) exactly as the E blocks are assembled from $E_{nj}^{(mj)}$. The compensated blocks E_{-ij}^{H} and $E_{-ij}^{H,(ij)}$ are assembled from ε^H in the same way; by (A.4),

$$\tilde{E}_{-ij} = E_{-ij}^{H} - \mathbf{1}\,(e_{-ij} \odot \eta_{-ij})^\top, \qquad \tilde{E}_{-ij}^{(ij)} = E_{-ij}^{H,(ij)} - \mathbf{1}\,(e_{ij} \odot \eta_{ij})^\top, \tag{A.5}$$

where $e_{-ij} = (e_{nj,k})_{n \neq i,k}$, $e_{ij} = (e_{ij,k})_k$, and $\mathbf{1}$ is a vector of ones.

Invertibility. In market j, let $z = (z_{nj,k})_{n \neq i,k}$ be any nonzero vector and extend it by $z_{ij,k} = 0$. Writing $\mathcal{E}_j(\tilde{P}_j, V)$ for the expenditure function, the compensated demand $Q^H_{nj,k} = \partial \mathcal{E}_j / \partial \tilde{P}_{nj,k}$ is homogeneous of degree zero in prices, and under A1 depends on within-industry prices only through the CES price index $\tilde{P}_{j,k} = (\sum_n \tilde{P}^{1-\sigma_k}_{nj,k})^{1/(1-\sigma_k)}$. Differentiating along $d \ln \tilde{P}_{nj,k} = z_{nj,k}$ gives

$$\sum_{n,m,k,g} e_{nj,k} z_{nj,k} \varepsilon^{H,(mj,g)}_{nj,k} z_{mj,g} = \frac{1}{Y_j} \sum_{k,g} \frac{\partial^2 \mathcal{E}_j}{\partial \tilde{P}_{j,k} \partial \tilde{P}_{j,g}} \tilde{P}_{j,k} \tilde{P}_{j,g} \bar{z}_k \bar{z}_g - \sum_k \sigma_k e_{j,k} \Big[\sum_n \lambda_{nj,k} z^2_{nj,k} - \bar{z}^2_k \Big], \tag{A.6}$$

with $\bar{z}_k \equiv \sum_n \lambda_{nj,k} z_{nj,k}$. The first term is nonpositive by concavity of $\mathcal{E}_j$ in prices; the bracket is a variance, which is strictly positive because $z_{ij,k} = 0$ and $z \neq 0$ while $\lambda_{ij,k} > 0$. Hence the quadratic form is negative definite on the subspace $\{z_{ij,k} = 0\}$, so E^H_{-ij} is invertible. The same argument on the subspace $\{z_{nj,k} = 0,\ n \neq i\}$ shows that E^H_{ij} is invertible. The Marshallian blocks $\tilde{E}_{-ij}$ and $\tilde{E}_{ij}$ differ from these by a rank-one matrix (A.5); we assume, as the paper does in Definition (D1), that they are invertible at the optimum.

A.2. Proof of Theorem 2 (restricted entry)

Environment. Entry is restricted: the number of firms $\bar{M}_{n,k}$ in every industry and country is fixed. By equation (4) with $\gamma_k/(\gamma_k - 1) = 1 + \mu_k$, producer prices are $P_{nj,k} = (1 + \mu_k) \varrho'_{nj,k} w_n$ with $\varrho'_{nj,k} \equiv \tau_{nj,k} \bar{a}_{n,k} \bar{M}^{-\mu_k}_{n,k}$ (equation (6)), so that $P_{nj,k}$ is proportional to the wage w_n and $d \ln P_{nj,k} = d \ln w_n$. Writing $R_{n,k} \equiv \sum_j P_{nj,k} Q_{nj,k}$ for industry revenue, labor-market clearing and profits in country n are

$$w_n L_n = \sum_k \frac{R_{n,k}}{1 + \mu_k}, \qquad \Pi_n = \sum_k \frac{\mu_k}{1 + \mu_k} R_{n,k}, \tag{A.7}$$

and total income is $\grave{w}_n L_n \equiv w_n L_n + \Pi_n = \sum_k R_{n,k}$, where $\grave{w}_n = (1 + \bar{\mu}_n) w_n$ and $\bar{\mu}_n$ is the sales-weighted average markup of equation (5), equivalently $1/(1 + \bar{\mu}_n) = \sum_k \rho_{n,k}/(1 + \mu_k)$. Economy i applies import taxes $t_{ni,k}$ and export subsidies $x_{ij,k}$, with $s_{i,k} = 0$, and foreign governments impose no taxes, so consumer prices are

$$\tilde{P}_{ni,k} = (1 + t_{ni,k}) P_{ni,k}, \qquad \tilde{P}_{ii,k} = P_{ii,k}, \qquad \tilde{P}_{ij,k} = \frac{P_{ij,k}}{1 + x_{ij,k}}, \qquad \tilde{P}_{nj,k} = P_{nj,k} \quad (n \neq i,\ j \neq i). \tag{A.8}$$

Tax revenue, $\mathcal{R}_i = \sum_{n \neq i,k} (\tilde{P}_{ni,k} - P_{ni,k}) Q_{ni,k} + \sum_{j \neq i,k} (\tilde{P}_{ij,k} - P_{ij,k}) Q_{ij,k}$, is rebated lump sum, so $Y_i = \grave{w}_i L_i + \mathcal{R}_i$ and $Y_n = \grave{w}_n L_n$ for $n \neq i$.

The paper's derivation of Theorem 2 (online Appendix I, with footnotes 20 and 29 of the paper) holds relative profit-adjusted foreign wages $\grave{w}_n / \grave{w}_m$, $n, m \neq i$, fixed. We adopt the same stipulation. Choosing one foreign $\grave{w}_n$ as the unit of account fixes every $\grave{w}_n$ and every foreign income $Y_n = \grave{w}_n L_n$, $n \neq i$, while the wages $w_n = \grave{w}_n / (1 + \bar{\mu}_n)$ remain endogenous, because $\bar{\mu}_n$ responds to the composition of foreign sales. By Lemma 1 of the paper, scaling $1 + t_i$, $1 + x_i$, and $\grave{w}_i$ by a

common factor $a > 0$ leaves the equilibrium allocation and welfare unchanged. Each such family of tax vectors contains exactly one member with $w_i = 1$, and we work with that representative; in it $P_{ii,k} = \tilde{P}_{ii,k}$ is constant. The uniform shifter $1 + \bar{t}_i$ in the statement of the theorem indexes the family.

The planning problem. Since the government controls the consumer prices $\tilde{P}_{ni,k}$ ($n \neq i$) and $\tilde{P}_{ij,k}$ ($j \neq i$) one for one through its taxes, the optimal policy solves

$$\max_{\{\tilde{P}_{ni,k}\}_{n\neq i}, \{\tilde{P}_{ij,k}\}_{j\neq i}, Y_i, \{w_n\}_{n\neq i}} V_i(Y_i, \tilde{P}_i) \tag{A.9}$$
$$\text{s.t.} \quad w_n L_n - \sum_{j,k} \frac{P_{nj,k} Q_{nj,k}}{1+\mu_k} = 0 \ \ \forall n, \qquad \sum_{j,k} P_{nj,k} Q_{nj,k} - Y_n = 0 \ \ \forall n \neq i,$$

where V_i is the indirect utility of the representative consumer, $Q_{nj,k} = \mathcal{D}_{nj,k}(Y_j, \tilde{P}_j)$, uncontrolled consumer prices follow (A.8), and Y_n ($n \neq i$) are the fixed foreign incomes. The constraints are labor-market clearing in every country and the income identity in every foreign country. The home budget constraint and balanced trade are implied: adding up the budget identities $\sum_{n,k} \tilde{P}_{nj,k} Q_{nj,k} = Y_j$ over j gives $\sum_j Y_j = \sum_n \sum_{j,k} P_{nj,k} Q_{nj,k} + \mathcal{R}_i = \sum_n \tilde{w}_n L_n + \mathcal{R}_i$ by (A.7), and subtracting the foreign income identities leaves $Y_i = \tilde{w}_i L_i + \mathcal{R}_i$. Conversely, any feasible point of (A.9) is implemented by $1 + t_{ni,k} = \tilde{P}_{ni,k}/P_{ni,k}$ and $1 + x_{ij,k} = P_{ij,k}/\tilde{P}_{ij,k}$. Hence (A.9) is equivalent to the government's problem.

First-order conditions. Let $\lambda_i \equiv \partial V_i / \partial Y_i > 0$, and attach the multipliers $\lambda_i u_n$ to the labor-market constraints and $\lambda_i \nu_n$ ($n \neq i$) to the foreign income constraints. Because $P_{nj,k} Q_{nj,k}$ varies as $d \ln w_n + d \ln Q_{nj,k}$, the labor constraint of n varies by $-\sum_{j,k} P_{nj,k} Q_{nj,k} \, d \ln Q_{nj,k} / (1+\mu_k)$ when it holds, and the income constraint by $Y_n d \ln w_n + \sum_{j,k} P_{nj,k} Q_{nj,k} \, d \ln Q_{nj,k}$. With Roy's identity, $\partial V_i / \partial \tilde{P}_{ni,k} = -\lambda_i Q_{ni,k}$, the variation of the Lagrangian is

$$\frac{d\mathcal{L}_i}{\lambda_i} = dY_i - \sum_{n\neq i,k} \tilde{P}_{ni,k} Q_{ni,k} \, d \ln \tilde{P}_{ni,k} + \sum_{n\neq i} \nu_n Y_n \, d \ln w_n - \sum_{n,j,k} \left(\frac{u_n}{1+\mu_k} - \nu_n \mathbf{1}_{n\neq i} \right) P_{nj,k} Q_{nj,k} \, d \ln Q_{nj,k}, \tag{A.10}$$

where, in market i, $d \ln Q_{ni,g} = \eta_{ni,g} d \ln Y_i + \sum_{m\neq i,k} \varepsilon^{(mi,k)}_{ni,g} d \ln \tilde{P}_{mi,k}$, and in market $j \neq i$, $d \ln Q_{nj,g} = \sum_k \varepsilon^{(ij,k)}_{nj,g} d \ln \tilde{P}_{ij,k} + \sum_{m\neq i,k} \varepsilon^{(mj,k)}_{nj,g} d \ln w_m$. Define the *import valuations*

$$\phi_{ni,k} \equiv \frac{u_n/(1+\mu_k) - \nu_n}{1 + t_{ni,k}} \qquad (n \neq i), \tag{A.11}$$

and use $P_{ni,k} Q_{ni,k} / Y_i = e_{ni,k}/(1+t_{ni,k})$, $P_{ii,k} Q_{ii,k}/Y_i = e_{ii,k}$, $P_{ij,k} Q_{ij,k}/Y_j = (1+x_{ij,k}) e_{ij,k}$, and $P_{nj,k} Q_{nj,k}/Y_j = e_{nj,k}$ for $n, j \neq i$. Setting the coefficients of dY_i, $d \ln \tilde{P}_{ji,k}$, $d \ln \tilde{P}_{ij,k}$, and $d \ln w_m$ in

(A.10) to zero gives, for all $j, m \neq i$ and all k,

$$1 = \sum_g \frac{u_i}{1+\mu_g} e_{ii,g}\eta_{ii,g} + \sum_{n\neq i,g} \phi_{ni,g} e_{ni,g}\eta_{ni,g}, \tag{A.12}$$

$$0 = e_{ji,k} + \sum_g \frac{u_i}{1+\mu_g} e_{ii,g}\varepsilon^{(ji,k)}_{ii,g} + \sum_{n\neq i,g} \phi_{ni,g} e_{ni,g}\varepsilon^{(ji,k)}_{ni,g}, \tag{A.13}$$

$$0 = \sum_g \frac{u_i(1+x_{ij,g})}{1+\mu_g} e_{ij,g}\varepsilon^{(ij,k)}_{ij,g} + \sum_{n\neq i,g} \Big(\frac{u_n}{1+\mu_g} - \nu_n\Big) e_{nj,g}\varepsilon^{(ij,k)}_{nj,g}, \tag{A.14}$$

$$\nu_m Y_m = u_i \sum_{\ell\neq i,g,k} \frac{P_{i\ell,g}Q_{i\ell,g}}{1+\mu_g}\varepsilon^{(m\ell,k)}_{i\ell,g} + \sum_{n\neq i,\ell\neq i,g,k} \Big(\frac{u_n}{1+\mu_g} - \nu_n\Big) P_{n\ell,g}Q_{n\ell,g}\varepsilon^{(m\ell,k)}_{n\ell,g}. \tag{A.15}$$

Condition (A.13) is the first-order condition for the import price $\tilde{P}_{ji,k}$, (A.14) for the export price $\tilde{P}_{ij,k}$, and (A.15) for the foreign wage w_m; the latter collects the effect of w_m on the foreign consumer prices $\tilde{P}_{m\ell,k}$, $\ell \neq i$ (its effect on $\tilde{P}_{mi,k}$ is absorbed by the tariff). Together with the constraints of (A.9), (A.12)–(A.15) are the complete set of necessary conditions at a regular interior optimum, where regular means that the constraint gradients are linearly independent. We now solve them.

Step 1: the import block. Divide (A.13) by $e_{ji,k}$ and use the weighted transposes (A.4), $(\tilde{E}^{(ni)}_{ji})_{kg} = (e_{ni,g}/e_{ji,k})\varepsilon^{(ji,k)}_{ni,g}$. Stacking over (j,k) with $j \neq i$, and writing $\phi \equiv (\phi_{ni,g})_{n\neq i,g}$ and $c \equiv (1/(1+\mu_g))_g$,

$$\mathbf{1} + \tilde{E}^{(ii)}_{-ii}\, u_i c + \tilde{E}_{-ii}\,\phi = 0. \tag{A.16}$$

Cournot aggregation (A.2), divided by $e_{ji,k}$, reads $\tilde{E}_{-ii}\mathbf{1} + \tilde{E}^{(ii)}_{-ii}\mathbf{1} = -\mathbf{1}$, so $-\tilde{E}^{-1}_{-ii}\mathbf{1} = \mathbf{1} + \tilde{E}^{-1}_{-ii}\tilde{E}^{(ii)}_{-ii}\mathbf{1}$. Solving (A.16) for ϕ and using this identity,

$$\phi = \mathbf{1} + \tilde{E}^{-1}_{-ii}\tilde{E}^{(ii)}_{-ii}\Big(1 - \frac{u_i}{1+\mu_k}\Big)_k. \tag{A.17}$$

Equation (A.17) is equivalent to (A.13) because $\tilde{E}_{-ii}$ is invertible.

Step 2: the domestic multiplier. Substitute (A.17) into (A.12) and write $m_{ii} \equiv (e_{ii,g}\eta_{ii,g})_g$, $m_{-ii} \equiv (e_{ni,g}\eta_{ni,g})_{n\neq i,g}$. Engel aggregation gives $m^\top_{-ii}\mathbf{1} = 1 - m^\top_{ii}\mathbf{1}$, so (A.12) becomes

$$0 = u_i\, m^\top_{ii} c - m^\top_{ii}\mathbf{1} + m^\top_{-ii}\tilde{E}^{-1}_{-ii}\tilde{E}^{(ii)}_{-ii}(\mathbf{1} - u_i c) = u_i \sum_k \frac{\rho^c_{i,k}}{1+\mu_k} - \sum_k \rho^c_{i,k}, \qquad \rho^c_i \equiv m_{ii} - \big(\tilde{E}^{-1}_{-ii}\tilde{E}^{(ii)}_{-ii}\big)^\top m_{-ii}. \tag{A.18}$$

Hence, provided $\sum_k \rho^c_{i,k}/(1+\mu_k) \neq 0$,

$$u_i = 1 + \tilde{\mu}_i, \qquad \tilde{\mu}_i \equiv \frac{\sum_k \rho^c_{i,k}\,\mu_k/(1+\mu_k)}{\sum_k \rho^c_{i,k}/(1+\mu_k)}. \tag{A.19}$$

Given (A.17), (A.19) is equivalent to (A.12). Under homothetic preferences, $\eta \equiv 1$ and $\tilde{E} = E$ by (A.4), so $\rho^c_i = e_{ii} - (E^{-1}_{-ii}E^{(ii)}_{-ii})^\top e_{-ii} = v_i$, the weight vector of Theorem 2 in the text. With

nonhomothetic preferences the shares are scaled by income elasticities and the E blocks are replaced by their weighted transposes, as stated in the text. Only the ratios of the weights matter for $\tilde{\mu}_i$.

Step 3: tariffs and export subsidies. Define the vector $\Omega_{-ii} = (\omega_{ni,k})_{n\neq i,k}$ and the scalar $\bar{t}_i$ by

$$(1+\bar{t}_i)(1+\omega_{ni,k}) \equiv \frac{u_n}{1+\mu_k} - \nu_n \qquad (n \neq i). \tag{A.20}$$

The split between $1+\bar{t}_i$ and $\mathbf{1}+\Omega_{-ii}$ is a normalization; only the product enters the policy formulas, and we fix it below. By (A.11), $1+t_{ni,k} = (1+\bar{t}_i)(1+\omega_{ni,k})/\phi_{ni,k}$, so (A.17) and (A.19) give the import tariff of Theorem 2:

$$1+t^{**}_{-ii} = (1+\bar{t}_i)(\mathbf{1}+\Omega_{-ii}) \oslash \Big[\mathbf{1} + \tilde{E}^{-1}_{-ii}\tilde{E}^{(ii)}_{-ii}\Big(1 - \frac{1+\tilde{\mu}_i}{1+\mu_k}\Big)_k\Big]. \tag{A.21}$$

For the export subsidy, divide (A.14) by $e_{ij,k}$ and stack over k: $\tilde{E}_{ij}\big[u_i(1+x_{ij,g})/(1+\mu_g)\big]_g + \tilde{E}^{(-ij)}_{ij}\big[u_n/(1+\mu_g) - \nu_n\big]_{n\neq i,g} = 0$. Since $\tilde{E}_{ij}$ is invertible, this is equivalent to

$$1+x^{**}_{ij} = -(1+\bar{t}_i)\Big[\tilde{E}^{-1}_{ij}\tilde{E}^{(-ij)}_{ij}(\mathbf{1}+\Omega_{-ii})\Big] \odot \Big(\frac{1+\mu_k}{1+\tilde{\mu}_i}\Big)_k. \tag{A.22}$$

Under homotheticity, (A.21)–(A.22) are the displays of Theorem 2 in the text.

Step 4: the foreign multipliers and Ω. It remains to determine $(u_n, \nu_n)_{n\neq i}$, which pin down Ω_{-ii} through (A.20). Two sets of conditions involve them. First, multiply (A.11) by $P_{ni,k}Q_{ni,k}/Y_n = r_{ni,k}\rho_{n,k}$ (recall $Y_n = \sum_{\ell,g} P_{n\ell,g}Q_{n\ell,g}$), use $\tilde{P}_{ni,k}Q_{ni,k} = e_{ni,k}Y_i$, and sum over k:

$$u_n \sum_k \frac{r_{ni,k}\rho_{n,k}}{1+\mu_k} - \nu_n \sum_k r_{ni,k}\rho_{n,k} = \frac{Y_i}{Y_n}\sum_k e_{ni,k}\,\phi_{ni,k} \qquad (n \neq i), \tag{A.23}$$

with ϕ given by (A.17) at $u_i = 1+\tilde{\mu}_i$. This condition values country n's exports to i at the optimal import valuations. Second, divide (A.15) by Y_m and use $P_{n\ell,g}Q_{n\ell,g} = Y_n r_{n\ell,g}\rho_{n,g}$ and $P_{i\ell,g}Q_{i\ell,g} = \tilde{w}_i L_i\, r_{i\ell,g}\rho_{i,g}$:

$$\nu_m - \sum_{n\neq i}\frac{Y_n}{Y_m}\sum_{\ell\neq i,g,k}\Big(\frac{u_n}{1+\mu_g} - \nu_n\Big) r_{n\ell,g}\rho_{n,g}\varepsilon^{(m\ell,k)}_{n\ell,g} = \frac{(1+\tilde{\mu}_i)\tilde{w}_i L_i}{Y_m}\sum_{\ell\neq i,g,k}\frac{r_{i\ell,g}\rho_{i,g}}{1+\mu_g}\varepsilon^{(m\ell,k)}_{i\ell,g} \qquad (m \neq i). \tag{A.24}$$

Equations (A.23)–(A.24) are $2(N-1)$ linear equations in the $2(N-1)$ unknowns $(u_n, \nu_n)_{n\neq i}$, with coefficients that depend only on sales and expenditure shares, markups, demand elasticities, and the income ratios Y_i/Y_n and $\tilde{w}_i L_i/Y_n$. This is the linear system referred to in the text. At a regular optimum it has a unique solution, and (A.20) then defines Ω_{-ii} up to the common factor $1+\bar{t}_i$. Within each exporting economy n,

$$\omega_{ni,k} - \omega_{ni,g} = \frac{u_n}{1+\bar{t}_i}\Big(\frac{1}{1+\mu_k} - \frac{1}{1+\mu_g}\Big), \tag{A.25}$$

so $\omega_{ni,k}$ is affine in $1/(1+\mu_k)$ across industries and does not vary across industries when markups are uniform.

Step 5: the Lerner family and the converse. Scaling $1+t_i$, $1+x_i$, and w_i (hence $\dot{w}_i$ and Y_i) by $a>0$ leaves quantities, shares, elasticities, and all foreign variables unchanged (Lemma 1). In (A.23)–(A.24) the coefficient matrix is unchanged and the right-hand side scales by a, so $(u_n, v_n)_{n\neq i}$ scale by a, u_i and $\tilde{\mu}_i$ are unchanged, and (A.21)–(A.22) hold along the family with $1+\bar{t}_i$ scaled by a and Ω_{-ii} unchanged. This is the uniform tax shifter of the theorem. Conversely, let a tax vector satisfy (A.21)–(A.22) with $\tilde{\mu}_i$ from (A.19) and $(u_n, v_n)_{n\neq i}$ solving (A.23)–(A.24). Then (A.17) holds by construction, so (A.13) holds; (A.22) gives (A.14); (A.19) with (A.17) gives (A.12); and (A.24) is (A.15). Hence the formulas are equivalent to the full set of first-order conditions (A.12)–(A.15). These conditions are necessary at any regular interior optimum, and sufficient for a local optimum when the Hessian of the Lagrangian is negative definite on the tangent space of the constraints; the theorem does not assert that the optimum is unique. □

Remark (the published statement). Published Theorem 2 is (A.21)–(A.22) with two substitutions: the sales-weighted markup $\bar{\mu}_i$ replaces $\tilde{\mu}_i$, and $\omega_{ni,k}$ is evaluated with equation (10). The derivation in online Appendix I obtains the $\bar{\mu}_i$ version by setting the term $\tilde{\Delta}'_i$ in its equation (I.3) to zero, which is exact to first order around $\mu_k \approx \bar{\mu}_i$ (footnote 29 there), and by treating income elasticities as unitary. In the matrix display of Theorem 2 on p. 2778 the domestic ratio is printed as $(1+\mu_k)/(1+\bar{\mu}_i)$; the Cobb–Douglas display on p. 2779 and equation (I.8) of the online appendix carry the orientation of (A.21), which is the one implied by the first-order conditions.-

Cobb–Douglas preferences. With Cobb–Douglas preferences across industries and CES within, $\eta \equiv 1$, $\tilde{E} = E$, and

$$\varepsilon^{(mi,g)}_{ni,k} = \mathbf{1}_{g=k}\big[-\sigma_k \mathbf{1}_{n=m} + (\sigma_k - 1)\lambda_{mi,k}\big], \tag{A.26}$$

so all elasticity blocks are block diagonal across industries. Write $d_{i,k} \equiv 1 + (\sigma_k - 1)\lambda_{ii,k}$. In industry k, the block of E_{-ii} is $-\sigma_k I + (\sigma_k - 1)\mathbf{1}\lambda^{\top}_{-ii,k}$ with $\lambda_{-ii,k} = (\lambda_{mi,k})_{m\neq i}$, and the corresponding column of $E^{(ii)}_{-ii}$ is $(\sigma_k - 1)\lambda_{ii,k}\mathbf{1}$. Since $\sum_{m\neq i}\lambda_{mi,k} = 1 - \lambda_{ii,k}$, the block maps $\alpha\mathbf{1}$ to $-\alpha d_{i,k}\mathbf{1}$, and therefore

$$\big[E^{-1}_{-ii}E^{(ii)}_{-ii}\big]_{(n,g),k} = -\mathbf{1}_{g=k}\frac{(\sigma_k - 1)\lambda_{ii,k}}{d_{i,k}}. \tag{A.27}$$

Substituting (A.27) into (A.17) gives $\phi_{ni,k} = \big[1 + \frac{1+\tilde{\mu}_i}{1+\mu_k}(\sigma_k - 1)\lambda_{ii,k}\big]/d_{i,k}$, and (A.21) becomes the import tariff of the Cobb–Douglas display in the text, with $1 + t^*_{ji,k} = (1+\omega_{ji,k})(1+\bar{t}_i)$. The weights (A.18) are

$$v_{i,k} = e_{i,k}\lambda_{ii,k} + e_{i,k}(1-\lambda_{ii,k})\frac{(\sigma_k - 1)\lambda_{ii,k}}{d_{i,k}} = \frac{e_{i,k}\lambda_{ii,k}\sigma_k}{d_{i,k}} > 0, \tag{A.28}$$

so by (A.19), $\tilde{\mu}_i = \sum_k \vartheta_{i,k}\mu_k / \sum_k \vartheta_{i,k}$ with $\vartheta_{i,k} = v_{i,k}/(1+\mu_k)$, as in the text. For exports, the own elasticity of i's goods in market j is $-[1 + (\sigma_k - 1)(1 - \lambda_{ij,k})]$ and the cross elasticities with respect

to origin $n \neq i$ are $(\sigma_k - 1)\lambda_{nj,k}$, so (A.22) gives $1 + x^{**}_{ij,k} = \frac{1+\mu_k}{1+\bar{\mu}_i}(1 + x^{*}_{ij,k})$ with $1 + x^{*}_{ij,k}$ as in the text.

Two countries. Let j be the only foreign country. The two equations (A.23)–(A.24) determine (u_j, v_j). A shorter route to (A.24) uses (A.14) to eliminate the export terms from (A.15): with (A.26), $\varepsilon^{(jj,k)}_{jj,k} - \varepsilon^{(ij,k)}_{jj,k}\varepsilon^{(jj,k)}_{ij,k}/\varepsilon^{(ij,k)}_{ij,k} = -\sigma_k/[1 + (\sigma_k - 1)\lambda_{jj,k}]$, and (A.15) becomes

$$-v_j = \sum_k w_{j,k}\Big(\frac{u_j}{1+\mu_k} - v_j\Big), \qquad w_{j,k} \equiv \frac{e_{j,k}\sigma_k\lambda_{jj,k}}{1 + (\sigma_k - 1)\lambda_{jj,k}}. \tag{A.29}$$

We normalize the shifter so that the revenue-weighted average of $\omega_{ji,k}$ across industries is zero, $\sum_k \rho_{j,k}\omega_{ji,k} = 0$. By (A.20) and $\sum_k \rho_{j,k}/(1+\mu_k) = 1/(1+\bar{\mu}_j)$, this means $1 + \bar{t}_i = u_j/(1+\bar{\mu}_j) - v_j$, and then

$$\omega_{ji,k} = \frac{u_j}{1+\bar{t}_i}\Big(\frac{1}{1+\mu_k} - \frac{1}{1+\bar{\mu}_j}\Big) = -\Big(1 - \frac{1+\bar{\mu}_j}{1+\mu_k}\Big)\frac{u_j}{(1+\bar{t}_i)(1+\bar{\mu}_j)}. \tag{A.30}$$

Dividing (A.29) by $1 + \bar{t}_i$ and writing $\theta_{j,k} \equiv 1 - (1+\bar{\mu}_j)/(1+\mu_k)$, so that $1 + \omega_{ji,k} = 1 - \theta_{j,k}u_j/[(1+\bar{t}_i)(1+\bar{\mu}_j)]$ and $u_j/[(1+\bar{t}_i)(1+\bar{\mu}_j)] = 1 + v_j/(1+\bar{t}_i)$, gives

$$\frac{u_j}{(1+\bar{t}_i)(1+\bar{\mu}_j)} = \frac{1 - \sum_g w_{j,g}}{1 - \sum_g w_{j,g}\theta_{j,g}} > 0, \tag{A.31}$$

which with (A.30) is the closed form of $\omega_{ji,k}$ in the text. The factor is positive because $w_{j,k} < e_{j,k}$ (as $\lambda_{jj,k} < 1$), so $\sum_g w_{j,g} < 1$, and $\theta_{j,g} < 1$, so $\sum_g w_{j,g}\theta_{j,g} < \sum_g w_{j,g}$. With uniform markups $\theta_{j,k} = 0$ and $\omega_{ji,k} = 0$. Inserting $\omega_{ji,k}$ and (A.27) into (A.21)–(A.22), with $\lambda_{nj,k} = \lambda_{jj,k}$ the only foreign share in market j and $1 - \lambda_{ij,k} = \lambda_{jj,k}$, gives the two-country display of the text.

A.3. Proof of Theorem 2′ (free entry)

Environment. Entry is free in every industry and country, and labor L_n is fixed in supply and mobile across industries. Firms pay the entry cost f^e_k in labor and charge the markup $1 + \mu_k = \gamma_k/(\gamma_k - 1)$ over marginal cost, with $\gamma_k \geq \sigma_k > 1$ (Assumption A1). Free entry equates industry profits, $\mu_k R_{n,k}/(1+\mu_k)$, to entry spending, $w_n f^e_k M_{n,k}$, so that industry employment, variable plus entry labor, is $L_{n,k} = R_{n,k}/w_n$ (footnote 14 of the paper) and $M_{n,k} = \mu_k L_{n,k}/[(1+\mu_k)f^e_k]$. Substituting into equation (4),

$$P_{nj,k} = \tau_{nj,k}\pi_{n,k}, \qquad \pi_{n,k} \equiv \frac{w_n}{B_{n,k}L_{n,k}^{\mu_k}}, \qquad \frac{1}{B_{n,k}} \equiv (1+\mu_k)\bar{a}_{n,k}\Big[\frac{\mu_k}{(1+\mu_k)f^e_k}\Big]^{-\mu_k}, \tag{A.32}$$

so that $d\ln P_{nj,k} = d\ln w_n - \mu_k d\ln L_{n,k}$: a larger industry has lower prices through entry. Industry output, inclusive of trade costs, is $Q_{n,k} \equiv \sum_j \tau_{nj,k}Q_{nj,k} = R_{n,k}/\pi_{n,k} = B_{n,k}L_{n,k}^{1+\mu_k}$. Profits net of entry costs are zero, so $\bar{\mu}_n = 0$, $\tilde{w}_n = w_n$, $Y_n = w_n L_n$ for $n \neq i$, and $Y_i = w_i L_i + \mathcal{R}_i$. Taxes and consumer prices are as in (A.8). No restriction is placed on foreign wages. As in Section A.2 we work with the Lerner representative $w_i = 1$.

The planning problem. The optimal policy solves

$$\max_{\{\tilde{P}_{ni,k}\}_{n\neq i},\{\tilde{P}_{ij,k}\}_{j\neq i},Y_i,\{L_{n,k}\},\{w_n\}_{n\neq i}} V_i(Y_i,\tilde{P}_i) \tag{A.33}$$
$$\text{s.t.}\quad B_{n,k}L_{n,k}^{1+\mu_k}-\sum_j \tau_{nj,k}Q_{nj,k}=0\ \ \forall n,k, \qquad L_n-\sum_k L_{n,k}=0\ \ \forall n,$$

where $Q_{nj,k}=\mathcal{D}_{nj,k}(Y_j,\tilde{P}_j)$, $Y_j=w_jL_j$ for $j\neq i$, uncontrolled consumer prices equal the producer prices (A.32), and $w_i=1$. The constraints are goods-market clearing in every industry and labor-market clearing in every country. At a feasible point, $R_{n,k}=\pi_{n,k}\mathcal{Q}_{n,k}=w_nL_{n,k}$ and $\sum_k R_{n,k}=w_nL_n$, so adding the budget identities over markets as in Section A.2 gives $Y_i=w_iL_i+\mathcal{R}_i$ and balanced trade; conversely, any feasible point is implemented by $1+t_{ni,k}=\tilde{P}_{ni,k}/P_{ni,k}$ and $1+x_{ij,k}=P_{ij,k}/\tilde{P}_{ij,k}$. Hence (A.33) is equivalent to the government's problem.

First-order conditions. Let $\lambda_i\equiv\partial V_i/\partial Y_i$, and attach the multipliers $\alpha_{n,k}$ to the goods-market constraints and ν_n to the labor-market constraints. Define the normalized multipliers and the *valuations*

$$u_{n,k}\equiv\frac{\alpha_{n,k}}{\lambda_i\pi_{n,k}}, \qquad h_n\equiv\frac{\nu_n}{\lambda_i w_n}, \qquad z_{nj,k}\equiv u_{n,k}\frac{P_{nj,k}}{\tilde{P}_{nj,k}}, \tag{A.34}$$

so that $\alpha_{n,k}\tau_{nj,k}Q_{nj,k}=\lambda_i u_{n,k}P_{nj,k}Q_{nj,k}=\lambda_i z_{nj,k}\tilde{P}_{nj,k}Q_{nj,k}$, and $z_{ni,k}=u_{n,k}/(1+t_{ni,k})$, $z_{ij,k}=u_{i,k}(1+x_{ij,k})$, $z_{ii,k}=u_{i,k}$, $z_{nj,k}=u_{n,k}$ otherwise. Let $\mathbb{X}_{nj,k}\equiv\tilde{P}_{nj,k}Q_{nj,k}=e_{nj,k}Y_j$ and define, for every origin n, market j, and industry k, the residual

$$b_{nj,k}\equiv\mathbf{1}_{j=i}\mathbb{X}_{nj,k}+\sum_{a,g}u_{a,g}P_{aj,g}Q_{aj,g}\,\varepsilon_{aj,g}^{(nj,k)}=\mathbb{X}_{nj,k}\Big[\mathbf{1}_{j=i}+\sum_{a,g}(\tilde{E}_{nj}^{(aj)})_{kg}z_{aj,g}\Big], \tag{A.35}$$

which measures the marginal value of a one-percent fall in the consumer price $\tilde{P}_{nj,k}$ through the goods-market constraints, net of its direct value to home consumers when $j=i$. Using Roy's identity for the controlled prices and for $\tilde{P}_{ii,k}$, the first-order conditions with respect to Y_i, $\tilde{P}_{ji,k}$ $(j\neq i)$, $\tilde{P}_{ij,k}$ $(j\neq i)$, $L_{n,k}$, and w_m $(m\neq i)$ are

$$1=\sum_{n,k}e_{ni,k}\eta_{ni,k}z_{ni,k}, \tag{A.36}$$
$$0=b_{ji,k}\quad(j\neq i), \tag{A.37}$$
$$0=b_{ij,k}\quad(j\neq i), \tag{A.38}$$
$$h_n=(1+\mu_k)u_{n,k}+\frac{\mu_k}{R_{n,k}}\sum_{j\in\mathcal{U}_n}b_{nj,k}, \tag{A.39}$$
$$0=\sum_{j\neq i,k}b_{mj,k}+Z_m, \qquad Z_m\equiv\sum_{a,g}u_{a,g}P_{am,g}Q_{am,g}\,\eta_{am,g}, \tag{A.40}$$

where $\mathcal{U}_i=\{i\}$ and $\mathcal{U}_n=\mathbb{C}\setminus\{i\}$ for $n\neq i$ are the markets in which origin n's consumer price is not controlled by i and therefore moves with $L_{n,k}$. In (A.39), the first term is the value of the marginal

product of labor through $\partial(B_{n,k}L_{n,k}^{1+\mu_k})/\partial L_{n,k} = (1+\mu_k)w_n/\pi_{n,k}$, and the second collects the price effect $d\ln P_{nj,k} = -\mu_k d\ln L_{n,k}$ in the markets $\mathcal{U}_n$ (for $n = i$ the direct effect on home consumers, $\lambda_i\mu_k\mathbb{X}_{ii,k}/L_{i,k}$, is the term $\mathbf{1}_{j=i}\mathbb{X}_{nj,k}$ in $b_{ii,k}$). Condition (A.40) collects the effect of w_m on the foreign consumer prices $\tilde{P}_{m\ell,k}$, $\ell \neq i$, and on foreign income $Y_m = w_m L_m$. There is no condition for w_i, which is the numeraire.

Step 1: the import block and the domestic valuations. Dividing (A.37) by $\mathbb{X}_{ji,k}$ and stacking over (j,k), $j \neq i$, gives $\mathbf{1} + \tilde{E}_{-ii}^{(ii)}u_i + \tilde{E}_{-ii}z_{-ii} = 0$, with $u_i = (u_{i,k})_k$ and $z_{-ii} = (u_{n,k}/(1+t_{ni,k}))_{n\neq i,k}$. Using the Cournot identity $\tilde{E}_{-ii}\mathbf{1} + \tilde{E}_{-ii}^{(ii)}\mathbf{1} = -\mathbf{1}$ as in Section A.2,

$$z_{-ii} = \mathbf{1} + \tilde{E}_{-ii}^{-1}\tilde{E}_{-ii}^{(ii)}(\mathbf{1}-u_i), \qquad \text{hence} \qquad 1 + t_{-ii} = u_{-ii} \oslash \left[\mathbf{1} + \tilde{E}_{-ii}^{-1}\tilde{E}_{-ii}^{(ii)}(\mathbf{1}-u_i)\right], \tag{A.41}$$

where $u_{-ii} \equiv (u_{n,k})_{n\neq i,k}$. Substituting (A.41) into (A.36) and using Engel aggregation exactly as in (A.18) gives

$$v_i^\top(u_i - \mathbf{1}) = 0, \qquad v_i \equiv m_{ii} - \left(\tilde{E}_{-ii}^{-1}\tilde{E}_{-ii}^{(ii)}\right)^\top m_{-ii}, \tag{A.42}$$

with m_{ii} and m_{-ii} as in Section A.2; under homotheticity $v_i = e_{ii} - (E_{-ii}^{-1}E_{-ii}^{(ii)})^\top e_{-ii}$, as in the text.

Step 2: the domestic scale wedge. For $n = i$, (A.39) involves $b_{ii,k} = \mathbb{X}_{ii,k}[1 + (\tilde{E}_{ii}u_i)_k + (\tilde{E}_{ii}^{(-ii)}z_{-ii})_k]$, where $\tilde{E}_{ii} \equiv \tilde{E}_{ii}^{(ii)}$ and $\tilde{E}_{ii}^{(-ii)} \equiv [\tilde{E}_{ii}^{(mi)}]_{m\neq i}$. Substituting (A.41) and the Cournot identity for the domestic rows, $\tilde{E}_{ii}\mathbf{1} + \tilde{E}_{ii}^{(-ii)}\mathbf{1} = -\mathbf{1}$,

$$b_{ii,k} = -\mathbb{X}_{ii,k}\left[S_i(\mathbf{1}-u_i)\right]_k, \qquad S_i \equiv \tilde{E}_{ii} - \tilde{E}_{ii}^{(-ii)}\tilde{E}_{-ii}^{-1}\tilde{E}_{-ii}^{(ii)}. \tag{A.43}$$

Since $\mathbb{X}_{ii,k}/R_{i,k} = r_{ii,k}$, the home conditions (A.39) read $h_i = (1+\mu_k)u_{i,k} - \mu_k r_{ii,k}[S_i(\mathbf{1}-u_i)]_k$ for every k. With $\Delta_i \equiv \mathbf{1} - u_i$, $\tilde{\mu}_i \equiv h_i - 1$, and $H_i \equiv \mathrm{diag}(\mathbf{1}+\mu) + \mathrm{diag}(\mu \odot r_{ii})S_i$, this is

$$H_i\Delta_i = \mu - \tilde{\mu}_i\mathbf{1}. \tag{A.44}$$

Equations (A.42) and (A.44) are $K+1$ linear equations in $(\Delta_i, \tilde{\mu}_i)$. When H_i is invertible and $v_i^\top H_i^{-1}\mathbf{1} \neq 0$, their unique solution is

$$u_i = \mathbf{1} - H_i^{-1}(\mu - \tilde{\mu}_i\mathbf{1}), \qquad \tilde{\mu}_i = \frac{v_i^\top H_i^{-1}\mu}{v_i^\top H_i^{-1}\mathbf{1}}, \tag{A.45}$$

the scale wedge and average scale elasticity of Theorem 2′. If scale elasticities are uniform, $\mu_k = \mu$, then $\tilde{\mu}_i = \mu$ and $u_i = \mathbf{1}$.

Step 3: export subsidies. Dividing (A.38) by $\mathbb{X}_{ij,k}$ and stacking over k: $\tilde{E}_{ij}z_{ij} + \tilde{E}_{ij}^{(-ij)}u_{-ii} = 0$ with $z_{ij} = u_i \odot (1+x_{ij})$, so

$$1 + x_{ij} = -\left[\tilde{E}_{ij}^{-1}\tilde{E}_{ij}^{(-ij)}u_{-ii}\right] \oslash u_i. \tag{A.46}$$

Step 4: the foreign block and Ω. The foreign valuations u_{-ii} and $h_{-i} \equiv (h_n)_{n\neq i}$ are determined by the foreign conditions (A.39) ($n \neq i$) and (A.40). In market $j \neq i$, (A.46) eliminates z_{ij} from the residuals of the foreign origins, $b_{-ij} \equiv (b_{nj,k})_{n\neq i,k}$, and from Z_j:

$$\begin{aligned} b_{-ij} &= \operatorname{diag}(\mathbb{X}_{-ij})\, S_j^F\, u_{-ii}, \qquad S_j^F \equiv \tilde{E}_{-ij} - \tilde{E}_{-ij}^{(ij)} \tilde{E}_{ij}^{-1} \tilde{E}_{ij}^{(-ij)}, \\ Z_j &= d_j^\top u_{-ii}, \qquad d_j \equiv \mathbb{X}_{-ij} \odot \eta_{-ij} - \left(\tilde{E}_{ij}^{-1} \tilde{E}_{ij}^{(-ij)}\right)^\top (\mathbb{X}_{ij} \odot \eta_{ij}). \end{aligned} \tag{A.47}$$

Let $B_i^F \equiv \sum_{j\neq i} \operatorname{diag}(\mathbb{X}_{-ij}) S_j^F$, let C be the $(N-1)K \times (N-1)$ incidence matrix with $C_{(n,k),a} = \mathbf{1}_{n=a}$, let $R_{-i} \equiv (R_{n,k})_{n\neq i,k}$ and $\mu_{-i} \equiv (\mu_k)_{n\neq i,k}$, and let D_i^F be the $(N-1) \times (N-1)K$ matrix with rows $d_j^\top$. Then the foreign conditions are the homogeneous linear system

$$\begin{aligned} A_i^F u_{-ii} &= C h_{-i}, \qquad W_i^F u_{-ii} = 0, \\ A_i^F &\equiv \operatorname{diag}(\mathbf{1} + \mu_{-i}) + \operatorname{diag}(\mu_{-i} \oslash R_{-i}) B_i^F, \qquad W_i^F \equiv C^\top B_i^F + D_i^F. \end{aligned} \tag{A.48}$$

Its coefficients are scale elasticities, revenue shares ($\mathbb{X}_{nj,k}/R_{n,k} = r_{nj,k}$), expenditure shares, and demand elasticities. One equation is redundant: by homogeneity (A.2), $\sum_{n,k} b_{nj,k} = -Z_j$ in every market $j \neq i$, and $b_{ij,k} = 0$ by (A.38), so $\mathbf{1}^\top W_i^F = 0$ identically. Hence (A.48) has $(N-1)K + N - 2$ independent equations in $(N-1)K + N - 1$ unknowns and, at a regular optimum, a one-dimensional solution set: u_{-ii} is determined up to a positive scalar. We write

$$u_{-ii} = (1+\bar{t}_i)(\mathbf{1} + \Omega_{-ii}), \qquad h_{-i} = (1+\bar{t}_i)\xi_{-i}, \qquad q^\top(\mathbf{1} + \Omega_{-ii}) = 1, \qquad q \equiv R_{-i}/\mathbf{1}^\top R_{-i}, \tag{A.49}$$

so that $\mathbf{1} + \Omega_{-ii}$ is the solution of (A.48) normalized to a revenue-weighted mean of one and $1 + \bar{t}_i$ is the uniform shifter; $\Omega_{-ii} = (\Omega_{ji})_{j\neq i}$ stacks the blocks $\Omega_{ji} = (\omega_{ji,k})_k$ of the text. With this notation, (A.41) and (A.46) are the displays of Theorem 2′ in the text, with u_i from (A.45).

Step 5: the Lerner family and the converse. Steps 1–4 solve the first-order conditions (A.36)–(A.40) completely: (A.42)–(A.44) determine (u_i, h_i), (A.48) determines (u_{-ii}, h_{-i}) up to the scalar $1 + \bar{t}_i$, and (A.41) and (A.46) then determine the policies. Scaling (u_{-ii}, h_{-i}) by $a > 0$ leaves (u_i, h_i) unchanged and multiplies every $1 + t_{ni,k}$ and every $1 + x_{ij,k}$ by a: a uniform import tariff combined with a uniform export subsidy at the same rate, which is neutral by Lemma 1 of the paper. The critical points of (A.33) therefore form one Lerner family, indexed by $\bar{t}_i$, and the normalization in (A.49) fixes the representative. Conversely, given a solution (u, h) of (A.42), (A.44), and (A.48), the multipliers $\alpha_{n,k} = \lambda_i \pi_{n,k} u_{n,k}$ and $\nu_n = \lambda_i w_n h_n$ and the policies (A.41) and (A.46) satisfy every first-order condition of (A.33). The derivation uses the invertibility of $\tilde{E}_{-ii}$, $\tilde{E}_{ij}$, and H_i, the condition $v_i^\top H_i^{-1} \mathbf{1} \neq 0$, and the rank condition on (A.48); these hold generically. As in Section A.2, the first-order conditions are necessary at an optimum satisfying the constraint qualification and sufficient for a local optimum under the second-order conditions; the theorem characterizes the optimum and does not assert its uniqueness. □

Cobb–Douglas preferences. Under the preferences of (A.26), income elasticities are unitary, so $\tilde{E} = E$ and the blocks are those computed in Section A.2: by (A.27), $E_{-ii}^{-1}E_{-ii}^{(ii)}$ is block diagonal across industries and equals $-(\sigma_k-1)\lambda_{ii,k}/d_{i,k}$ on the industry-k block, with $d_{i,k} \equiv 1+(\sigma_k-1)\lambda_{ii,k}$. Then (A.41) gives $z_{ni,k} = [1+(\sigma_k-1)\lambda_{ii,k}u_{i,k}]/d_{i,k}$ and

$$1+t_{ni,k} = \frac{1+(\sigma_k-1)\lambda_{ii,k}}{1+(\sigma_k-1)\lambda_{ii,k}u_{i,k}}\,u_{n,k}, \qquad 1+x_{ij,k} = \frac{(\sigma_k-1)\sum_{n\neq i}\lambda_{nj,k}u_{n,k}}{[1+(\sigma_k-1)(1-\lambda_{ij,k})]\,u_{i,k}}, \tag{A.50}$$

the second from (A.46) with $E_{ij} = -[1+(\sigma_k-1)(1-\lambda_{ij,k})]$ and $E_{ij}^{(-ij)} = (\sigma_k-1)\lambda_{nj,k}$ on industry k. With $u_{n,k} = (1+\bar{t}_i)(1+\omega_{ni,k})$, these are the displays of the text. For the domestic block, $E_{ii} = -\sigma_k+(\sigma_k-1)\lambda_{ii,k}$ and $E_{ii}^{(-ii)}E_{-ii}^{-1}E_{-ii}^{(ii)} = -(\sigma_k-1)^2\lambda_{ii,k}(1-\lambda_{ii,k})/d_{i,k}$ on industry k, and the identity $[-\sigma_k+(\sigma_k-1)\lambda_{ii,k}]\,d_{i,k}+(\sigma_k-1)^2\lambda_{ii,k}(1-\lambda_{ii,k}) = -\sigma_k$ gives

$$S_i = \operatorname{diag}\Big(-\frac{\sigma_k}{d_{i,k}}\Big)_k, \qquad H_i = \operatorname{diag}\big(1+\mu_k-\kappa_{i,k}\big)_k, \qquad \kappa_{i,k} \equiv \frac{\mu_k\sigma_k r_{ii,k}}{1+(\sigma_k-1)\lambda_{ii,k}}. \tag{A.51}$$

Since $r_{ii,k} \leq 1 \leq d_{i,k}$, we have $\kappa_{i,k} \leq \mu_k\sigma_k$, so $1+\mu_k-\kappa_{i,k} \geq (\gamma_k-\sigma_k)/(\gamma_k-1) > 0$ whenever $\sigma_k < \gamma_k$, and H_i is then invertible. Solving (A.44) industry by industry and using $v_{i,k} = e_{i,k}\lambda_{ii,k}\sigma_k/d_{i,k}$ from (A.28),

$$u_{i,k} = \frac{1+\tilde{\mu}_i-\kappa_{i,k}}{1+\mu_k-\kappa_{i,k}}, \qquad \tilde{\mu}_i = \frac{\sum_k \vartheta_{i,k}\mu_k}{\sum_k \vartheta_{i,k}}, \qquad \vartheta_{i,k} = \frac{e_{i,k}\lambda_{ii,k}\sigma_k}{[1+(\sigma_k-1)\lambda_{ii,k}](1+\mu_k-\kappa_{i,k})}, \tag{A.52}$$

the Cobb–Douglas scale wedge of the text. The weights $\vartheta_{i,k}$ are positive, so $\tilde{\mu}_i$ lies between the smallest and the largest scale elasticity; $u_{i,k} > 1$ in industries with $\mu_k < \tilde{\mu}_i$ and $u_{i,k} < 1$ in industries with $\mu_k > \tilde{\mu}_i$. If $\mu_k = \mu$ for all k, then $\tilde{\mu}_i = \mu$ and $u_i = \mathbf{1}$ whatever the $\kappa_{i,k}$; as $\mu \to 0$, $\kappa_{i,k} \to 0$ and $u_i \to \mathbf{1}$.

Two countries. Let $N = 2$ and call the foreign country j. In market j the only foreign origin is j itself, and the same algebra as in (A.51), with $\lambda_{ij,k} = 1-\lambda_{jj,k}$, gives $S_j^F = \operatorname{diag}(-\sigma_k/d_{j,k})_k$ with $d_{j,k} \equiv 1+(\sigma_k-1)\lambda_{jj,k}$. The foreign employment conditions (A.39), in which $\mathcal{U}_j = \{j\}$ and $\mathbb{X}_{jj,k}/R_{j,k} = r_{jj,k}$, read

$$h_j = (1+\mu_k-\kappa_{j,k})\,u_{j,k}, \qquad \kappa_{j,k} \equiv \frac{\mu_k\sigma_k r_{jj,k}}{1+(\sigma_k-1)\lambda_{jj,k}}, \tag{A.53}$$

and the foreign wage condition (A.40) holds identically: d_j in (A.47) has entries $Y_j e_{j,k}\lambda_{jj,k}\sigma_k/d_{j,k} = -b_{jj,k}/u_{j,k}$, which is the redundant equation of (A.48). Hence $u_{j,k} = h_j/(1+\mu_k-\kappa_{j,k})$, the free scalar h_j is the uniform component, and with the weights $q = (\rho_{j,k})_k$ the normalization (A.49) gives

$$1+\omega_{ji,k} = \frac{1/(1+\mu_k-\kappa_{j,k})}{\sum_g \rho_{j,g}/(1+\mu_g-\kappa_{j,g})}, \qquad 1+\bar{t}_i = h_j\sum_g \frac{\rho_{j,g}}{1+\mu_g-\kappa_{j,g}}. \tag{A.54}$$

The two-country display of the text absorbs the normalizing sum into the uniform component and writes $1+\bar{t}_i$ for h_j; relative tariffs and relative export subsidies across industries are the same under either convention. Substituting $u_{j,k} = (1+\bar{t}_i)/(1+\mu_k-\kappa_{j,k})$ into (A.50), with $\lambda_{nj,k} = \lambda_{jj,k}$ and $1-\lambda_{ij,k} = \lambda_{jj,k}$, gives the two-country formulas of the text. Three checks follow. With one industry, $\tilde{\mu}_i = \mu$, $u_i = 1$, and $1/(1+\mu-\kappa_j)$ is common to the tariff and the export subsidy, so scale economies enter only through the uniform component and the optimal policy is the terms-of-trade policy with export tax factor $(\sigma-1)\lambda_{jj}/[1+(\sigma-1)\lambda_{jj}]$. With uniform scale elasticities, $u_i = \mathbf{1}$, but $\omega_{ji,k}$ does not vanish unless $\kappa_{j,k}$ is constant across industries: the foreign scale response varies with σ_k, $r_{jj,k}$, and $\lambda_{jj,k}$ even when μ_k does not. As $\mu_k \to 0$ for all k, $u_i \to \mathbf{1}$ and $\omega_{ji,k} \to 0$.

Remark (restricted entry and the published statement). Dropping the domestic scale feedback $\operatorname{diag}(\mu \odot r_{ii})S_i$ from H_i gives $H_i = \operatorname{diag}(\mathbf{1}+\mu)$ and $u_{i,k} = (1+\tilde{\mu}_i)/(1+\mu_k)$, and (A.41) and (A.46) become the import and export displays of Theorem 2 in Section A.2, with Ω from the respective foreign system. Under free entry the term $\operatorname{diag}(\mu \odot r_{ii})S_i$ records that a tariff-induced reallocation of domestic demand changes domestic industry scale and hence domestic prices, which does not happen with a fixed mass of firms. In the Cobb–Douglas case the same feedback is the term $\kappa_{i,k}$ in (A.52), and setting $\kappa_{i,k} = 0$ there gives $u_{i,k} = (1+\tilde{\mu}_i)/(1+\mu_k)$. Replacing $\tilde{\mu}_i$ by $\bar{\mu}_i$ and evaluating $\omega_{ji,k}$ with equation (10) of the paper, as in the Remark of Section A.2, recovers Theorem 2 as published.

A.4. Proof of Theorem 3 (restricted entry)

Environment. The environment is that of Section A.2 with export taxes unavailable: $x_{ij,k} = 0$ for all $j \neq i$ and k, so that $\tilde{P}_{ij,k} = P_{ij,k}$ in (A.8). Two features differ. First, every foreign wage and income adjusts: the foreign incomes Y_n, $n \neq i$, are variables, determined jointly with the foreign wages by the labor-market clearing conditions and income identities of the foreign economies. With two countries this coincides with the stipulation of Section A.2, because fixing the single foreign effective wage, or the home wage, is only a choice of the unit of account; with three or more countries the two environments differ. Second, there is no Lerner family: scaling $1+t_i$ alone changes the allocation (the neutrality of Lemma 1 of the paper requires $1+t_i$ and $1+x_i$ to be scaled together), so no uniform shifter is left free and the level of the tariff vector is determined. We take the home wage as the unit of account, $w_i = 1$, so that $P_{ii,k} = \tilde{P}_{ii,k}$ and $P_{ij,k}$ are constant; every other nominal variable is measured relative to it.

The planning problem. The government chooses the home consumer prices of imports, home income, and the foreign wages and incomes, subject to labor-market clearing in every economy

and the income identity of every foreign economy:

$$\max_{\{\tilde{P}_{ni,k}\}_{n\neq i},\, Y_i,\, \{w_n, Y_n\}_{n\neq i}} V_i(Y_i, \tilde{P}_i) \tag{A.55}$$
$$\text{s.t.} \quad w_n L_n - \sum_{j,k} \frac{P_{nj,k} Q_{nj,k}}{1+\mu_k} = 0 \;\; \forall n, \qquad \sum_{j,k} P_{nj,k} Q_{nj,k} - Y_n = 0 \;\; \forall n \neq i.$$

The number of controls exceeds the number of constraints by $(N-1)K$, the number of import tariffs. As in Section A.2, the home budget constraint is implied by the others and is omitted, and the problem is equivalent to choosing the tariffs, with $1 + t_{ni,k} = \tilde{P}_{ni,k}/P_{ni,k}$.

First-order conditions. Let $\lambda_i \equiv \partial V_i/\partial Y_i$ and write the multipliers as $\lambda_i u_n$ and $\lambda_i \nu_n$. Using Roy's identity, $d\ln P_{nj,k} = d\ln w_n$ for $n \neq i$, and the constraints, the variation of the Lagrangian is

$$\frac{d\mathcal{L}_i}{\lambda_i} = dY_i - \sum_{n\neq i,k} \tilde{P}_{ni,k} Q_{ni,k}\, d\ln \tilde{P}_{ni,k} + \sum_{n\neq i} \nu_n \left(Y_n\, d\ln w_n - dY_n\right) - \sum_{n,j,k} \psi_{n,k} P_{nj,k} Q_{nj,k}\, d\ln Q_{nj,k}, \tag{A.56}$$

where

$$\psi_{n,k} \equiv \frac{u_n}{1+\mu_k} - \nu_n \mathbf{1}_{n\neq i} \tag{A.57}$$

collects the multipliers attached to a unit of sales of economy n in industry k. Relative to (A.10), the term $-\nu_n dY_n$ appears because the foreign incomes are variables. In the home market, $d\ln Q_{ni,g} = \eta_{ni,g}\, d\ln Y_i + \sum_{m\neq i,k} \varepsilon^{(mi,k)}_{ni,g}\, d\ln \tilde{P}_{mi,k}$, since $\tilde{P}_{ii,k}$ is constant. In a foreign market j, where there are no taxes and $P_{ij,k}$ is constant, $d\ln Q_{nj,g} = \eta_{nj,g}\, d\ln Y_j + \sum_{m\neq i,k} \varepsilon^{(mj,k)}_{nj,g}\, d\ln w_m$. The first-order conditions for Y_i and $\tilde{P}_{ji,k}$ are (A.12) and (A.13), with $\phi_{ni,k} \equiv \psi_{n,k}/(1+t_{ni,k})$ as in (A.11). Those for the income and the wage of a foreign economy m are, using $P_{nm,g} Q_{nm,g}/Y_m = e_{nm,g}$ for every n,

$$-\nu_m = \sum_{n,g} \psi_{n,g}\, e_{nm,g}\, \eta_{nm,g}, \tag{A.58}$$
$$\nu_m Y_m = \sum_{\ell\neq i} Y_\ell \sum_{n,g,k} \psi_{n,g}\, e_{n\ell,g}\, \varepsilon^{(m\ell,k)}_{n\ell,g}. \tag{A.59}$$

Equation (A.59) is (A.15) with $x_{ij,k} = 0$; the effect of w_m on the home consumer prices $\tilde{P}_{mi,k}$ is absorbed by the tariffs. There is no first-order condition for export prices. At a regular interior optimum (the constraint gradients linearly independent, as in Section A.2) these conditions are necessary, and the steps below solve them completely.

Steps 1–3: the import tariff given the foreign multipliers. Conditions (A.12) and (A.13) are those of Section A.2, so Steps 1–3 there apply verbatim, including the nonhomothetic case: (A.17) gives ϕ, (A.19) gives $u_i = 1 + \tilde{\mu}_i$ with the same demand-weighted average markup as in Theorem 2, and, writing

$$(1+\bar{\tau}^*_i)(1+\omega_{ni,k}) \equiv \psi_{n,k} = \frac{u_n}{1+\mu_k} - \nu_n \qquad (n \neq i) \tag{A.60}$$

in place of (A.20), the tariff (A.21) becomes

$$1 + t^{***}_{-ii} = (1 + \bar{\tau}^*_i)(\mathbf{1} + \Omega_{-ii}) \oslash \Big[\mathbf{1} + \tilde{E}^{-1}_{-ii}\tilde{E}^{(ii)}_{-ii}\Big(1 - \frac{1+\tilde{\mu}_i}{1+\mu_k}\Big)_k\Big], \tag{A.61}$$

the display of Theorem 3 (with homothetic preferences the $\tilde{E}$ blocks are the E matrices of the paper). The first-order conditions determine the product in (A.60); its split into a uniform shifter and the inverse export supply elasticities is a normalization of Ω_{-ii}, fixed below in the two-country case as in Section A.2.

Step 4: the foreign multipliers and the level of the tariff. Given $u_i = 1 + \tilde{\mu}_i$, conditions (A.58) and (A.59) for $m \neq i$ are $2(N-1)$ linear equations in the $2(N-1)$ unknowns $(u_n, v_n)_{n\neq i}$, with coefficients that depend only on expenditure shares, markups, demand elasticities, and the income ratios Y_ℓ / Y_m:

$$-v_m - \sum_{n\neq i,g}\Big(\frac{u_n}{1+\mu_g} - v_n\Big)e_{nm,g}\eta_{nm,g} = (1+\tilde{\mu}_i)\sum_g \frac{e_{im,g}\eta_{im,g}}{1+\mu_g}, \tag{A.62}$$

$$v_m - \sum_{\ell\neq i}\frac{Y_\ell}{Y_m}\sum_{n\neq i,g,k}\Big(\frac{u_n}{1+\mu_g} - v_n\Big)e_{n\ell,g}\varepsilon^{(m\ell,k)}_{n\ell,g} = (1+\tilde{\mu}_i)\sum_{\ell\neq i}\frac{Y_\ell}{Y_m}\sum_{g,k}\frac{e_{i\ell,g}\varepsilon^{(m\ell,k)}_{i\ell,g}}{1+\mu_g}. \tag{A.63}$$

This is the linear system referred to in the text. Its coefficient matrix is the transpose of the Jacobian of the foreign equilibrium conditions in (A.55), the labor-market clearing conditions and income identities of the foreign economies, with respect to the foreign wages and incomes; the right-hand side collects the derivatives of the home labor-market clearing condition with respect to the same variables, multiplied by u_i. We assume that this matrix is nonsingular. This is the condition under which the foreign equilibrium is a smooth function of the home policy, and it holds in every case we have computed. The system then determines $(u_n, v_n)_{n\neq i}$, and (A.60)–(A.61) determine the tariff vector, including its level. Within each exporting economy n,

$$\omega_{ni,k} - \omega_{ni,g} = \frac{u_n}{1+\bar{\tau}^*_i}\Big(\frac{1}{1+\mu_k} - \frac{1}{1+\mu_g}\Big), \tag{A.64}$$

as in (A.25): the product in (A.60) is affine in $1/(1+\mu_k)$, and it does not vary across industries when markups are uniform.

Step 5: the converse. Conversely, let $u_i = 1 + \tilde{\mu}_i$, let $(u_n, v_n)_{n\neq i}$ solve (A.62)–(A.63), and let the tariffs be given by (A.61). Then (A.13) holds by construction, (A.12) holds by (A.19), and (A.58)–(A.59) are (A.62)–(A.63): every first-order condition of (A.55) is satisfied. As in Section A.2, the first-order conditions are necessary at a regular optimum and sufficient for a local optimum when the Hessian of the Lagrangian is negative definite on the tangent space of the constraints. The theorem determines the tariff vector at any such optimum, including its level; it does not assert that the optimum is unique. □

Remark (the published statement). Theorem 3 of the paper (p. 2780) is (A.61) with $\bar{\mu}_i$ in place of $\tilde{\mu}_i$, with $\omega_{ji,k}$ evaluated with equation (10), and with the matrix display oriented as discussed in the Remark of Section A.2, multiplied by the uniform shifter of equation (J.5) of the online appendix,

$$1+\bar{\tau}_i^{*}=1+\frac{1}{\bar{e}_i},\qquad \bar{e}_i\equiv-\sum_{j\neq i}\sum_{g}\chi_{ij,g}\Big(1+\sum_{s}\varepsilon_{ij,g}^{(ij,s)}\Big), \tag{A.65}$$

where $\chi_{ij,g}$ is the share of destination j and industry g in economy i's exports and $\bar{e}_i$ is the elasticity of international demand for economy i's labor. In the main-text display of the paper the unit term of (A.65) is summed over s as well as g; the matrix form of (J.5), $(I_K+E_{ij})\mathbf{1}_K$, counts it once, and that is the form used here. Appendix J obtains (J.5) by evaluating the response of foreign demand to the home wage at fixed foreign wages and incomes and by treating home income as proportional to the home wage. Here these two steps are replaced by (A.58)–(A.59), in which every foreign wage and income adjusts, and the product $(1+\bar{\tau}_i^{*})(1+\omega_{ji,k})$ is determined jointly rather than as the product of a shifter computed at fixed foreign variables and the $\omega_{ji,k}$ of Theorem 2. The uniqueness noted after the published statement is the absence of the Lerner family, which fixes the level of the tariff vector; it is not a claim that the planning problem has a unique solution, and none is made here. With two countries and uniform markups the published formula is exact, as shown below.

Cobb–Douglas preferences. With Cobb–Douglas preferences, (A.27) gives $\phi_{ni,k}=\big[1+\frac{1+\tilde{\mu}_i}{1+\mu_k}(\sigma_k-1)\lambda_{ii,k}\big]/d_{i,k}$ as in Section A.2, and (A.61) reads

$$1+t_{ji,k}^{***}=(1+\bar{\tau}_i^{*})(1+\omega_{ji,k})\,\frac{1+(\sigma_k-1)\lambda_{ii,k}}{1+\frac{1+\tilde{\mu}_i}{1+\mu_k}(\sigma_k-1)\lambda_{ii,k}}, \tag{A.66}$$

with $\tilde{\mu}_i$ as in Section A.2, the weights (A.28). This is the Cobb–Douglas display of the text.

Two countries. Let j be the only foreign country, so that (A.62)–(A.63) are two equations in (u_j,v_j). Consider first uniform markups, $\mu_k=\mu$, with any homothetic preferences, so that $\eta=1$. Then $u_i=1+\mu$, hence $\phi=\mathbf{1}$ by (A.17), and $\psi_{j,k}=u_j/(1+\mu)-v_j\equiv\psi_j$ does not depend on k: the tariff is uniform, $1+t_{ji,k}^{***}=\psi_j$. Writing $S_{ij}\equiv\sum_g e_{ij,g}$ for the share of economy j's expenditure that falls on imports, (A.58) and (A.59) read

$$-v_j=S_{ij}+\psi_j(1-S_{ij}),\qquad v_j=\sum_{g,k}e_{ij,g}\varepsilon_{ij,g}^{(jj,k)}+\psi_j\sum_{g,k}e_{jj,g}\varepsilon_{jj,g}^{(jj,k)}. \tag{A.67}$$

By the Cournot aggregation in (A.2), the two sums in the second equation add up to $-(1-S_{ij})$, and by the homogeneity condition there, $\sum_k\varepsilon_{ij,g}^{(jj,k)}=-\big(1+\sum_s\varepsilon_{ij,g}^{(ij,s)}\big)$, so that the first sum equals $S_{ij}\bar{e}_{ij}$, with $\bar{e}_{ij}$ the two-country value of (A.65) and $\chi_{ij,g}=e_{ij,g}/S_{ij}$. Adding the two equations in (A.67) gives $S_{ij}(1+\bar{e}_{ij})=\psi_jS_{ij}\bar{e}_{ij}$, hence, provided $\bar{e}_{ij}\neq0$,

$$1+t_{ji,k}^{***}=1+\frac{1}{\bar{e}_{ij}}\qquad\text{for every } k, \tag{A.68}$$

the published formula, with $\omega_{ji,k} = 0$ and the shifter of (A.65).

With Cobb–Douglas preferences and industry-specific markups, (A.26) gives

$$\sum_k \varepsilon_{ij,g}^{(jj,k)} = (\sigma_g - 1)\lambda_{jj,g}, \qquad \sum_k \varepsilon_{jj,g}^{(jj,k)} = -\big[1 + (\sigma_g - 1)\lambda_{ij,g}\big],$$

and (A.62)–(A.63) are

$$-\nu_j = (1+\tilde{\mu}_i)\sum_g \frac{e_{j,g}\lambda_{ij,g}}{1+\mu_g} + \sum_g \Big(\frac{u_j}{1+\mu_g} - \nu_j\Big) e_{j,g}\lambda_{jj,g}, \tag{A.69}$$

$$\nu_j = (1+\tilde{\mu}_i)\sum_g \frac{e_{j,g}\lambda_{ij,g}(\sigma_g - 1)\lambda_{jj,g}}{1+\mu_g} - \sum_g \Big(\frac{u_j}{1+\mu_g} - \nu_j\Big) e_{j,g}\lambda_{jj,g}\big[1 + (\sigma_g - 1)\lambda_{ij,g}\big]. \tag{A.70}$$

With $S_{ij} = \sum_g e_{j,g}\lambda_{ij,g}$ as before, let $\chi_{ij,g} \equiv e_{j,g}\lambda_{ij,g}/S_{ij}$ and

$$\begin{aligned} \bar{c}_{ij} &\equiv \sum_g \frac{\chi_{ij,g}}{1+\mu_g}, \qquad \bar{e}_{ij} \equiv \sum_g \chi_{ij,g}(\sigma_g - 1)\lambda_{jj,g}, \\ f_{ij} &\equiv \sum_g \frac{\chi_{ij,g}(\sigma_g - 1)\lambda_{jj,g}}{1+\mu_g}, \qquad q_j \equiv \frac{1}{S_{ij}}\sum_g \frac{e_{j,g}\lambda_{jj,g}}{1+\mu_g}, \end{aligned} \tag{A.71}$$

where $\bar{e}_{ij}$ is (A.65) under Cobb–Douglas preferences, since $1 + \varepsilon_{ij,g}^{(ij,g)}$ equals $-(\sigma_g - 1)\lambda_{jj,g}$. Using $\sum_g e_{j,g}\lambda_{jj,g} = 1 - S_{ij}$ and $\sum_g e_{j,g}\lambda_{jj,g}(\sigma_g - 1)\lambda_{ij,g} = S_{ij}\bar{e}_{ij}$, (A.69) and (A.70) become $\nu_j = -(1 + \tilde{\mu}_i)\bar{c}_{ij} - u_j q_j$ and $\nu_j(1 - \bar{e}_{ij}) = (1+\tilde{\mu}_i)f_{ij} - u_j(q_j + f_{ij})$, whose solution is

$$u_j = (1+\tilde{\mu}_i)\Lambda_{ij}, \qquad \nu_j = -(1+\tilde{\mu}_i)\big(\bar{c}_{ij} + \Lambda_{ij}q_j\big), \qquad \Lambda_{ij} \equiv \frac{\bar{c}_{ij}(1-\bar{e}_{ij}) + f_{ij}}{f_{ij} + q_j\bar{e}_{ij}}. \tag{A.72}$$

Hence

$$(1+\bar{\tau}_i^*)(1+\omega_{ji,k}) = \frac{u_j}{1+\mu_k} - \nu_j = (1+\tilde{\mu}_i)\Big[\bar{c}_{ij} + \Lambda_{ij}\Big(\frac{1}{1+\mu_k} + q_j\Big)\Big], \tag{A.73}$$

which is the two-country display of the text once (A.71) is substituted, and the tariff is (A.66). We normalize as in Section A.2, $\sum_k \rho_{j,k}\omega_{ji,k} = 0$; by $\sum_k \rho_{j,k}/(1+\mu_k) = 1/(1+\bar{\mu}_j)$, this gives $1 + \bar{\tau}_i^* = u_j/(1+\bar{\mu}_j) - \nu_j$, the right-hand side of (A.73) with $1/(1+\bar{\mu}_j)$ in place of $1/(1+\mu_k)$, and then $\omega_{ji,k} = \frac{u_j}{1+\bar{\tau}_i^*}\big(\frac{1}{1+\mu_k} - \frac{1}{1+\bar{\mu}_j}\big)$, the form of (A.30). With uniform markups, $\bar{c}_{ij} = 1/(1+\mu)$, $f_{ij} = \bar{e}_{ij}/(1+\mu)$, $q_j = (1 - S_{ij})/[S_{ij}(1+\mu)]$, and $\Lambda_{ij} = S_{ij}/\bar{e}_{ij}$, so that (A.73) equals $1 + 1/\bar{e}_{ij}$ in every industry, in agreement with (A.68). We have verified (A.73), and (A.61) with three countries and with CES and Stone–Geary preferences, against the numerically optimized third best to machine precision.

A.5. Proof of Theorem 3′ (free entry)

Environment. The environment is that of Section A.3 with export taxes unavailable: $x_{ij,k} = 0$ for all $j \neq i$ and k, so that the consumer price of economy i's exports is the producer price (A.32) in

every foreign market. As in Section A.4 there is no Lerner family, and the level of the tariff vector is determined. We take the home wage as the unit of account, $w_i = 1$.

The planning problem. The government chooses the home consumer prices of imports, home income, industry employment in every economy, and the foreign wages:

$$\max_{\{\tilde{P}_{ni,k}\}_{n\neq i}, Y_i, \{L_{n,k}\}, \{w_n\}_{n\neq i}} V_i(Y_i, \tilde{P}_i) \tag{A.74}$$
$$\text{s.t.} \quad B_{n,k} L_{n,k}^{1+\mu_k} - \sum_j \tau_{nj,k} Q_{nj,k} = 0 \ \ \forall n,k, \qquad L_n - \sum_k L_{n,k} = 0 \ \ \forall n,$$

the problem (A.33) without the export-price controls. The number of controls exceeds the number of constraints by $(N-1)K$, the number of import tariffs, and the problem is equivalent to the government's problem with $1 + t_{ni,k} = \tilde{P}_{ni,k}/P_{ni,k}$.

First-order conditions. Define the valuations $u_{n,k}$, h_n, and $z_{nj,k}$ as in (A.34) and the residuals $b_{nj,k}$ as in (A.35); now $z_{ij,k} = u_{i,k}$ for $j \neq i$. The first-order conditions for Y_i, $\tilde{P}_{ji,k}$, and w_m are (A.36), (A.37), and (A.40). There is no condition (A.38). The condition for $L_{n,k}$ is (A.39) with $\mathcal{U}_i = \mathbb{C}$ and $\mathcal{U}_n = \mathbb{C} \setminus \{i\}$ for $n \neq i$: the home producer price moves with $L_{i,k}$ in every export market as well as at home, and the home government cannot offset the movement with an export tax. At a regular interior optimum these conditions are necessary, and the steps below solve them completely.

Step 1: the import block. Conditions (A.36) and (A.37) are those of Section A.3, so Step 1 there applies verbatim: the tariff is (A.41), with u_{-ii} in levels, and the home income identity is (A.42). Writing

$$(1 + \bar{\tau}_i^*)(\mathbf{1} + \Omega_{-ii}) \equiv u_{-ii}, \tag{A.75}$$

with the normalization $q^\top(\mathbf{1} + \Omega_{-ii}) = 1$ of (A.49), the tariff (A.41) is the display of Theorem 3′. The first-order conditions determine u_{-ii} itself; the split into a shifter and the $\omega_{ji,k}$ is a normalization, as in Section A.4. With $u_{i,k}$ replaced by $(1 + \tilde{\mu}_i)/(1 + \mu_k)$, (A.41) is (A.61).

Step 2: the home free-entry conditions. For $n = i$, (A.39) now sums the residuals over every market. The home-market residual is (A.43), which uses only (A.41). The export-market residuals are

$$b_{ij,k} = \mathbb{X}_{ij,k}\left[\tilde{E}_{ij} u_i + \tilde{E}_{ij}^{(-ij)} u_{-ii}\right]_k \qquad (j \neq i), \tag{A.76}$$

the expression that (A.38) sets to zero in Section A.3, with u_i in place of z_{ij}. With $\mathbb{X}_{ij,k}/R_{i,k} = r_{ij,k}$, $\Delta_i \equiv \mathbf{1} - u_i$, $\tilde{\mu}_i \equiv h_i - 1$, and H_i as in Section A.3, the home conditions read

$$H_i \Delta_i = \mu - \tilde{\mu}_i \mathbf{1} + g_i, \qquad g_{i,k} \equiv \mu_k \sum_{j\neq i} r_{ij,k}\left[\tilde{E}_{ij} u_i + \tilde{E}_{ij}^{(-ij)} u_{-ii}\right]_k, \tag{A.77}$$

which is (A.44) with the export-market terms g_i added. Because g_i depends on u_{-ii}, (A.42) and (A.77) no longer determine (u_i, h_i) on their own.

Step 3: the foreign block. In market $j \neq i$, the residuals of the foreign origins and the income term of (A.40) are

$$b_{-ij} = \operatorname{diag}(\mathbb{X}_{-ij})\big[\tilde{E}_{-ij}u_{-ii} + \tilde{E}^{(ij)}_{-ij}u_i\big], \qquad Z_j = (\mathbb{X}_{-ij} \odot \eta_{-ij})^\top u_{-ii} + (\mathbb{X}_{ij} \odot \eta_{ij})^\top u_i, \tag{A.78}$$

the expressions behind (A.47) with u_i in place of z_{ij}. Let $B^F_i \equiv \sum_{j\neq i} \operatorname{diag}(\mathbb{X}_{-ij})\tilde{E}_{-ij}$ and $B^H_i \equiv \sum_{j\neq i} \operatorname{diag}(\mathbb{X}_{-ij})\tilde{E}^{(ij)}_{-ij}$, let D^F_i and D^H_i be the matrices with rows $(\mathbb{X}_{-ij} \odot \eta_{-ij})^\top$ and $(\mathbb{X}_{ij} \odot \eta_{ij})^\top$, $j \neq i$, and let C, R_{-i}, and μ_{-i} be as in Section A.3. The foreign conditions (A.39) ($n \neq i$) and (A.40) are

$$\begin{aligned} \big[\operatorname{diag}(\mathbf{1} + \mu_{-i}) + \operatorname{diag}(\mu_{-i} \oslash R_{-i})B^F_i\big]u_{-ii} + \operatorname{diag}(\mu_{-i} \oslash R_{-i})B^H_i u_i &= Ch_{-i}, \\ \big(C^\top B^F_i + D^F_i\big)u_{-ii} + \big(C^\top B^H_i + D^H_i\big)u_i &= 0. \end{aligned} \tag{A.79}$$

Here B^F_i and D^F_i are the matrices of Section A.3 with $\tilde{E}_{-ij}$ in place of S^F_j and $\mathbb{X}_{-ij} \odot \eta_{-ij}$ in place of d_j, since z_{ij} is not eliminated; replacing u_i by $z_{ij} = -\tilde{E}^{-1}_{ij}\tilde{E}^{(-ij)}_{ij}u_{-ii}$ market by market, as (A.38) does, returns (A.48). The redundancy of Section A.3 is gone: by homogeneity (A.2), $\sum_{n,k} b_{nj,k} = -Z_j$ in every market $j \neq i$, so the left-hand sides of (A.40) sum over $m \neq i$ to $-\sum_{j\neq i,k} b_{ij,k}$, which vanishes identically under (A.38) but is an equation here. Hence (A.42), (A.77), and (A.79) are $NK + N$ linear equations in the $NK + N$ unknowns $(u_i, u_{-ii}, h_i, h_{-i})$, with coefficients that are scale elasticities, revenue shares ($\mathbb{X}_{nj,k}/R_{n,k} = r_{nj,k}$), expenditure shares, and demand elasticities. This is the linear system referred to in the text. We assume that its coefficient matrix is nonsingular; it is in every case we have computed. The system then determines $(u_i, u_{-ii}, h_i, h_{-i})$, and (A.41) determines the tariff vector, including its level.

Step 4: the converse. Conversely, let $(u_i, u_{-ii}, h_i, h_{-i})$ solve (A.42), (A.77), and (A.79), and let the tariffs be given by (A.41). Then (A.37) holds by construction, (A.36) holds by (A.42), and (A.39) and (A.40) are (A.77) and (A.79): with the multipliers $\alpha_{n,k} = \lambda_i \pi_{n,k} u_{n,k}$ and $\nu_n = \lambda_i w_n h_n$, every first-order condition of (A.74) is satisfied. The derivation uses the invertibility of $\tilde{E}_{-ii}$ and the nonsingularity of the system. As in Section A.2, the first-order conditions are necessary at a regular optimum and sufficient for a local optimum under the second-order conditions. The theorem determines the tariff vector at any such optimum, including its level; it does not assert that the optimum is unique. □

Cobb–Douglas preferences. Under the preferences of (A.26), S_i and H_i are as in (A.51), $v_{i,k} = e_{i,k}\lambda_{ii,k}\sigma_k/d_{i,k}$ as in (A.28), and, with the elasticities used in (A.50), the export-market term of (A.76) is

$$\big[\tilde{E}_{ij}u_i + \tilde{E}^{(-ij)}_{ij}u_{-ii}\big]_k = -\big[1 + (\sigma_k - 1)(1 - \lambda_{ij,k})\big]u_{i,k} + (\sigma_k - 1)\sum_{n\neq i} \lambda_{nj,k}u_{n,k}. \tag{A.80}$$

The tariff (A.41) is the first display of (A.50),

$$1+t^{***}_{ni,k}=\frac{1+(\sigma_k-1)\lambda_{ii,k}}{1+(\sigma_k-1)\lambda_{ii,k}u_{i,k}}\,u_{n,k}, \tag{A.81}$$

and with $u_{n,k}=(1+\bar\tau^*_i)(1+\omega_{ni,k})$ it is the Cobb–Douglas display of the text.

Two countries. Let j be the only foreign country, so that $\lambda_{nj,k}=\lambda_{jj,k}$ and $1-\lambda_{ij,k}=\lambda_{jj,k}$ in (A.80), and the export-market term is $-[u_{i,k}+(\sigma_k-1)\lambda_{jj,k}(u_{i,k}-u_{j,k})]$. With S_i and H_i from (A.51), the home conditions (A.77) read

$$h_i=(1+\mu_k-\kappa_{i,k})u_{i,k}+\kappa_{i,k}-\mu_k r_{ij,k}\big[u_{i,k}+(\sigma_k-1)\lambda_{jj,k}(u_{i,k}-u_{j,k})\big]. \tag{A.82}$$

In market j, the residual of origin j is $b_{jj,k}=\mathbb{X}_{jj,k}\{[-\sigma_k+(\sigma_k-1)\lambda_{jj,k}]u_{j,k}+(\sigma_k-1)\lambda_{ij,k}u_{i,k}\}=\mathbb{X}_{jj,k}[-u_{j,k}+(\sigma_k-1)\lambda_{ij,k}(u_{i,k}-u_{j,k})]$, and (A.79) reads, with $\mathbb{X}_{jj,k}/R_{j,k}=r_{jj,k}=1-r_{ji,k}$ and $\mathbb{X}_{nj,k}/Y_j=e_{j,k}\lambda_{nj,k}$,

$$\begin{aligned} h_j&=(1+\mu_k r_{ji,k})u_{j,k}+\mu_k r_{jj,k}(\sigma_k-1)\lambda_{ij,k}(u_{i,k}-u_{j,k}),\\ 0&=\sum_k e_{j,k}\lambda_{ij,k}\big[u_{i,k}+(\sigma_k-1)\lambda_{jj,k}(u_{i,k}-u_{j,k})\big]. \end{aligned} \tag{A.83}$$

The second line is the single wage condition (A.40), written as $-\sum_k b_{ij,k}/Y_j=0$ by homogeneity (A.2). With (A.42), $\sum_k v_{i,k}(u_{i,k}-1)=0$, these are the $2K+2$ equations of the text. Three checks follow. With one industry, (A.42) gives $u_i=1$, the wage condition gives $u_j=1+1/[(\sigma-1)\lambda_{jj}]$, and (A.81) gives $1+t^{***}_{ji}=u_j$, the value of (A.66) with (A.73) at $K=1$: scale economies leave the one-industry tariff unchanged. As $\mu_k\to 0$ for all k, $\kappa_{i,k}\to 0$, (A.82) and (A.83) give $u_{i,k}=h_i$ and $u_{j,k}=h_j$ for every k, (A.42) gives $h_i=1$, and the wage condition gives $h_j=1+1/\bar e_{ij}$ with $\bar e_{ij}$ as in (A.71); the tariff is $1+1/\bar e_{ij}$ in every industry, as in (A.68). Replacing $u_{i,k}$ by $z_{ij,k}$ in the terms in brackets and in (A.83), and setting the brackets to zero as (A.38) does, returns (A.45) and (A.53), and the wage condition holds identically, as in Section A.3. We have verified (A.82)–(A.83), and (A.41) with three countries and with CES and Stone–Geary preferences, against the numerically optimized third best to machine precision.

A.6. Proof of Theorem 4 (input–output linkages, free entry)

Environment. Countries are indexed by $n,m,j\in\mathbb{C}$ and industries by k,g,s; country i is the policy maker. Entry is free in every industry and country, labor L_n is fixed in supply and mobile across industries, and production uses labor and intermediate inputs. Industry n,k combines labor and intermediates into a bundle whose unit cost is $\mathrm{C}_{n,k}(w_n,\tilde P_n)$, homogeneous of degree one in the wage w_n and in the vector $\tilde P_n\equiv(\tilde P_{mn,g})_{m,g}$ of purchaser prices in market n; the bundle $\mathcal{I}_{n,k}$ produces the industry's output inclusive of trade costs, $\mathcal{Q}_{n,k}\equiv\sum_j\tau_{nj,k}Q_{nj,k}=B_{n,k}\mathcal{I}_{n,k}^{1+\mu_k}$, with $\tau_{nj,k}$ the iceberg trade cost ($\bar a_{nj,k}$ in the paper's notation) and $\mu_k\geq 0$ the scale elasticity of free entry (Section A.3).

Producer prices are $P_{nj,k} = \tau_{nj,k}\pi_{n,k}$. Zero profits equate the industry's gross revenue $\mathcal{Y}_{n,k}$ to its sales and to its costs, $\mathcal{Y}_{n,k} = \pi_{n,k}\mathcal{Q}_{n,k} = \mathrm{C}_{n,k}\mathcal{I}_{n,k}$, so that

$$\pi_{n,k} = B_{n,k}^{-\frac{1}{1+\mu_k}}\,\mathrm{C}_{n,k}(w_n, \tilde{P}_n)\,\mathcal{Q}_{n,k}^{-\frac{\mu_k}{1+\mu_k}}, \tag{A.84}$$

which is equation (13) of the paper with $\bar{\rho}_{nj,k} = \tau_{nj,k}B_{n,k}^{-1/(1+\mu_k)}$. Eliminating $\mathcal{I}_{n,k}$ instead gives the free-entry condition in revenue form,

$$\pi_{n,k}B_{n,k}\mathcal{Y}_{n,k}^{\mu_k} = \mathrm{C}_{n,k}(w_n, \tilde{P}_n)^{1+\mu_k}, \tag{A.85}$$

which is the form the proof uses. Cost shares are $\alpha_{n,k}^{m,g} \equiv \partial \ln \mathrm{C}_{n,k}/\partial \ln \tilde{P}_{mn,g}$ for inputs and $\beta_{n,k} \equiv \partial \ln \mathrm{C}_{n,k}/\partial \ln w_n$ for labor, with $\sum_{m,g}\alpha_{n,k}^{m,g} + \beta_{n,k} = 1$ by homogeneity. As in Appendix K of the paper, the labor share is constant,[6] so the input shares depend on $\tilde{P}_n$ alone and are homogeneous of degree zero. By Shephard's lemma, industry n,g demands $\mathcal{I}_{n,g}\,\partial \mathrm{C}_{n,g}/\partial \tilde{P}_{mn,k}$ units of good mn,k and $\mathcal{I}_{n,g}\,\partial \mathrm{C}_{n,g}/\partial w_n$ units of labor; with $\mathcal{I}_{n,g} = \mathcal{Y}_{n,g}/\mathrm{C}_{n,g}$,

$$Q_{mn,k}^{I(g)} = \frac{\alpha_{n,g}^{m,k}\,\mathcal{Y}_{n,g}}{\tilde{P}_{mn,k}}, \qquad L_{n,g} = \frac{\beta_{n,g}\mathcal{Y}_{n,g}}{w_n}. \tag{A.86}$$

Total demand for good mn,k is final plus intermediate demand,

$$Q_{mn,k} = Q_{mn,k}^{C}(\tilde{P}_n, Y_n) + \sum_g Q_{mn,k}^{I(g)}, \tag{A.87}$$

where $Q_{mn,k}^{C}$ is the Marshallian demand of market n's representative consumer at income Y_n. Labor-market clearing is $\sum_g \beta_{n,g}\mathcal{Y}_{n,g} = w_n L_n$. Taxes apply to a good irrespective of its use, so one purchaser price $\tilde{P}_{mn,k}$ serves both uses. Country i taxes imports, domestic sales, and exports, and no other country taxes; as in the paper's definition of consumer prices,

$$\tilde{P}_{ni,k} = (1+t_{ni,k})P_{ni,k}\ \ (n \neq i), \qquad \tilde{P}_{ii,k} = \frac{P_{ii,k}}{1+s_{i,k}}, \qquad \tilde{P}_{ij,k} = \frac{P_{ij,k}}{(1+x_{ij,k})(1+s_{i,k})}\ \ (j \neq i), \tag{A.88}$$

and $\tilde{P}_{mj,k} = P_{mj,k}$ for $m,j \neq i$. Profits net of entry costs are zero, so $Y_j = w_j L_j$ for $j \neq i$ and $Y_i = w_i L_i + \mathcal{R}_i$, where the tax revenue is $\mathcal{R}_i = \sum_{n,k}(\tilde{P}_{ni,k} - P_{ni,k})Q_{ni,k} + \sum_{j\neq i,k}(\tilde{P}_{ij,k} - P_{ij,k})Q_{ij,k}$.

[6] Appendix K of the paper: "we impose no restrictions on how intermediate inputs are aggregated in the production process. We, however, assume that the share of labor in production is constant and equal to $1 - \bar{\alpha}_{i,k}$." Equivalently $\mathrm{C}_{n,k} = w_n^{\beta_{n,k}}\Phi_{n,k}(\tilde{P}_n)^{1-\beta_{n,k}}$ with $\Phi_{n,k}$ homogeneous of degree one, so that the input shares $\alpha_{n,k}^{m,g}$ depend on $\tilde{P}_n$ only. With a variable labor share the labor-market constraints below depend on prices and their multipliers enter the price conditions. Steps 1 and 3, and with them the domestic and the import lines, are unchanged; the residual $b_{mn,k}$ of (A.94) acquires the term $h_n \sum_g \mathcal{Y}_{n,g}\,\partial\beta_{n,g}/\partial \ln \tilde{P}_{mn,k}$, with h_n the value of market n's labor income defined in (A.102), which changes the export line and the system for ν.

Expenditure and revenue shares are

$$e_{mn,k} \equiv \frac{\tilde{P}_{mn,k}Q_{mn,k}}{\mathcal{E}_n}, \quad \mathcal{E}_n \equiv \sum_{m,s} \tilde{P}_{mn,s}Q_{mn,s}, \qquad r_{nj,k} \equiv \frac{P_{nj,k}Q_{nj,k}}{\mathcal{Y}_{n,k}}, \qquad \beta_{mn,kg} \equiv \frac{\alpha^{m,k}_{n,g}\mathcal{Y}_{n,g}}{\tilde{P}_{mn,k}Q_{mn,k}} = \frac{Q^{I(g)}_{mn,k}}{Q_{mn,k}}, \tag{A.89}$$

so that $\sum_j r_{nj,k} = 1$ at an equilibrium and $\sum_g \beta_{mn,kg} = 1 - Q^C_{mn,k}/Q_{mn,k}$. The equilibrium system is homogeneous of degree one in nominal variables, and we work with the Lerner representative $w_i = 1$. No restriction is placed on foreign wages. Regularity conditions are collected at the end of the proof.

The planning problem. The optimal policy solves

$$\begin{aligned} &\max\ V_i(\tilde{P}_i, Y_i) \quad \text{over} \quad \{\tilde{P}_{ni,k}\}_{n,k},\ \{\tilde{P}_{ij,k}\}_{j\neq i,k},\ Y_i,\ \{\mathcal{Y}_{n,k}\}_{n,k},\ \{\pi_{n,k}\}_{n,k},\ \{w_m\}_{m\neq i} \\ &\text{s.t.} \quad (\mathrm{G})_{n,k}:\ \mathcal{Y}_{n,k} - \pi_{n,k}\sum_j \tau_{nj,k}Q_{nj,k} = 0, \qquad (\mathrm{F})_{n,k}:\ \pi_{n,k}B_{n,k}\mathcal{Y}^{\mu_k}_{n,k} - \mathrm{C}_{n,k}(w_n, \tilde{P}_n)^{1+\mu_k} = 0, \\ &\qquad (\mathrm{L})_n:\ w_nL_n - \sum_k \beta_{n,k}\mathcal{Y}_{n,k} = 0, \qquad \text{for all } n,k, \end{aligned} \tag{A.90}$$

where V_i is the indirect utility of country i's consumer, the quantities are the total demands (A.87) with the input demands (A.86), uncontrolled purchaser prices equal producer prices, $\tilde{P}_{mj,k} = \tau_{mj,k}\pi_{m,k}$ for $m, j \neq i$, foreign incomes are $Y_j = w_jL_j$, and $w_i = 1$. The constraints are goods-market clearing in value (industry revenue equals sales at producer prices), free entry (A.85), and labor-market clearing. Two identities hold at every feasible point. First, market j's purchases are $\sum_{m,k} \tilde{P}_{mj,k}Q_{mj,k} = Y_j + \sum_g (1-\beta_{j,g})\mathcal{Y}_{j,g}$, by the consumer's budget and (A.86); for $j \neq i$ this equals $\sum_g \mathcal{Y}_{j,g}$ by $(\mathrm{L})_j$, and by $(\mathrm{G})_{j,g}$ it equals $\sum_{g,n} P_{jn,g}Q_{jn,g}$. Summing over $j \neq i$ and cancelling the sales between foreign countries, which appear on both sides at the common price $\tilde{P}_{mj,k} = P_{mj,k}$, gives balanced trade,

$$\sum_{j\neq i}\sum_k \tilde{P}_{ij,k}Q_{ij,k} = \sum_{m\neq i}\sum_k P_{mi,k}Q_{mi,k}. \tag{A.91}$$

Second, the same computation for market i gives

$$Y_i - w_iL_i - \mathcal{R}_i = \sum_{m\neq i}\sum_k P_{mi,k}Q_{mi,k} - \sum_{j\neq i}\sum_k \tilde{P}_{ij,k}Q_{ij,k} = 0,$$

so the home budget is implied by the constraints and is not imposed. Conversely, any feasible point is a free-entry equilibrium under the taxes defined by (A.88), $1 + t_{ni,k} = \tilde{P}_{ni,k}/P_{ni,k}$, $1 + s_{i,k} = P_{ii,k}/\tilde{P}_{ii,k}$, and $1 + x_{ij,k} = P_{ij,k}/[(1+s_{i,k})\tilde{P}_{ij,k}]$. Hence (A.90) is equivalent to the government's problem of choosing taxes.

First-order conditions. The Lagrangian of (A.90) is

$$\mathcal{L} = V_i(\tilde{P}_i, Y_i) + \sum_{n,k} a_{n,k}\Big[\mathcal{Y}_{n,k} - \pi_{n,k}\sum_j \tau_{nj,k}Q_{nj,k}\Big] + \sum_{n,k}\varphi_{n,k}\Big[\pi_{n,k}B_{n,k}\mathcal{Y}_{n,k}^{\mu_k} - \mathrm{C}_{n,k}^{1+\mu_k}\Big] + \sum_n \psi_n\Big[w_n L_n - \sum_k \beta_{n,k}\mathcal{Y}_{n,k}\Big]. \tag{A.92}$$

Let $\lambda_i \equiv \partial V_i/\partial Y_i$ and define the normalized multipliers

$$u_{n,k} \equiv \frac{a_{n,k}}{\lambda_i}, \qquad \nu_{n,k} \equiv \frac{\varphi_{n,k}\,\mathrm{C}_{n,k}^{1+\mu_k}}{\lambda_i \mathcal{Y}_{n,k}}, \qquad h_n \equiv \frac{\psi_n}{\lambda_i}, \qquad z_{mn,g} \equiv u_{m,g}\frac{P_{mn,g}}{\tilde{P}_{mn,g}}. \tag{A.93}$$

Here $u_{n,k}$ is the value to country i of a unit of revenue of industry n,k (the multiplier of its revenue identity), $\nu_{n,k}$ the value of its free-entry condition per unit of revenue (the $\nu_{n,k}$ of the theorem), h_n the value of a unit of labor income in country n, and $z_{mn,g}$ the value, per unit of spending on good mn,g at purchaser prices, of the producer revenue that spending generates. The symbols u, h and z play the roles they play in Section A.3; the goods-market and labor multipliers are written $a_{n,k}$ and ψ_n rather than $\alpha_{n,k}$ and ν_n because α denotes a cost share and ν the free-entry value here. Three derivative facts are used throughout. Roy's identity, $\partial V_i/\partial \tilde{P}_{mi,g} = -\lambda_i Q^{\mathrm{C}}_{mi,g}$. Shephard's lemma in the form $\partial \mathrm{C}_{n,k}^{1+\mu_k}/\partial \tilde{P}_{mn,g} = (1+\mu_k)\mathrm{C}_{n,k}^{1+\mu_k}\alpha_{n,k}^{m,g}/\tilde{P}_{mn,g}$ and $\partial \mathrm{C}_{n,k}^{1+\mu_k}/\partial w_n = (1+\mu_k)\mathrm{C}_{n,k}^{1+\mu_k}\beta_{n,k}/w_n$, which with (A.85) and (A.93) give $\varphi_{n,k}\,\partial \mathrm{C}_{n,k}^{1+\mu_k}/\partial \tilde{P}_{mn,g} = \lambda_i(1+\mu_k)\nu_{n,k}Q^{I(k)}_{mn,g}$ and $\varphi_{n,k}B_{n,k}\mathcal{Y}_{n,k}^{\mu_k} = \lambda_i \nu_{n,k}\mathcal{Y}_{n,k}/\pi_{n,k}$ at feasible points. And the dependence of demand: $Q_{mn,g}$ depends on the purchaser prices of market n, on Y_n through final demand, and on $\mathcal{Y}_{n,g}$ through (A.86), with $\partial Q_{mn,k}/\partial \mathcal{Y}_{n,g} = \alpha_{n,g}^{m,k}/\tilde{P}_{mn,k}$; derivatives of demand with respect to prices are taken at fixed Y_n and $\mathcal{Y}_n$, which are separate variables of (A.90). Under constant labor shares the labor-market constraints do not depend on prices.

For a foreign market $j \neq i$, the derivative of $\mathcal{L}$ with respect to the purchaser price $\tilde{P}_{mj,k}$, at fixed Y_j and $\mathcal{Y}_j$, collects the goods-market terms of all industries selling in j and the free-entry terms of the industries of j:

$$\frac{\partial \mathcal{L}}{\partial \tilde{P}_{mj,k}} = -\frac{\lambda_i\, b_{mj,k}}{\tilde{P}_{mj,k}}, \qquad b_{mj,k} \equiv \sum_{m',s} z_{m'j,s}\,\tilde{P}_{m'j,s}Q_{m'j,s}\frac{\partial \ln Q_{m'j,s}}{\partial \ln \tilde{P}_{mj,k}} + \sum_g (1+\mu_g)\,\nu_{j,g}\,\alpha_{j,g}^{m,k}\mathcal{Y}_{j,g}. \tag{A.94}$$

In units of the good's sales, with the elasticity blocks of the theorem and the shares (A.89),

$$\Gamma_{mj,k} \equiv \frac{b_{mj,k}}{\tilde{P}_{mj,k}Q_{mj,k}} = \sum_{m',s}\big[E_{mj}^{(m'j)}\big]_{k,s}\,z_{m'j,s} + \sum_g (1+\mu_g)\,\beta_{mj,kg}\,\nu_{j,g}. \tag{A.95}$$

The first-order conditions, divided by λ_i, and those for producer prices multiplied by $\pi_{n,k}$, are

$$(Y_i) \quad 1 = \sum_{n,k} z_{ni,k}\, \tilde{P}_{ni,k} \frac{\partial Q^C_{ni,k}}{\partial Y_i}, \tag{A.96}$$

$$(\tilde{P}_{mi,k}) \quad Q^C_{mi,k} + \sum_{n,s} z_{ni,s}\, \tilde{P}_{ni,s} \frac{\partial Q_{ni,s}}{\partial \tilde{P}_{mi,k}} + \sum_g (1+\mu_g)\, \nu_{i,g}\, Q^{I(g)}_{mi,k} = 0 \qquad \text{for all } m,k, \tag{A.97}$$

$$(\mathcal{Y}_{n,k}) \quad u_{n,k} + \mu_k \nu_{n,k} = \sum_{m,g} z_{mn,g}\, \alpha^{m,g}_{n,k} + h_n \beta_{n,k} \qquad \text{for all } n,k, \tag{A.98}$$

$$(\pi_{i,k}) \quad u_{i,k} = \nu_{i,k}, \tag{A.99}$$

$$(\pi_{m,k}) \quad \nu_{m,k}\mathcal{Y}_{m,k} = u_{m,k}\mathcal{Y}_{m,k} + \sum_{j \neq i} b_{mj,k} \qquad (m \neq i), \tag{A.100}$$

$$(\tilde{P}_{ij,k}) \quad b_{ij,k} = 0 \qquad (j \neq i), \tag{A.101}$$

$$(w_m) \quad h_m = \bar{v}_m + \sum_g \ell_{m,g}(1+\mu_g)\nu_{m,g} \qquad (m \neq i), \tag{A.102}$$

where

$$\bar{v}_m \equiv \sum_{n,g} z_{nm,g}\, \tilde{P}_{nm,g} \frac{\partial Q^C_{nm,g}}{\partial Y_m}, \qquad \ell_{m,g} \equiv \frac{\beta_{m,g}\mathcal{Y}_{m,g}}{w_m L_m}$$

value market m's marginal spending and give the employment share of industry m,g. Their derivation. Y_i enters V_i and, through final demand in market i, the revenue identities: $\lambda_i - \sum_{n,k} a_{n,k} P_{ni,k}\, \partial Q^C_{ni,k}/\partial Y_i = 0$, which is (A.96) since $a_{n,k} P_{ni,k} = \lambda_i z_{ni,k} \tilde{P}_{ni,k}$. A home purchaser price $\tilde{P}_{mi,k}$ enters V_i (Roy), the revenue identities through $Q_{ni,s}$, and the free-entry conditions of the home industries through $\mathrm{C}_{i,g}$ (Shephard); dividing by $-\lambda_i$ gives (A.97). A revenue $\mathcal{Y}_{n,k}$ enters its own identity with coefficient $a_{n,k}$, the identities of all sellers in market n through input demand, $-\sum_{m,g} a_{m,g} P_{mn,g} \alpha^{m,g}_{n,k} / \tilde{P}_{mn,g}$, its free-entry condition with coefficient $\varphi_{n,k}\mu_k \pi_{n,k} B_{n,k} \mathcal{Y}^{\mu_k - 1}_{n,k} = \lambda_i \mu_k \nu_{n,k}$, and the labor market of n with coefficient $-\psi_n \beta_{n,k}$; this is (A.98). A home producer price $\pi_{i,k}$ enters $(\mathrm{G})_{i,k}$, with derivative $-a_{i,k}\mathcal{Q}_{i,k} = -a_{i,k}\mathcal{Y}_{i,k}/\pi_{i,k}$, and $(\mathrm{F})_{i,k}$, with derivative $\varphi_{i,k} B_{i,k} \mathcal{Y}^{\mu_k}_{i,k} = \lambda_i \nu_{i,k} \mathcal{Y}_{i,k}/\pi_{i,k}$; the purchaser prices of good i,k are controls, so nothing else depends on it, and (A.99) follows. A foreign producer price $\pi_{m,k}$ has the same two terms and in addition sets the purchaser prices $\tilde{P}_{mj,k} = \tau_{mj,k}\pi_{m,k}$ in every foreign market, so by (A.94) and $\tau_{mj,k}/\tilde{P}_{mj,k} = 1/\pi_{m,k}$ its derivative gains $\sum_{j \neq i} \tau_{mj,k}\, \partial \mathcal{L}/\partial \tilde{P}_{mj,k} = -\lambda_i \sum_{j \neq i} b_{mj,k}/\pi_{m,k}$; multiplying by $\pi_{m,k}/\lambda_i$ gives (A.100). The export prices $\tilde{P}_{ij,k}$ are controls that do not enter V_i, so (A.94) gives (A.101). A foreign wage w_m enters foreign income $Y_m = w_m L_m$, hence final demand in market m in the revenue identities, the free-entry conditions of the industries of m through $\mathrm{C}_{m,g}$ (Shephard), and $(\mathrm{L})_m$ with coefficient $\psi_m L_m$; dividing by $\lambda_i L_m$ gives (A.102). There is no condition for w_i, which is the numéraire.

Step 1: the Home block and the domestic subsidy. Conditions (A.96), (A.97), and (A.98) for $n = i$ with $u_{i,k} = \nu_{i,k}$ from (A.99) involve only the multipliers attached to market i: the NK values $z_{mi,k}$, the K values $\nu'_{i,g} \equiv (1+\mu_g)\nu_{i,g}$, and h_i. Written with the unknowns on the right, they form the

square linear system of $NK + K + 1$ equations

$$\begin{aligned} 1 &= \sum_{n,k} z_{ni,k}\, \tilde{P}_{ni,k} \frac{\partial Q^{C}_{ni,k}}{\partial Y_i}, \\ -Q^{C}_{mi,k} &= \sum_{n,s} z_{ni,s}\, \tilde{P}_{ni,s} \frac{\partial Q_{ni,s}}{\partial \tilde{P}_{mi,k}} + \sum_{g} \nu'_{i,g}\, Q^{I(g)}_{mi,k} \qquad \text{for all } m,k, \\ 0 &= \sum_{m,g} z_{mi,g}\, \alpha^{m,g}_{i,k} + h_i \beta_{i,k} - \nu'_{i,k} \qquad \text{for all } k, \end{aligned} \tag{A.103}$$

whose coefficient matrix we call M_i. The vector $(z_{\cdot i}, \nu'_i, h_i) = (1,1,1)$ solves it. The first equation at $z = 1$ is Engel aggregation, $\sum_{n,k} \tilde{P}_{ni,k}\, \partial Q^{C}_{ni,k}/\partial Y_i = 1$. For the second, Cournot aggregation gives $\sum_{n,s} \tilde{P}_{ni,s}\, \partial Q^{C}_{ni,s}/\partial \tilde{P}_{mi,k} = -Q^{C}_{mi,k}$, and since input spending $\sum_{n,s} \tilde{P}_{ni,s} Q^{I(g)}_{ni,s} = (1 - \beta_{i,g}) \mathcal{Y}_{i,g}$ does not depend on prices under constant labor shares,

$$\sum_{n,s} \tilde{P}_{ni,s} \frac{\partial Q^{I(g)}_{ni,s}}{\partial \tilde{P}_{mi,k}} = -Q^{I(g)}_{mi,k} \qquad \text{for all } g, \tag{A.104}$$

so at $z = 1$ the first sum equals $-Q^{C}_{mi,k} - \sum_g Q^{I(g)}_{mi,k}$ and the equation holds with $\nu' = 1$. The third equation at $(z, h_i) = (1,1)$ is $\sum_{m,g} \alpha^{m,g}_{i,k} + \beta_{i,k} = 1$, the homogeneity of $\mathrm{C}_{i,k}$. When M_i is nonsingular this is the only solution (the singular case that arises when tax items separate final from intermediate use is treated in Step 4). Hence

$$z_{mi,k} = 1 \;\; \text{for all } m,k, \qquad (1 + \mu_g)\, \nu_{i,g} = 1, \qquad h_i = 1. \tag{A.105}$$

By (A.93) and (A.88), $z_{mi,g} = 1$ means $u_{m,g} = \tilde{P}_{mi,g}/P_{mi,g} = 1 + t_{mi,g}$ for $m \neq i$ and $u_{i,g} = \tilde{P}_{ii,g}/P_{ii,g} = 1/(1 + s_{i,g})$. Combined with $u_{i,k} = \nu_{i,k} = 1/(1 + \mu_k)$,

$$1 + s^{*}_{i,k} = 1 + \mu_k, \tag{A.106}$$

the domestic line of the theorem at $\bar{s}^{\mathcal{C}}_i = 0$. The remaining values of z follow from u and the wedges: $z_{ij,g} = u_{i,g} P_{ij,g}/\tilde{P}_{ij,g} = 1 + x_{ij,g}$ for $j \neq i$, and $z_{mj,g} = u_{m,g} = 1 + t_{mi,g}$ for $m, j \neq i$, where $\tilde{P}_{mj,g} = P_{mj,g}$. Substituting into (A.95),

$$\Gamma_{mj,k} = \sum_{s} \left[E^{(ij)}_{mj}\right]_{k,s} (1 + x_{ij,s}) + \sum_{m' \neq i} \sum_{s} \left[E^{(m'j)}_{mj}\right]_{k,s} (1 + t_{m'i,s}) + \sum_{g} (1 + \mu_g)\, \beta_{mj,kg}\, \nu_{j,g}, \tag{A.107}$$

which in the stacked notation of the theorem is $\Gamma_{mj} = E^{(ij)}_{mj}(1 + x_{ij}) + E^{(-ij)}_{mj}(1 + t_i) + \tilde{\Lambda}_{mj}(1 + \bar{t}_i)$, with $\tilde{\Lambda}_{mj,k}(1 + \bar{t}_i) \equiv \sum_g (1 + \mu_g) \beta_{mj,kg} \nu_{j,g}$ as in the theorem at $\bar{s}^{\mathcal{C}}_i = 0$.

Step 2: the export line and the revenue values. Condition (A.101) says $b_{ij,k} = 0$, that is $\Gamma_{ij,k} = 0$, for every export good. With (A.107) at $m = i$ and $E_{ij} \equiv E_{ij}^{(ij)}$,

$$\sum_g \left[E_{ij}\right]_{k,g}(1 + x_{ij,g}) + \sum_{m \neq i} \sum_g \left[E_{ij}^{(mj)}\right]_{k,g}(1 + t_{mi,g}) + \tilde{\Lambda}_{ij,k}(1 + \bar{t}_i) = 0 \qquad \text{for all } j \neq i, k, \tag{A.108}$$

or, stacked over k, $E_{ij}(1 + x_{ij}) + E_{ij}^{(-ij)}(1 + t_i) + \tilde{\Lambda}_{ij}(1 + \bar{t}_i) = 0$. When E_{ij} is invertible this is the export line of the theorem at $\bar{s}_i^{\mathcal{C}} = 0$. The blocks appear in the expenditure-weighted orientation because $b_{ij,k}$ weights the response of each good mj, g by the spending on that good and (A.95) divides by the spending on the good whose price changes, which gives the ratio $e_{mj,g}/e_{ij,k}$.

The values $\nu_{m,k}$ of the foreign free-entry conditions follow from (A.100). In a foreign market $\tilde{P}_{mj,k} = P_{mj,k}$, so $b_{mj,k} = P_{mj,k}Q_{mj,k}\Gamma_{mj,k} = r_{mj,k}\mathcal{Y}_{m,k}\Gamma_{mj,k}$. Dividing (A.100) by $\mathcal{Y}_{m,k}$ and using $u_{m,k} = 1 + t_{mi,k}$,

$$\nu_{m,k} = (1 + t_{mi,k}) + \sum_{j \neq i} r_{mj,k}\,\Gamma_{mj,k} \qquad (m \neq i), \tag{A.109}$$

the system of the theorem. Since $\Gamma_{mj,k}$ contains $\nu_{j,g}$ through its last term, (A.109) is linear in ν:

$$(I - M)\,\nu = (1 + t_{\cdot i}) + \delta, \qquad M_{(m,k),(j,g)} \equiv r_{mj,k}(1 + \mu_g)\beta_{mj,kg} = \frac{(1 + \mu_g)\,\alpha_{j,g}^{m,k}\mathcal{Y}_{j,g}}{\mathcal{Y}_{m,k}}, \tag{A.110}$$

where $(1 + t_{\cdot i})$ has entries $1 + t_{mi,k}$, M collects the $(1 + \mu_g)$-weighted shares of industry m,k's revenue earned from input sales to industry j,g, and $\delta_{m,k} \equiv \sum_{j \neq i} r_{mj,k}\big[\sum_s [E_{mj}^{(ij)}]_{k,s}(1 + x_{ij,s}) + \sum_{m' \neq i,s} [E_{mj}^{(m'j)}]_{k,s}(1 + t_{m'i,s})\big]$ collects the demand responses. When $I - M$ is invertible, $\nu = (I - M)^{-1}[(1 + t_{\cdot i}) + \delta]$ is determined by the equilibrium and the taxes, and so is $\tilde{\Lambda}_{ij}$.

Step 3: the import line. Let $\xi \equiv (\mathcal{Y}_{m,k}, \pi_{m,k}, w_m)_{m \neq i,k}$ collect the $(N-1)(2K+1)$ foreign variables, $q \equiv (Q_{ji,k})_{j \neq i,k}$ the import quantities of country i, $p \equiv (\tilde{P}_{ij,k})_{j \neq i,k}$ its export prices, and $\Phi(\xi; q, p)$ the $(N-1)(2K+1)$ foreign constraints $(\mathrm{G})_{m,k}$, $(\mathrm{F})_{m,k}$, $(\mathrm{L})_m$, $m \neq i$. The home variables enter Φ only through q, the sales to market i in $(\mathrm{G})_{j,k}$, and through p, the export prices in foreign demands and foreign unit costs. When $\partial\Phi/\partial\xi$ is nonsingular, the implicit function theorem defines the foreign equilibrium $\xi(q, p)$ with $\Phi(\xi(q, p); q, p) \equiv 0$. The derivatives in $\tilde{\omega}_{ji,k}$ are those of $\xi(q, p)$ and of the export demands $Q_{in,g}(\xi(q, p), p)$ with respect to $Q_{ji,k}$ at fixed p and fixed other import quantities; we write $d/dQ_{ji,k}$ for them.

Split the Lagrangian as $\mathcal{L} = \mathcal{L}^H + \mathcal{L}^F$, where

$$\mathcal{L}^F \equiv \sum_{m \neq i} \sum_k \left[a_{m,k}(\mathrm{G})_{m,k} + \varphi_{m,k}(\mathrm{F})_{m,k}\right] + \sum_{m \neq i} \psi_m(\mathrm{L})_m$$

collects the foreign constraints with their multipliers and $\mathcal{L}^H$ the rest. Differentiating the identity

$\Phi(\xi(q,p);q,p) \equiv 0$ with respect to $Q_{ji,k}$, multiplying each component by its multiplier and adding,

$$\frac{\partial \mathcal{L}^F}{\partial \xi}\,\frac{d\xi}{dQ_{ji,k}} + \frac{\partial \mathcal{L}^F}{\partial Q_{ji,k}} = 0, \tag{A.111}$$

where the partial derivatives treat $\mathcal{L}^F$ as a function of its arguments. This holds at every feasible point and for arbitrary multipliers. From $(\mathrm{G})_{j,k}$, $\partial \mathcal{L}^F/\partial Q_{ji,k} = -a_{j,k}P_{ji,k}$. The home part depends on ξ only through the export demands $Q_{in,g}$, $n \neq i$, in $(\mathrm{G})_{i,g}$, so $\partial \mathcal{L}^H/\partial \xi = -\sum_{n\neq i,g} a_{i,g}P_{in,g}\,\partial Q_{in,g}/\partial \xi$. At an optimum the foreign conditions (A.98) for $n \neq i$, (A.100) and (A.102) say $\partial \mathcal{L}/\partial \xi = 0$, that is $\partial \mathcal{L}^F/\partial \xi = -\partial \mathcal{L}^H/\partial \xi$. Substituting into (A.111) and dividing by λ_i,

$$u_{j,k}P_{ji,k} = \sum_{n\neq i}\sum_{g} u_{i,g}P_{in,g}\,\frac{dQ_{in,g}}{dQ_{ji,k}}.$$

By (A.105) and (A.88), $u_{j,k} = 1 + t_{ji,k}$ and $u_{i,g}P_{in,g} = (1 + x_{in,g})\tilde{P}_{in,g}$. Multiplying by $Q_{ji,k}$,

$$(1 + t_{ji,k})\,P_{ji,k}Q_{ji,k} = \sum_{n\neq i}\sum_{g}(1 + x_{in,g})\,\tilde{P}_{in,g}Q_{in,g}\,\frac{\partial \ln Q_{in,g}}{\partial \ln Q_{ji,k}}, \tag{A.112}$$

the balance form of Remark (a): the tax-inclusive cost of the marginal import equals the tax-inclusive value of the exports it generates through the foreign equilibrium. It contains neither $\bar{t}_i$ nor $\tilde{\omega}_{ji,k}$.

The trade-balance identity (A.91) holds at every point of the foreign manifold $\Phi = 0$, since its proof used only the foreign constraints and the demand identities. Differentiating it with respect to $Q_{ji,k}$ along $\xi(q,p)$, where $P_{mi,g} = \tau_{mi,g}\pi_{m,g}$, and multiplying by $Q_{ji,k}$,

$$\sum_{n\neq i}\sum_{g}\tilde{P}_{in,g}Q_{in,g}\,\frac{\partial \ln Q_{in,g}}{\partial \ln Q_{ji,k}} = \sum_{m\neq i}\sum_{g}P_{mi,g}Q_{mi,g}\,\frac{\partial \ln P_{mi,g}}{\partial \ln Q_{ji,k}} + P_{ji,k}Q_{ji,k}. \tag{A.113}$$

Fix any $\bar{t}_i$, multiply (A.113) by $1 + \bar{t}_i$, and subtract the result from (A.112):

$$\begin{aligned}\big[(1 + t_{ji,k}) - (1 + \bar{t}_i)\big]P_{ji,k}Q_{ji,k} = (1 + \bar{t}_i)\sum_{m\neq i,g}P_{mi,g}Q_{mi,g}\frac{\partial \ln P_{mi,g}}{\partial \ln Q_{ji,k}} \\ + \sum_{n\neq i,g}\big[(1 + x_{in,g}) - (1 + \bar{t}_i)\big]\tilde{P}_{in,g}Q_{in,g}\frac{\partial \ln Q_{in,g}}{\partial \ln Q_{ji,k}}.\end{aligned}$$

Dividing by $(1 + \bar{t}_i)P_{ji,k}Q_{ji,k}$ gives, with $\tilde{\omega}_{ji,k}$ as defined in the theorem,

$$\frac{1 + t_{ji,k}}{1 + \bar{t}_i} - 1 = \tilde{\omega}_{ji,k}, \qquad \text{that is} \qquad 1 + t^*_{ji,k} = (1 + \tilde{\omega}_{ji,k})(1 + \bar{t}_i), \tag{A.114}$$

the import line at $\bar{s}^{\mathcal{C}}_i = 0$, for every value of $\bar{t}_i$. The steps are reversible: since (A.113) is an identity, (A.114) for any one $\bar{t}_i$ implies (A.112).

Step 4: the Lerner family, the shifter $\bar{s}_i^{\mathcal{C}}$, and the converse. *Homogeneity.* The system formed by (A.106), (A.108), (A.109), and (A.114) is homogeneous of degree one in $(1+t_{\cdot i}, 1+x_{i\cdot}, \nu, 1+\bar{t}_i)$: the export line and the system for ν are linear in these quantities, $\tilde{\omega}_{ji,k}$ depends on them only through the ratios $(1+x_{in,g})/(1+\bar{t}_i)$, and the domestic line does not involve them. If (t, x, ν) solves the system at $\bar{t}_i$, then $(a(1+t)-1,\ a(1+x)-1,\ a\nu)$ solves it at $a(1+\bar{t}_i)-1$ for every $a>0$. This family consists of optima, not only of solutions of the first-order conditions, by the Lerner symmetry of (A.90). For $a>0$ let g_a divide every foreign nominal variable, $\pi_{m,k}$, $\mathcal{Y}_{m,k}$, and w_m for $m \neq i$, and every export price $\tilde{P}_{ij,k}$ by a, leaving Y_i, $\mathcal{Y}_{i,k}$, $\pi_{i,k}$, and the home purchaser prices $\tilde{P}_{mi,k}$ unchanged. Every price in a foreign market j then scales by a^{-1} ($P_{mj,k} = \tau_{mj,k}\pi_{m,k}$ for $m \neq i$, and $\tilde{P}_{ij,k}$ by construction), and so do foreign income $Y_j = w_j L_j$ and foreign revenue $\mathcal{Y}_{j,g}$; consumer demand is homogeneous of degree zero in $(\tilde{P}_j, Y_j)$ and input demand $\alpha_{j,g}^{m,k}\mathcal{Y}_{j,g}/\tilde{P}_{mj,k}$ of degree zero in $(\tilde{P}_j, \mathcal{Y}_{j,g})$, so no quantity changes in any market, and V_i is unchanged. The constraints $(\mathrm{G})_{m,k}$ and $(\mathrm{L})_m$ for $m \neq i$ scale by a^{-1} and $(\mathrm{F})_{m,k}$ by $a^{-1-\mu_k}$, since both $\pi_{m,k}B_{m,k}\mathcal{Y}_{m,k}^{\mu_k}$ and $\mathrm{C}_{m,k}^{1+\mu_k}$ do, by the homogeneity of $\mathrm{C}_{m,k}$; the home constraints are unchanged, because $w_i = 1$ and the home purchaser prices are. Hence g_a maps the feasible set onto itself ($g_a^{-1} = g_{1/a}$) and preserves welfare, so it maps optima to optima. In tax terms, by (A.88), $1+t_{mi,k} = \tilde{P}_{mi,k}/P_{mi,k}$ and $(1+x_{ij,k})(1+s_{i,k}) = P_{ij,k}/\tilde{P}_{ij,k}$ are multiplied by a while $1+s_{i,k}$ is unchanged: a uniform import tax combined with a uniform export subsidy at the same rate, which is neutral (Lemma 1 of the paper). The multipliers transform accordingly. Let θ collect the controls and Λ the multipliers, and let Λ_a multiply $a_{m,k}$ and ψ_m by a and $\varphi_{m,k}$ by $a^{1+\mu_k}$ for $m \neq i$, leaving the home multipliers unchanged. Then $\mathcal{L}(g_a\theta; \Lambda_a) = \mathcal{L}(\theta; \Lambda)$ for every θ, and differentiating, $D_\theta\mathcal{L}(g_a\theta; \Lambda_a)\, Dg_a = D_\theta\mathcal{L}(\theta; \Lambda)$ with Dg_a diagonal and invertible, so critical points map to critical points. By (A.93), λ_i and the home values are unchanged, the foreign values $u_{m,k}$, $v_{m,k}$, and h_m are multiplied by a, $z_{mi,g} = 1$ is unchanged, and $z_{mj,g}$ and $z_{ij,g}$ for $j \neq i$ are multiplied by a, in agreement with Step 1.

The normalization of $\bar{t}_i$. The set of optima is a union of orbits of $\{g_a\}_{a>0}$. Along an orbit the import-value weights $q_{ji,k} \equiv P_{ji,k}Q_{ji,k}/\sum_{m\neq i,g} P_{mi,g}Q_{mi,g}$ are constant and the import-weighted mean tax factor $\sum_{j\neq i,k} q_{ji,k}(1+t_{ji,k})$ is multiplied by a, so it takes every positive value exactly once. Writing

$$1+\bar{t}_i \equiv \sum_{j\neq i}\sum_k q_{ji,k}\,(1+t^*_{ji,k}) \tag{A.115}$$

selects a representative, as in the normalization (A.49); summing (A.114) with the weights $q_{ji,k}$ shows that (A.115) is equivalent to $\sum_{j\neq i,k} q_{ji,k}\,\tilde{\omega}_{ji,k} = 0$. The orbit also accounts for the degree of freedom in the first-order conditions. Along $a \mapsto g_a\theta$ from a feasible θ every constraint remains zero, so $\mathcal{L}(g_a\theta; \Lambda) = V_i(\theta)$ for every a and every Λ; differentiating at $a=1$,

$$\sum_{m\neq i}\sum_k \left[\pi_{m,k}\frac{\partial\mathcal{L}}{\partial\pi_{m,k}} + \mathcal{Y}_{m,k}\frac{\partial\mathcal{L}}{\partial\mathcal{Y}_{m,k}}\right] + \sum_{m\neq i} w_m\frac{\partial\mathcal{L}}{\partial w_m} + \sum_{j\neq i}\sum_k \tilde{P}_{ij,k}\frac{\partial\mathcal{L}}{\partial\tilde{P}_{ij,k}} = 0 \tag{A.116}$$

at every feasible point. When the foreign conditions (A.98), (A.100), and (A.102) hold, (A.116) reduces to $\sum_{j\neq i,k} b_{ij,k} = 0$: one export condition is implied by the others, the counterpart of the

redundant equation in Step 4 of Section A.3.

The shifter $\bar{s}_i^{\mathcal{C}}$. Step 1 assumed M_i nonsingular, in which case the Home block forces $h_i = 1$ and the domestic line holds with $\bar{s}_i^{\mathcal{C}} = 0$: when every taxed item can be used both in consumption and in production, the planner has no neutral consumption tax, because a uniform rise in all home purchaser prices changes input demands and unit costs. The shifter arises when the tax items separate final from intermediate use, the fragmentation described in footnote 38 of the paper: every industry k is split into a final version k', whose output is bought only by consumers, and an intermediate version k'', whose output is bought only by producers, each version with its own producer price, revenue, and taxes, so that $Q^C_{mn,k''} = 0$ and $\alpha^{m,k'}_{n,g} = 0$. Under this fragmentation let S_c, $c > 0$, multiply Y_i and every final purchaser price $\tilde{P}_{mi,k'}$ (all origins m, including $m = i$) by c, leaving the intermediate prices $\tilde{P}_{mi,k''}$, the export prices, and every producer-side variable unchanged. Consumer demand and V_i are homogeneous of degree zero in final prices and income, input demand and unit costs depend on the intermediate prices only, and no foreign variable changes, so S_c changes no quantity and no constraint: $\mathcal{L}(S_c\theta; \Lambda) = \mathcal{L}(\theta; \Lambda)$ identically in θ, and critical points map to critical points with the same multipliers Λ and the same welfare. (The home budget, implied by the constraints, continues to hold: the additional revenue $(c-1)\sum_{m,k'} \tilde{P}_{mi,k'} Q^C_{mi,k'} = (c-1)Y_i$ is the rise in Y_i.) In tax terms, by (A.88), $1 + t_{mi,k'}$ is multiplied by c, $1 + s_{i,k'}$ is divided by c, and, since $\tilde{P}_{ij,k'}$ and $P_{ij,k'}$ are unchanged, $1 + x_{ij,k'}$ is multiplied by c; the taxes on intermediate items are unchanged. This is a uniform consumption tax at rate $c - 1$ on final goods, rebated to consumers. By (A.93), since $\lambda_i = \partial V_i / \partial Y_i$ is divided by c, every value u, v, and h is multiplied by c, $z_{mi,k'} = u_{m,k'} P_{mi,k'} / \tilde{P}_{mi,k'}$ is unchanged, and $z_{mi,k''}$ is multiplied by c.

The Home block reflects this symmetry. Consumer demand does not depend on intermediate prices and input demand does not depend on final prices, so M_i is block triangular. The rows (A.96) and (A.97) for final items involve only $z_{\cdot i,k'}$; they are Engel and Cournot aggregation over final goods, solved by $z_{\cdot i,k'} = 1$, and by nothing else when the matrix $[\tilde{P}_{ni,k'} \partial Q^C_{ni,k'} / \partial \tilde{P}_{mi,k'}]$ of final-good demand derivatives is nonsingular (for CES demand within an industry the matrix of log derivatives is $-\sigma_k I + (\sigma_k - 1)\mathbf{1}\lambda^\top$, with determinant $(-\sigma_k)^N / \sigma_k \neq 0$). The remaining rows, (A.97) for intermediate items and (A.98) for the home industries, involve $(z_{\cdot i,k''}, v'_i, h_i)$ and are homogeneous, the former by (A.104) with $Q^C_{mi,k''} = 0$ and the latter by $\sum_{m,g} \alpha^{m,g}_{i,k} + \beta_{i,k} = 1$ with $\alpha^{m,g'}_{i,k} = 0$. They are solved by $(z_{\cdot i,k''}, v'_i, h_i) = (c, c, c)$ for every c, and by nothing else when their coefficient matrix has rank one less than its number of unknowns. The solution set of the Home block is then $(z_{\cdot i,k'}, z_{\cdot i,k''}, v'_i, h_i) = (1, c, c, c)$: the kernel of M_i is the direction in which S_c moves the values.

Let θ^* be an optimum whose Home-block solution is $(1, c, c, c)$, so that $c = h_i > 0$. The point $\hat{\theta} \equiv S_{1/c}\theta^*$ is an optimum with the same multipliers Λ, the same welfare, and Home-block solution $(1, 1, 1, 1)$. Steps 1–3 used only this solution, so they apply to $\hat{\theta}$ verbatim: its taxes $\hat{s}, \hat{t}, \hat{x}$ satisfy the domestic, export, and import lines with $\bar{s}_i^{\mathcal{C}} = 0$ and some $\hat{\bar{t}}_i$, and its values $\hat{v}$ solve (A.109). The taxes of $\theta^* = S_c\hat{\theta}$ follow from the transformation rule: for intermediate items they coincide with those of $\hat{\theta}$; for final items $1 + t^*_{mi,k'} = c(1 + \hat{t}_{mi,k'})$, $1 + s^*_{i,k'} = (1 + \mu_k)/c$, and $1 + x^*_{ij,k'} = c(1 + \hat{x}_{ij,k'})$.

These are the three lines of the theorem with $1+\bar{s}_i^{\mathcal{C}}=c$ for final items and $\bar{s}_i^{\mathcal{C}}=0$ for intermediate items, where $\bar{t}_i$, $\tilde{\omega}$, $\tilde{\Lambda}$, and ν are those of $\hat{\theta}$: the domestic line directly; the import line because $\tilde{\omega}_{ji,k}$, built from quantities, from derivatives through the foreign equilibrium, and from the ratios $(1+\hat{x}_{in,g})/(1+\hat{\bar{t}}_i)$, is the same at θ^* and at $\hat{\theta}$; the export line because the demand blocks vanish between final and intermediate items and $\tilde{\Lambda}_{ij,k'}=0$ for a final item ($\beta_{ij,k'g}=0$), so the final rows of $\hat{\theta}$'s export line, $\sum_{g'}[E_{ij}]_{k',g'}(1+\hat{x}_{ij,g'})+\sum_{m'\neq i,g'}[E_{ij}^{(m'j)}]_{k',g'}(1+\hat{t}_{m'i,g'})=0$, multiplied by c are the final rows at θ^*, while the intermediate rows, in which $\tilde{\Lambda}_{ij,k''}(1+\bar{t}_i)=\sum_g(1+\mu_g)\beta_{ij,k''g}\hat{v}_{j,g}$, are unchanged. This is the sense in which the theorem evaluates $\tilde{\Lambda}_{ij}$ and $\tilde{\omega}_{ji,k}$ at $\bar{s}_i^{\mathcal{C}}=0$: since S_c changes no quantity and no elasticity, it amounts to solving the system for ν with the tax factors of $\hat{\theta}$, that is, with the final-item factors $1+t^*_{mi,k'}$ and $1+x^*_{ij,k'}$ divided by $1+\bar{s}_i^{\mathcal{C}}$. Because S_c preserves welfare, c is not determined: $1+\bar{s}_i^{\mathcal{C}}$ is an arbitrary positive number, as the theorem states. Under the fragmentation, the nonsingularity of M_i in the list of regularity conditions is replaced by the rank condition on its two blocks.

The converse. Let an allocation be feasible, let its taxes satisfy the domestic line with $\bar{s}_i^{\mathcal{C}}=0$, the export line, and the import line for some $\bar{t}_i$, and let $\nu_{m,k}$, $m\neq i$, solve (A.109). Set $u_{m,k}=1+t_{mi,k}$ for $m\neq i$, $u_{i,k}=\nu_{i,k}=1/(1+\mu_k)$, $h_i=1$, define h_m by (A.102), the values z by (A.93), and the multipliers $a_{n,k}=\lambda_i u_{n,k}$, $\varphi_{n,k}=\lambda_i\nu_{n,k}\mathcal{Y}_{n,k}/\mathrm{C}_{n,k}^{1+\mu_k}$, $\psi_n=\lambda_i h_n$. The Home block holds because $(1,1,1)$ solves it, (A.99) holds because the domestic line gives $u_{i,k}=1/(1+s_{i,k})=1/(1+\mu_k)=\nu_{i,k}$, (A.101) is the export line, (A.100) is the system for ν, and (A.102) defines h_m. It remains to verify (A.98) for $m\neq i$. Let $\rho\equiv\partial\mathcal{L}/\partial\xi$ at these multipliers; its π- and w-components vanish by what was just shown. The identity (A.111) holds for any multipliers, and $\partial\mathcal{L}/\partial\xi=\partial\mathcal{L}^H/\partial\xi+\partial\mathcal{L}^F/\partial\xi$ with $\partial\mathcal{L}^H/\partial\xi=-\sum_{n\neq i,g}a_{i,g}P_{in,g}\,\partial Q_{in,g}/\partial\xi$, so

$$\rho\,\frac{d\xi}{dQ_{ji,k}}=-\sum_{n\neq i}\sum_g a_{i,g}P_{in,g}\frac{dQ_{in,g}}{dQ_{ji,k}}+a_{j,k}P_{ji,k}=\lambda_i\Big[u_{j,k}P_{ji,k}-\sum_{n\neq i}\sum_g u_{i,g}P_{in,g}\frac{dQ_{in,g}}{dQ_{ji,k}}\Big]=0,$$

where the last equality is the balance form (A.112), implied by the import line through the reversible steps of Step 3. Since the π- and w-components of ρ are zero, this says $\rho_{\mathcal{Y}}\,\partial\mathcal{Y}_{-i}/\partial q=0$, where $\partial\mathcal{Y}_{-i}/\partial q$ is the $(N-1)K$-square matrix of derivatives of foreign revenues with respect to the import quantities through the foreign equilibrium. When it is nonsingular, $\rho_{\mathcal{Y}}=0$ and every first-order condition holds. Under the fragmentation, the converse applies to the representative $\hat{\theta}$, and S_c extends it to any $\bar{s}_i^{\mathcal{C}}$.

Regularity. The proof uses the constraint qualification for (A.90); the nonsingularity of M_i, or, when the tax items separate final from intermediate use, the rank condition on its two blocks with kernel spanned by the consumption-tax direction; the nonsingularity of $\partial\Phi/\partial\xi$ in Step 3 and of $\partial\mathcal{Y}_{-i}/\partial q$ in the converse; and the invertibility of E_{ij} and of $I-M$ in the displays of the theorem. These hold generically. As in Sections A.2 and A.3, the first-order conditions are necessary at an optimum satisfying the constraint qualification and sufficient for a local optimum under the second-order conditions; the theorem characterizes the optimum and does not assert its uniqueness. □

A.7. Proofs of the remarks and of the Cobb–Douglas formulas

Remark (a). The balance form is (A.112), which Step 3 of Section A.6 derives from the foreign first-order conditions before $\bar{t}_i$ is introduced; the import line is its rearrangement by the differentiated trade balance (A.113), and Step 3 shows that the rearrangement is reversible. Its right-hand side uses $u_{i,g}P_{in,g} = (1 + x^*_{in,g})\tilde{P}_{in,g}$, which follows from $u_{i,g} = 1/(1 + s_{i,g})$ and (A.88).

Remark (b). For $n \neq i$, substitute into (A.98) the values of Step 1 of Section A.6, $u_{n,k} = 1 + t^*_{ni,k}$ and $z_{mn,g} = v_{mn,g}$, and h_n from (A.102), where $\ell_{n,k} = \beta_{n,k}\mathcal{Y}_{n,k}/(w_n L_n) = \beta_{n,k}\mathcal{Y}_{n,k}/Y_n$ because $Y_n = w_n L_n$; the result is the display of Remark (b). It is a first-order condition in its own right and holds at the optimum together with the balance form; the converse in Step 4 shows that, given the Home block, the export line, and the system for ν, the conditions (A.98) for $n \neq i$ are equivalent to the balance form whenever $\partial \mathcal{Y}_{-i}/\partial q$ is nonsingular. For the two-country check, with one industry and no inputs, $\alpha = 0$ and $\beta_j = \ell_j = 1$, so the display reads $1 + t^* = \bar{v}_j + \nu_j$; with Cobb–Douglas preferences $\bar{v}_j = (1 + x^*)\lambda_{ij} + (1 + t^*)\lambda_{jj}$, and $\nu_j = (1 + t^*) + \lambda_{jj}\sigma(x^* - t^*)$ by the two-country formulas below. Solving, $(1 + t^*)/(1 + x^*) = 1 + 1/[(\sigma - 1)\lambda_{jj}]$, in which μ cancels.

Remark (c). Let country i's shares in foreign markets vanish at a common rate ε: the revenue shares $r_{ni,k}$, the expenditure shares $e_{in,g}$, the cost shares $\alpha^{i,g}_{n,k}$, and the ratios of country i's trade flows to foreign revenues and incomes are $O(\varepsilon)$, while demand elasticities, cost-share derivatives, the ratios among country i's own trade flows, and $(\partial\Phi/\partial\xi)^{-1}$ stay bounded. (i) $\tilde{\omega}_{ji,k} \to 0$. In Step 3 of Section A.6, $Q_{ji,k}$ enters Φ only through $(\mathrm{G})_{j,k}$, with $\partial(\mathrm{G})_{j,k}/\partial \ln Q_{ji,k} = -P_{ji,k}Q_{ji,k} = -r_{ji,k}\mathcal{Y}_{j,k}$. Scaling each constraint by the corresponding foreign revenue or income, the forcing term of the implicit function theorem is $O(\varepsilon)$, so $d \ln \xi / d \ln Q_{ji,k} = O(\varepsilon)$ and with it $\partial \ln P_{mi,g}/\partial \ln Q_{ji,k} = d \ln \pi_{m,g}/d \ln Q_{ji,k}$ and $\partial \ln Q_{in,g}/\partial \ln Q_{ji,k}$, since $Q_{in,g}$ depends on ξ through Y_n, $\mathcal{Y}_n$, and the foreign prices $\tau_{mn,g}\pi_{m,g}$ at fixed $\tilde{P}_{in,g}$. The weights in $\tilde{\omega}_{ji,k}$ are ratios among country i's trade flows and stay bounded, so $\tilde{\omega}_{ji,k} \to 0$, and the import line gives $1 + t^*_{ji,k} \to (1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i)$. (ii) $\nu \to 0$ and $\tilde{\Lambda}_{ij} \to 0$. In $\Gamma_{nn',k}$ for $n, n' \neq i$, written as in (A.107), the terms with $m' = i$ carry the factors $e_{in',s}/e_{nn',k} = O(\varepsilon)$ and vanish, and the import-tax factors in the remaining terms converge to the common value $(1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i)$. The remaining coefficients sum to -1 by Marshallian adding-up for total demand: Cournot aggregation for consumption and (A.104) for each industry's input demand give $\sum_{m,s} \tilde{P}_{mn',s}\,\partial Q_{mn',s}/\partial \tilde{P}_{nn',k} = -Q_{nn',k}$ at fixed $Y_{n'}$ and $\mathcal{Y}_{n'}$, which is the first display of Remark (c). Hence $\Gamma_{nn',k} \to -(1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i) + \sum_g (1 + \mu_g)\beta_{nn',kg}\nu_{n',g}$. Substituting into (A.109) and using $\sum_{n'} r_{nn',k} = 1$,

$$\nu_{n,k} \to (1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i)\, r_{ni,k} + \sum_{n' \neq i} \sum_g r_{nn',k}\,(1 + \mu_g)\,\beta_{nn',kg}\,\nu_{n',g},$$

that is, $\nu = (1 + \bar{t}_i)(1 + \bar{s}^{\mathcal{C}}_i)\, r_{\cdot i} + M\nu$ with the matrix M of (A.110). The entries of M, which equal $(1 + \mu_g)\alpha^{n,k}_{n',g}\mathcal{Y}_{n',g}/\mathcal{Y}_{n,k}$, stay bounded. When $I - M$ is invertible, $\nu \to 0$ because $r_{\cdot i} \to 0$, and $\tilde{\Lambda}_{ij,k} \to 0$ because $\beta_{ij,kg} \leq 1$. Applying Marshallian adding-up to the rows ij,k of the export

line, $E_{ij}^{(-ij)}\mathbf{1} = -\mathbf{1} - E_{ij}\mathbf{1}$, so the export line becomes $E_{ij}(1+x_{ij}^*) = (1+\bar{t}_i)(1+\bar{s}_i^{\mathcal{C}})(\mathbf{1}+E_{ij}\mathbf{1})$, the small-open-economy form of Theorem 1 of the paper. With two countries and one industry, $M = (1+\mu)\alpha_j^j \to (1+\mu)\bar{\alpha}_j$, since the share of home inputs in $\bar{\alpha}_j$ vanishes.

Remark (d). Footnote 40 of the paper evaluates $\omega_{ji,k}$ at fixed wages, where without scale economies producer prices do not respond to import quantities and the published $\omega_{ji,k}$ vanishes. The second term of $\tilde{\omega}_{ji,k}$ does not. Through $(\mathrm{G})_{j,k}$, a unit increase in $Q_{ji,k}$ raises industry j,k's revenue by $P_{ji,k}$ and therefore its purchases of inputs from country i by $\alpha_{j,k}^{i,g}P_{ji,k}$ for each g, at fixed prices, wages, and income. This first-order effect contributes the term $\sum_g[(1+x_{ij,g}^*)/(1+\bar{t}_i)-1]\,\alpha_{j,k}^{i,g}$ to $\tilde{\omega}_{ji,k}$; the remaining effects propagate the revenue change through the input–output network, foreign income, and prices and do not cancel it in general. The term differs from zero when the export-subsidy factors differ from the mean import-tax factor, and it differs across import items with the exporting industry's use of home inputs, so the import taxes are not strictly uniform without scale economies.

Cobb–Douglas preferences (proof of (51)). With Cobb–Douglas spending shares $\theta_{j,k}$ for consumers and $\alpha_{j,g}^k$ per unit of revenue for industry j,g, and the same CES aggregator across origins (elasticity σ_k) for both uses, the input cost shares are $\alpha_{j,g}^{m,k} = \alpha_{j,g}^k\lambda_{mj,k}$, gross spending on industry k in market j is $\mathcal{E}_{j,k} = \theta_{j,k}Y_j + \sum_g \alpha_{j,g}^k\mathcal{Y}_{j,g}$, and $\tilde{P}_{mj,k}Q_{mj,k} = \lambda_{mj,k}\mathcal{E}_{j,k}$. Hence $e_{mj,k}/e_{m'j,k} = \lambda_{mj,k}/\lambda_{m'j,k}$, $\beta_{mj,kg} = \alpha_{j,g}^k\mathcal{Y}_{j,g}/\mathcal{E}_{j,k}$ independently of the origin m, and, at fixed Y_j and $\mathcal{Y}_j$, $\partial\ln Q_{mj,k}/\partial\ln\tilde{P}_{m'j,g} = \mathbf{1}_{g=k}[-\sigma_k\mathbf{1}_{m=m'} + (\sigma_k-1)\lambda_{m'j,k}]$, so that $[E_{mj}^{(m'j)}]_{k,g} = \mathbf{1}_{g=k}[-\sigma_k\mathbf{1}_{m=m'} + (\sigma_k-1)\lambda_{mj,k}]$. By (A.107),

$$\Gamma_{m'j,k} = -\sigma_k z_{m'j,k} + (\sigma_k-1)\sum_m \lambda_{mj,k}z_{mj,k} + \tilde{\Lambda}_{j,k}(1+\bar{t}_i), \qquad \tilde{\Lambda}_{j,k}(1+\bar{t}_i) \equiv \sum_g (1+\mu_g)\,\alpha_{j,g}^k\,\frac{\mathcal{Y}_{j,g}}{\mathcal{E}_{j,k}}\,\nu_{j,g},$$

with $z_{ij,k} = 1+x_{ij,k}^*$ and $z_{mj,k} = 1+t_{mi,k}^*$ for $m \neq i$. The export line $\Gamma_{ij,k} = 0$ gives $(1+x_{ij,k}^*)[\sigma_k - (\sigma_k-1)\lambda_{ij,k}] = (\sigma_k-1)\sum_{m\neq i}\lambda_{mj,k}(1+t_{mi,k}^*) + \tilde{\Lambda}_{j,k}(1+\bar{t}_i)$, the second line of (51) once the import line $1+t_{mi,k}^* = (1+\tilde{\omega}_{mi,k})(1+\bar{t}_i)$, its first line, is substituted; goods serve both uses, so $\bar{s}_i^{\mathcal{C}} = 0$. Subtracting the export line from $\Gamma_{jn,g}$ gives $\Gamma_{jn,g} = -\sigma_g(z_{jn,g} - z_{in,g}) = \sigma_g(x_{in,g}^* - t_{ji,g}^*)$, and (A.109) becomes the explicit formula for $\nu_{j,g}$.

Two countries, one industry (proof of (52)). Write $T \equiv 1+t^*$, $X \equiv 1+x^*$, and $A \equiv (1+\mu)\bar{\alpha}_j$. Labor-market clearing gives $Y_j = (1-\bar{\alpha}_j)\mathcal{Y}_j$, so $\mathcal{E}_j = Y_j + \bar{\alpha}_j\mathcal{Y}_j = \mathcal{Y}_j$, $r_{jj} = \lambda_{jj}\mathcal{E}_j/\mathcal{Y}_j = \lambda_{jj}$, and $\tilde{\Lambda}_j(1+\bar{t}_i) = A\nu_j$. The Cobb–Douglas formulas give $\nu_j = T + \lambda_{jj}\sigma(X-T)$ and $X[1+(\sigma-1)\lambda_{jj}] = (\sigma-1)\lambda_{jj}T + A\nu_j$. Eliminating ν_j, $X[1+(\sigma-1)\lambda_{jj} - A\sigma\lambda_{jj}] = T[(\sigma-1)\lambda_{jj} + A(1-\sigma\lambda_{jj})]$, hence

$$\frac{T}{X} - 1 = \frac{1-A}{(\sigma-1)\lambda_{jj} + A(1-\sigma\lambda_{jj})} = \frac{1-A}{(1-A)(\sigma-1)\lambda_{jj} + A\lambda_{ij}},$$

using $1-\sigma\lambda_{jj} = \lambda_{ij} - (\sigma-1)\lambda_{jj}$. The published ratio follows from the same export line with $A\nu_j$ replaced by the reimported fraction $\bar{\alpha}_j\lambda_{ij}$ of a unit of exports, valued at T: $X[1+(\sigma-1)\lambda_{jj}] = T[(\sigma-1)\lambda_{jj} + \bar{\alpha}_j\lambda_{ij}]$.

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